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(12) **United States Patent
Cottrell**

(10) **Patent No.: US 12,383,106 B2**
(45) **Date of Patent: Aug. 12, 2025**

(54) **WET DRY APPLIANCE**

(71) Applicant: **SHARKNINJA OPERATING LLC,**
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(72) Inventor: **Lee Cottrell,** Edenbridge (GB)

(73) Assignee: **SharkNinja Operating LLC,**
Needham, MA (US)

9,408,517 B2 8/2016 Fitzpatrick et al.
9,661,975 B2 5/2017 Charlton
9,687,133 B2 6/2017 Zhu et al.
9,833,117 B2 12/2017 Ludtke et al.
10,092,155 B2 10/2018 Xia et al.
10,182,697 B2 1/2019 Schultheis et al.
10,478,038 B1 11/2019 Gardner
10,512,383 B2 12/2019 Luyckx et al.
10,631,702 B2 4/2020 Xia et al.

(Continued)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 89 days.

FOREIGN PATENT DOCUMENTS

AU 2019101774 A4 3/2021
CA 3025513 A1 5/2019

(Continued)

(21) Appl. No.: **17/832,192**

(22) Filed: **Jun. 3, 2022**

(65) **Prior Publication Data**

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Related U.S. Application Data

(60) Provisional application No. 63/285,867, filed on Dec.
3, 2021.

(51) **Int. Cl.**
A47L 7/00 (2006.01)

(52) **U.S. Cl.**
CPC **A47L 7/0028** (2013.01)

(58) **Field of Classification Search**
CPC A47L 7/0028; A47L 7/0038; A47L 7/0023
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,491,005 B2 2/2009 Blom
9,131,824 B2 9/2015 Crawford et al.
9,320,406 B2 4/2016 Irwin et al.

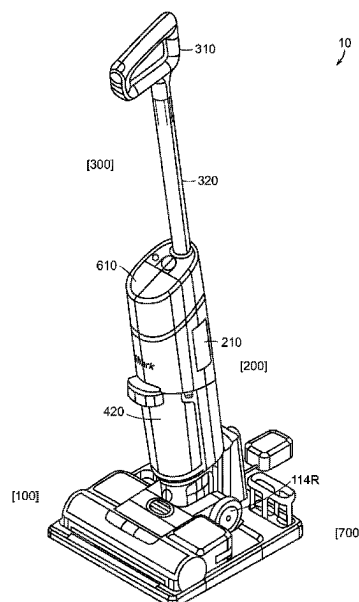
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(57) **ABSTRACT**

A cleaning device and methods for cleaning are provided. In one embodiment, the cleaning device includes a head assembly, a body assembly, and a handle assembly. The cleaning device also includes components that enable the cleaning device to operate in dry cleaning modes and wet cleaning modes. Dry cleaning modes can employ a vacuum assembly, including a motor, tubing, and a fluid recovery tank in order to draw in debris and waste into a fluid recovery tank. Wet cleaning modes can further employ a fluid supply tank, a pump, and tubing in order to supply fluid to a brushroll to aid in a cleaning process.

12 Claims, 52 Drawing Sheets



(56)

References Cited**U.S. PATENT DOCUMENTS**

10,682,034	B2	6/2020	Ribbe et al.	
10,820,769	B2	11/2020	Xia et al.	
D908,994	S	1/2021	Xia et al.	
D908,995	S	1/2021	Xia et al.	
10,925,455	B1	2/2021	Xia et al.	
10,966,586	B2	4/2021	Luyckx et al.	
10,973,383	B1	4/2021	Nguyen et al.	
11,006,809	B2	5/2021	Li	
11,013,389	B2	5/2021	Resch et al.	
11,013,392	B2	5/2021	Mcdonnell	
11,071,428	B2	7/2021	Resch et al.	
11,076,735	B2	8/2021	Resch et al.	
11,089,933	B2	8/2021	Xia et al.	
11,096,539	B2	8/2021	Xia et al.	
11,096,540	B2	8/2021	Xia et al.	
11,096,541	B2	8/2021	Xia et al.	
11,096,542	B2	8/2021	Xia et al.	
11,096,543	B2	8/2021	Xia et al.	
11,096,544	B2	8/2021	Nguyen et al.	
11,122,949	B2	9/2021	Xia et al.	
11,122,956	B2	9/2021	Harrington et al.	
11,147,426	B2	10/2021	Nguyen et al.	
11,147,428	B2	10/2021	Metzel et al.	
11,147,429	B2	10/2021	Mcdonnell	
11,185,205	B2	11/2021	Luyckx et al.	
11,202,549	B2	12/2021	Krebs et al.	
11,241,134	B2	2/2022	Xia et al.	
11,304,581	B2	4/2022	Resch et al.	
11,375,868	B2	7/2022	Zhu et al.	
2005/0273969	A1 *	12/2005	Watson	A47L 9/2857 15/344
2006/0150364	A1 *	7/2006	Morgan	A47L 7/0042 15/320
2017/0119225	A1 *	5/2017	Xia	A47L 11/302
2019/0082925	A1	3/2019	Conrad et al.	
2019/0125156	A1	5/2019	Gunter et al.	
2020/0060498	A1	2/2020	Bradley et al.	
2020/0214527	A1	7/2020	Resch et al.	
2020/0253447	A1	8/2020	Wang et al.	
2021/0022581	A1	1/2021	Piksa	
2021/0113047	A1	4/2021	Zhao et al.	
2021/0169294	A1	6/2021	Pruitt et al.	
2021/0228046	A1	7/2021	Nguyen et al.	
2021/0267428	A1	9/2021	Scholten et al.	
2021/0298554	A1	9/2021	Resch et al.	
2021/0321849	A1	10/2021	Resch et al.	
2021/0338033	A1	11/2021	Nguyen et al.	
2022/0015597	A1	1/2022	Nguyen et al.	
2022/0061622	A1	3/2022	Luyckx et al.	
2022/0071466	A1	3/2022	Xia et al.	

FOREIGN PATENT DOCUMENTS

CA	3028214	A1	6/2019
CA	3104286	A1	12/2019
CA	3104290	A1	12/2019
CA	3123772	A1	12/2019
CA	3104494	A1	7/2020
CA	3104496	A1	7/2020
CA	3104499	A1	7/2020
CA	3104501	A1	7/2020
CA	3066796	C	7/2021
CN	105816120	A	8/2016

CN	106108810	A	11/2016
CN	206434268	U	8/2017
CN	107348918	A	11/2017
CN	106108810	B	8/2018
CN	108720754	A	11/2018
CN	108903859	A	11/2018
CN	108937750	A	12/2018
CN	109124511	A	1/2019
CN	105816120	B	2/2019
CN	109431401	A	3/2019
CN	109938647	A	6/2019
CN	209463915	U	10/2019
CN	108720754	B	10/2020
CN	211933874	U	11/2020
CN	112469319	A	3/2021
CN	112512394	A	3/2021
CN	109124511	B	4/2021
CN	112806900	A	5/2021
CN	112842164	A	5/2021
CN	112914441	A	6/2021
CN	112932324	A	6/2021
CN	112956946	A	6/2021
CN	112971589	A	6/2021
CN	113057539	A	7/2021
CN	213850446	U	8/2021
CN	213883040	U	8/2021
CN	213883041	U	8/2021
CN	113455960	A	10/2021
CN	112469319	B	4/2022
EP	3491991	A1	6/2019
EP	3466311	B1	2/2020
EP	3491986	B1	6/2020
EP	3501363	B1	10/2020
EP	3763269	A1	1/2021
EP	3162262	B1	2/2021
EP	3785594	A1	3/2021
EP	3787468	A1	3/2021
EP	3788922	A1	3/2021
EP	3793420	A1	3/2021
EP	3834689	A1	6/2021
EP	3834690	A1	6/2021
EP	3766397	B8	11/2021
EP	3838096	B1	11/2021
EP	3788932	B1	12/2021
EP	3847940	B1	2/2022
EP	3888519	B1	2/2022
EP	3679845	B1	3/2022
EP	3815592	B1	3/2022
EP	3875013	B1	3/2022
WO	2019246476	A1	12/2019
WO	2022116343	A1	6/2022
WO	2022116344	A1	6/2022

OTHER PUBLICATIONS

U.S. Appl. No. 17/832,155, filed Jun. 3, 2022, Wet Dry Appliance.
U.S. Appl. No. 17/832,136, filed Jun. 3, 2022, Wet Dry Appliance.
Lechler USA (2021) "Flat Fan Nozzles", <https://web.archive.org/web/20211130173710/https://www.lechlerusa.com/en/products/product-by-type/flat-fan-nozzles>, pp. 1-5.
U.S. Appl. No. 18/887,643, filed Sep. 17, 2024, Wet Dry Appliance.
International Search Report and Written Opinion received for PCT Patent Application No. PCT/US2024/024011, mailed on Aug. 9, 2024, 11 pages.

* cited by examiner

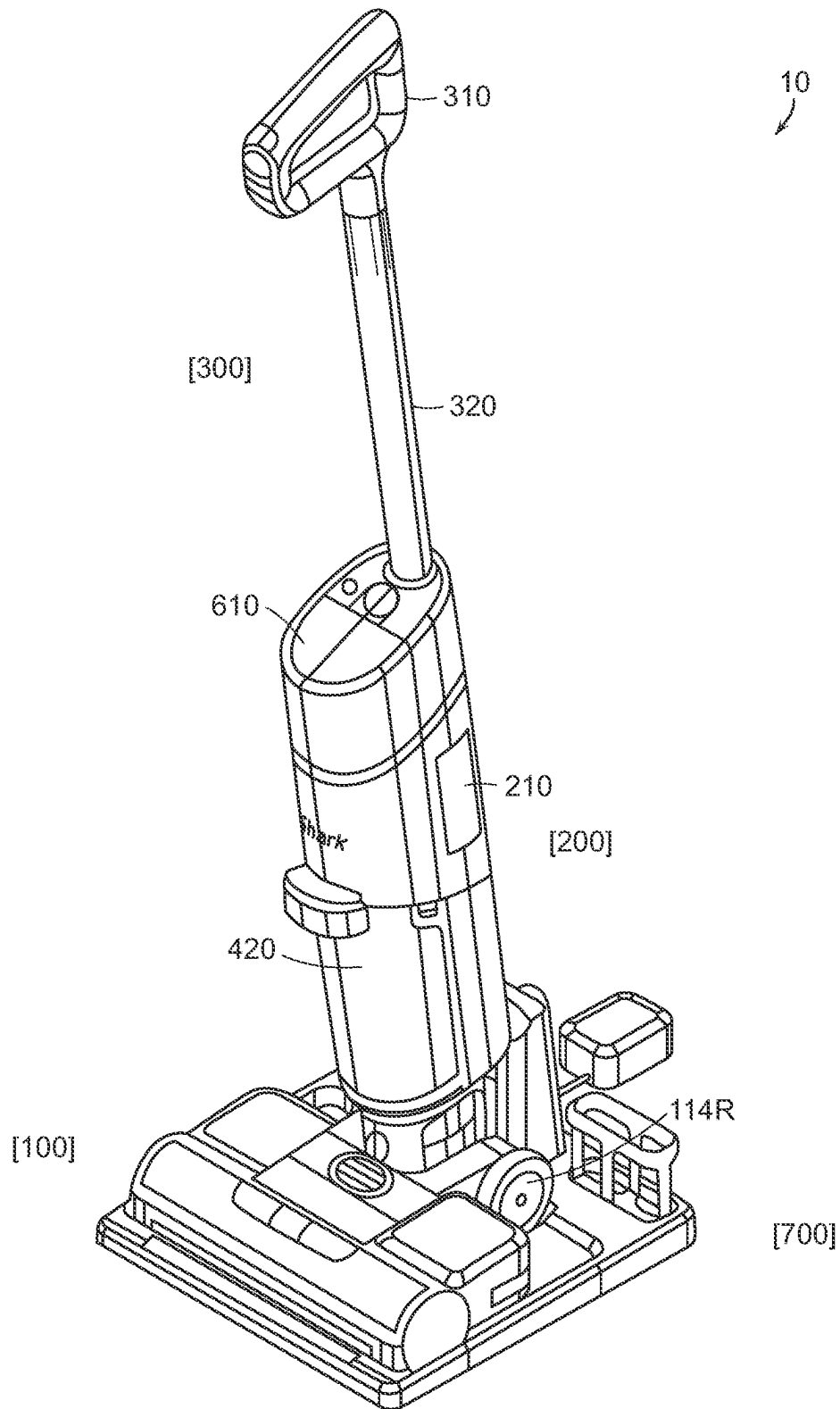


FIG. 1A

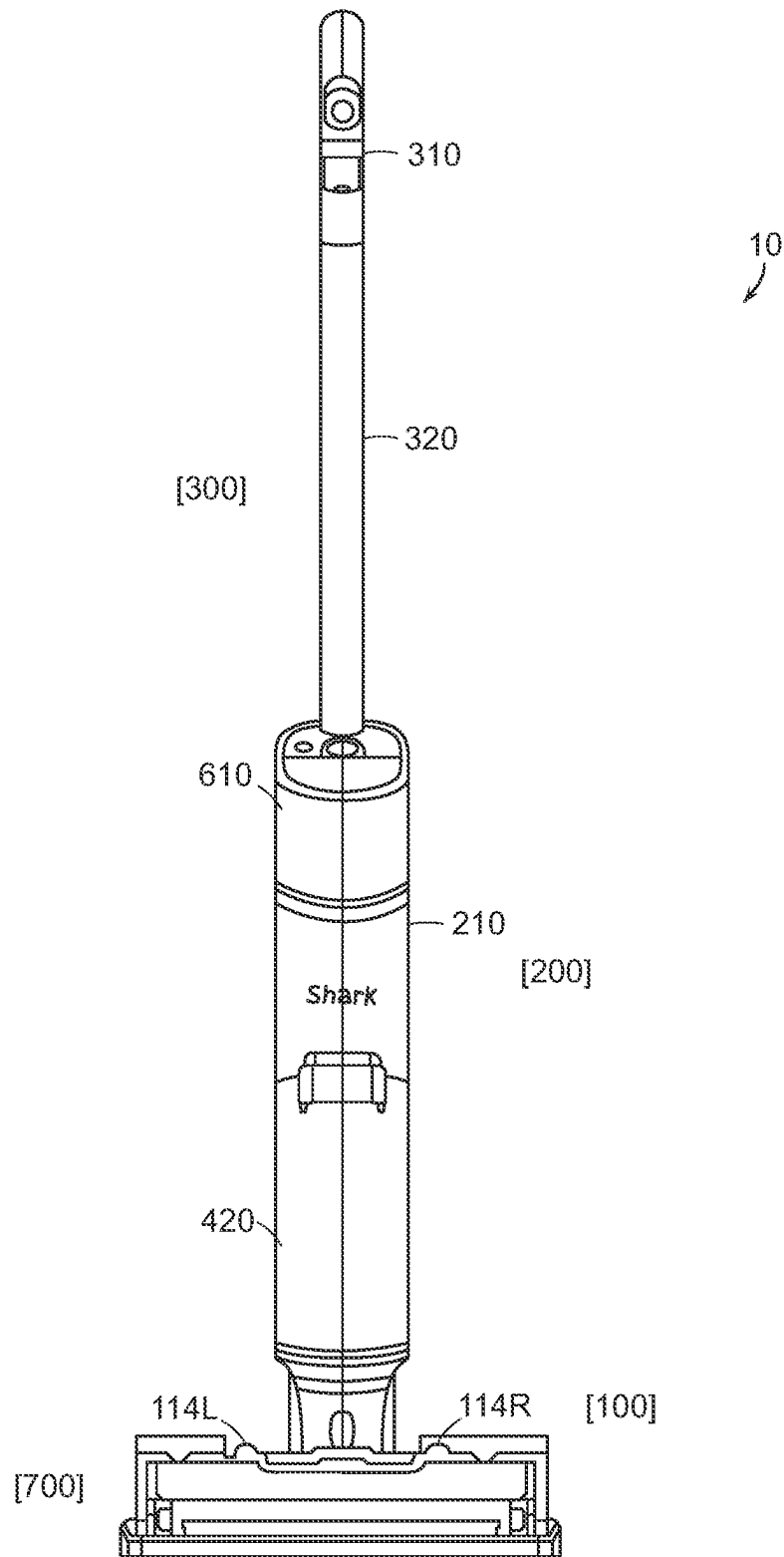


FIG. 1B

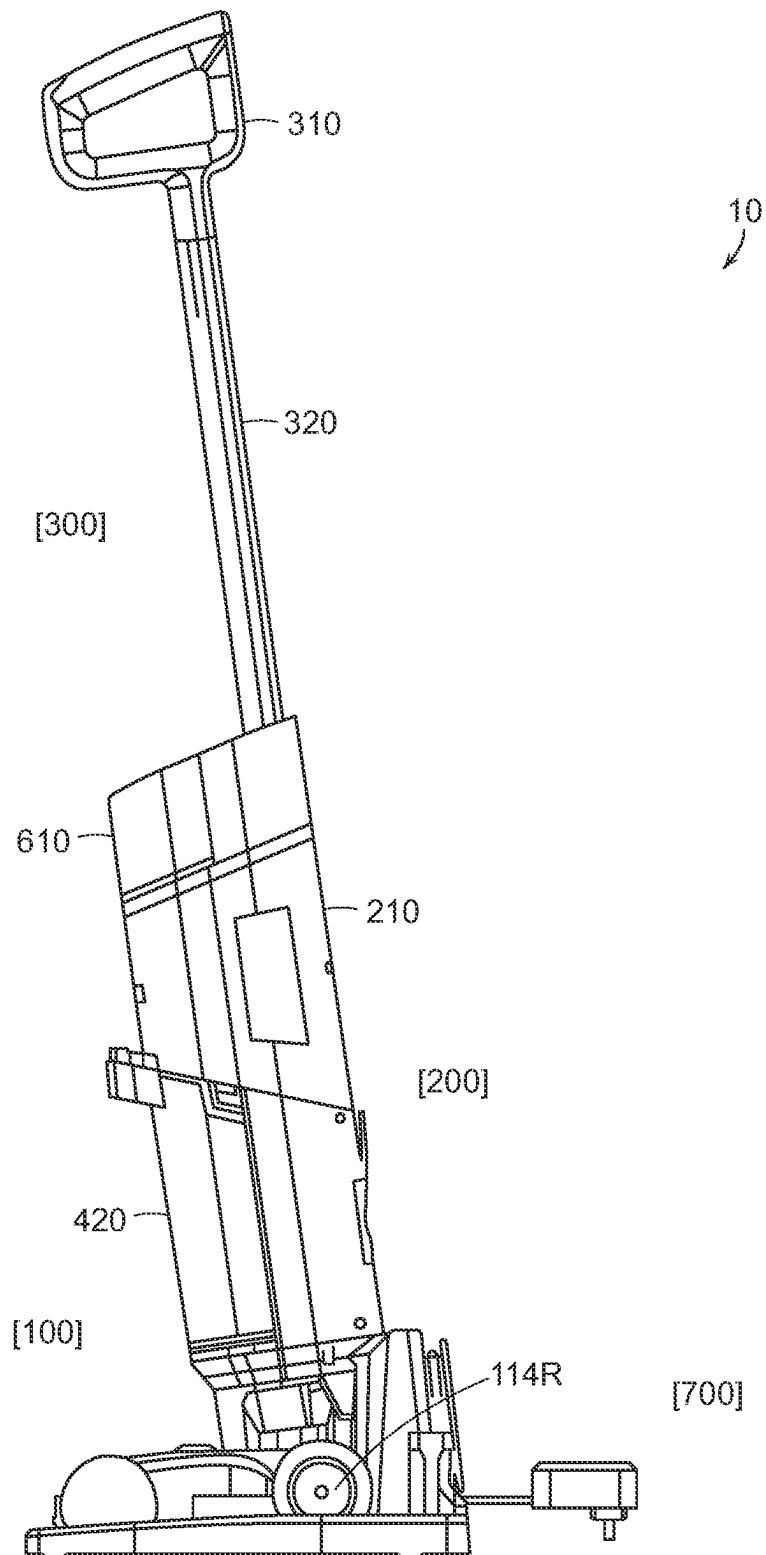


FIG. 1C

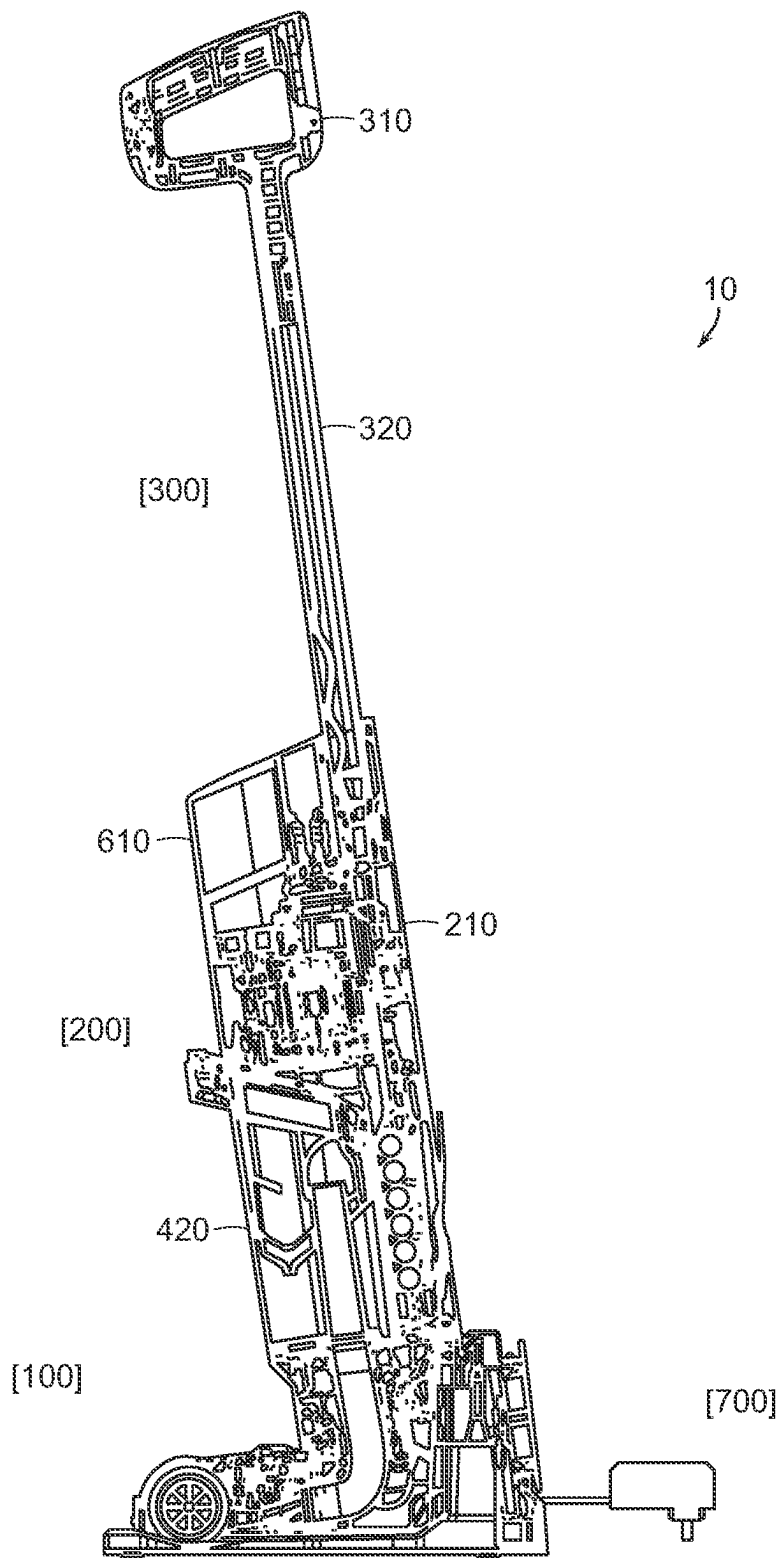


FIG. 1D

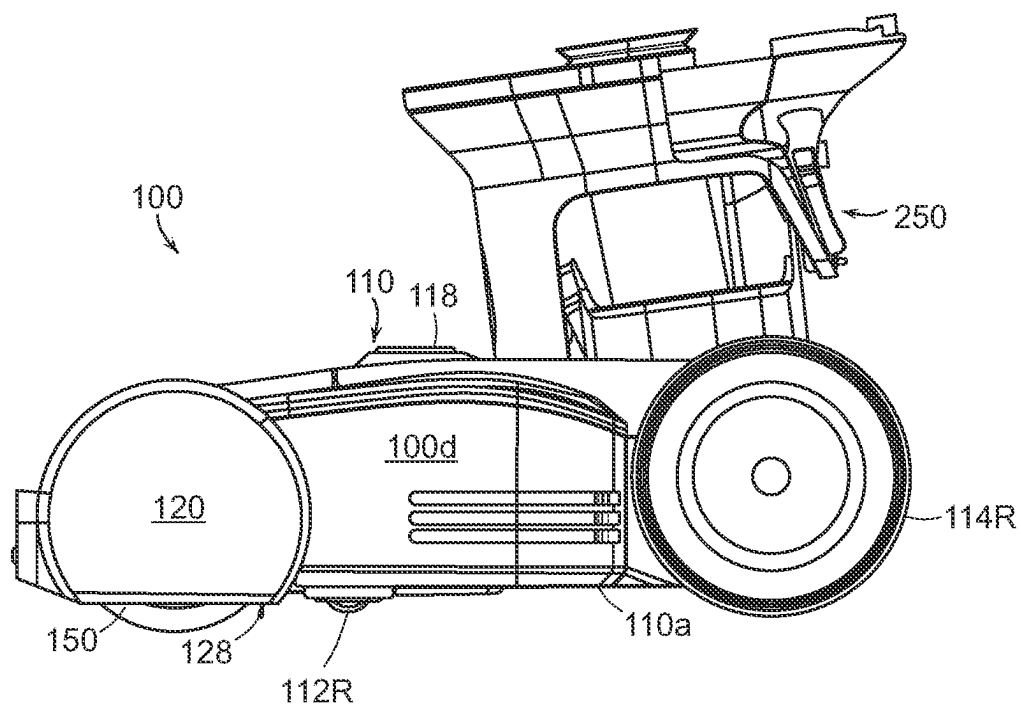
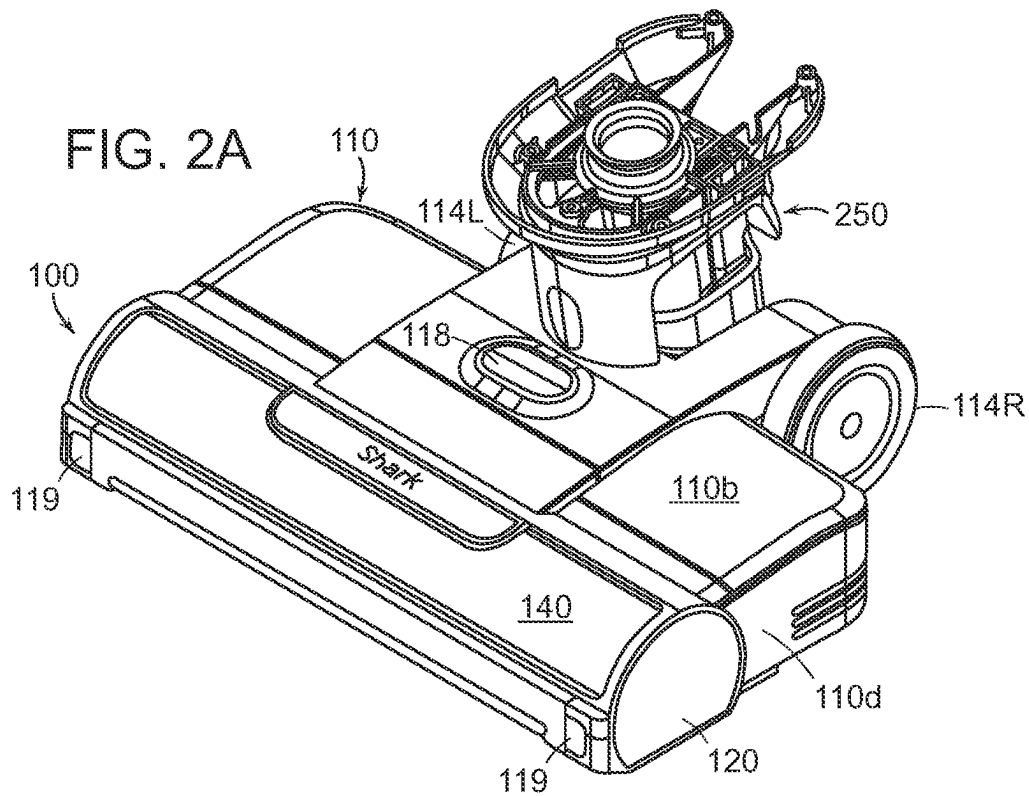


FIG. 2B

FIG. 2C

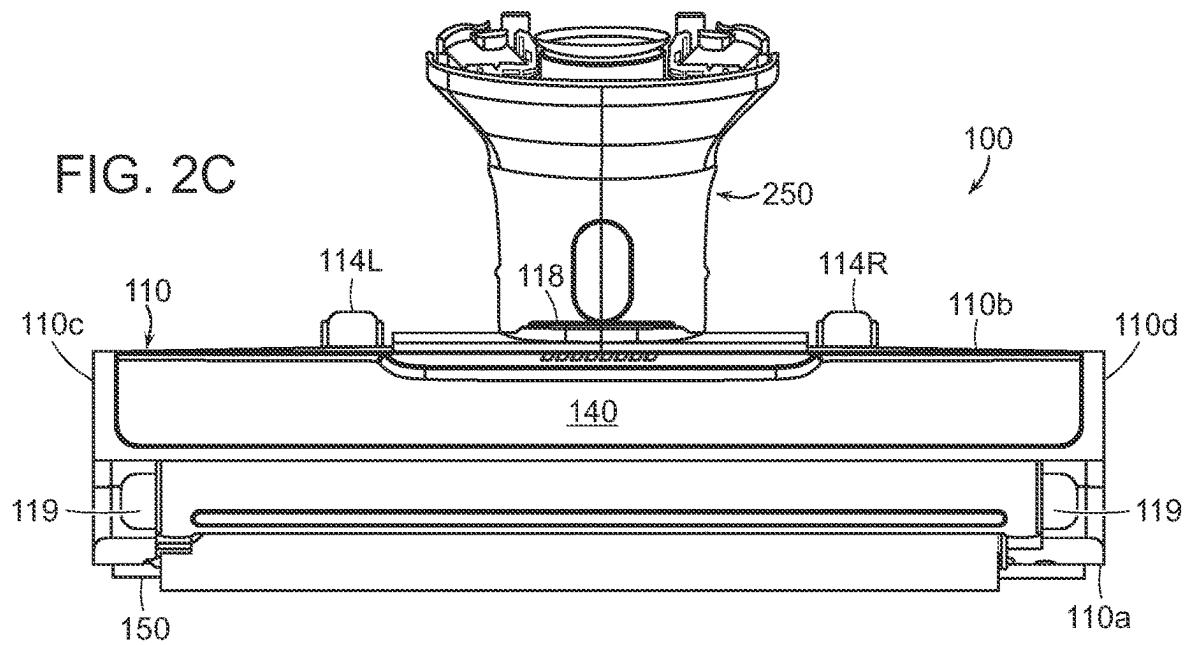


FIG. 2D

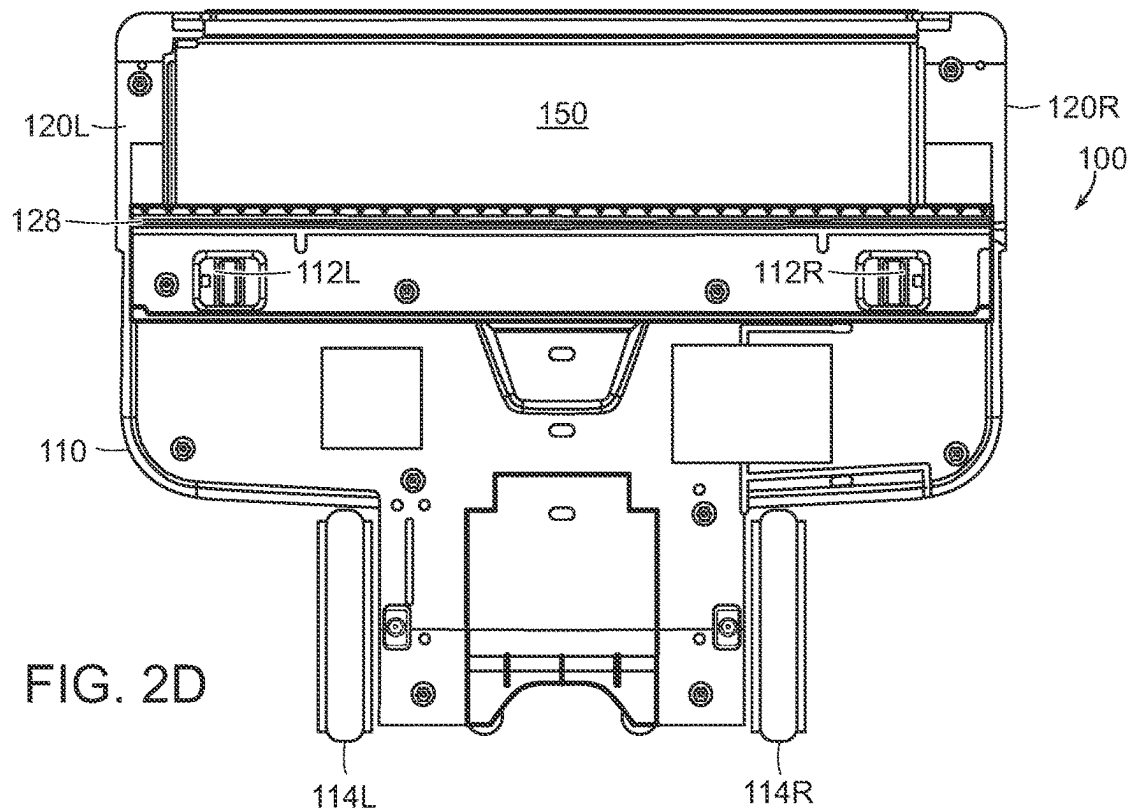


FIG. 2E

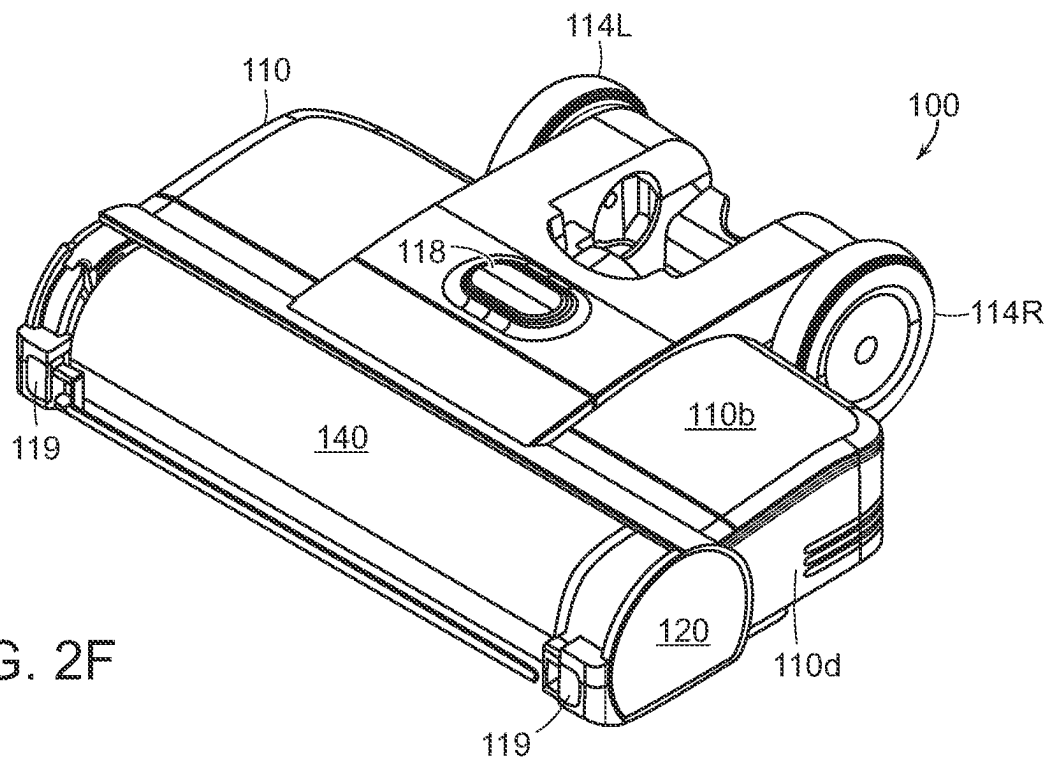
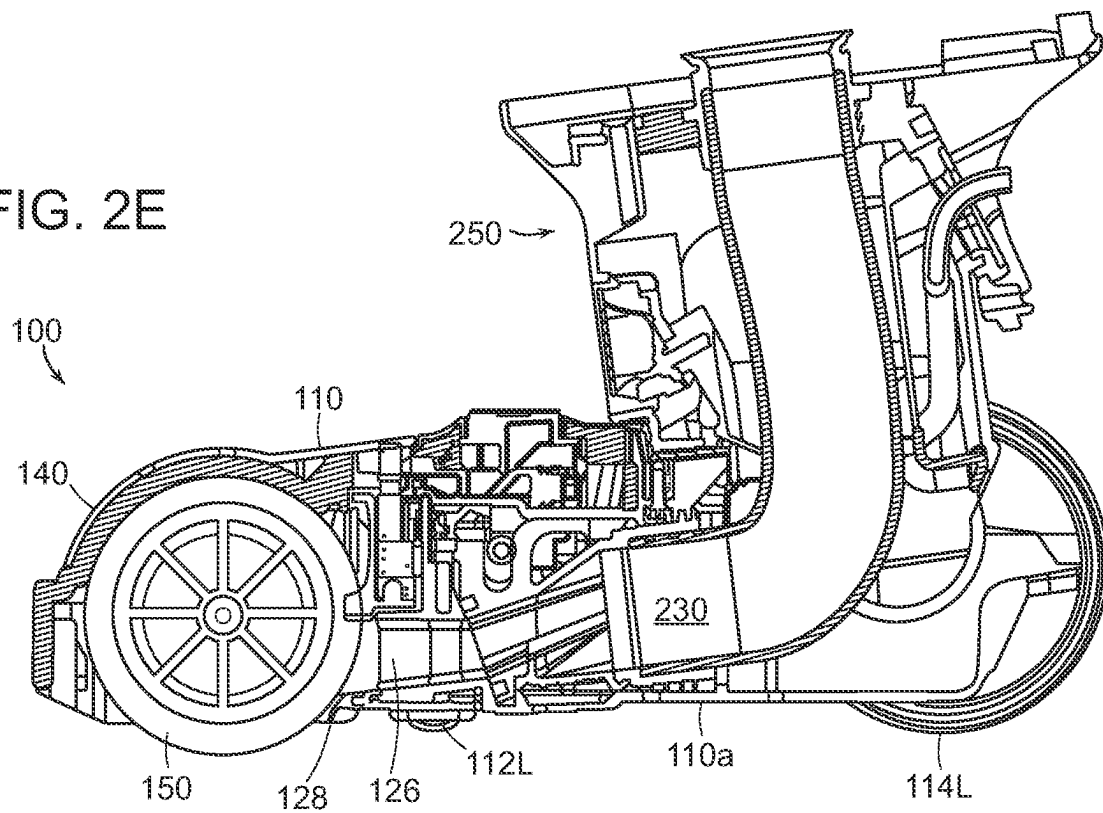
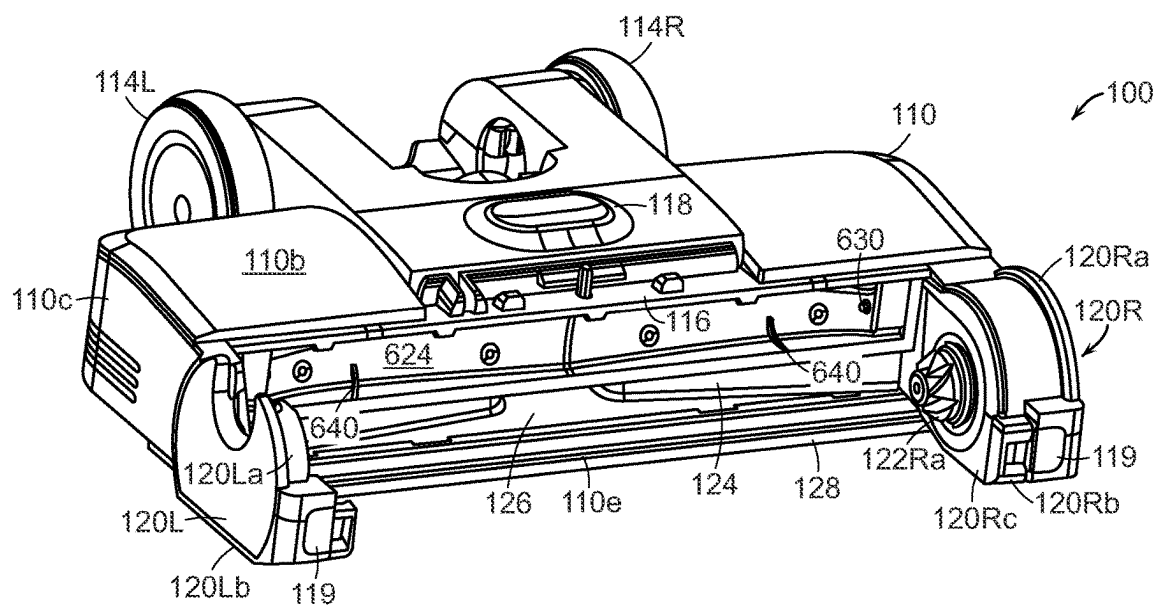
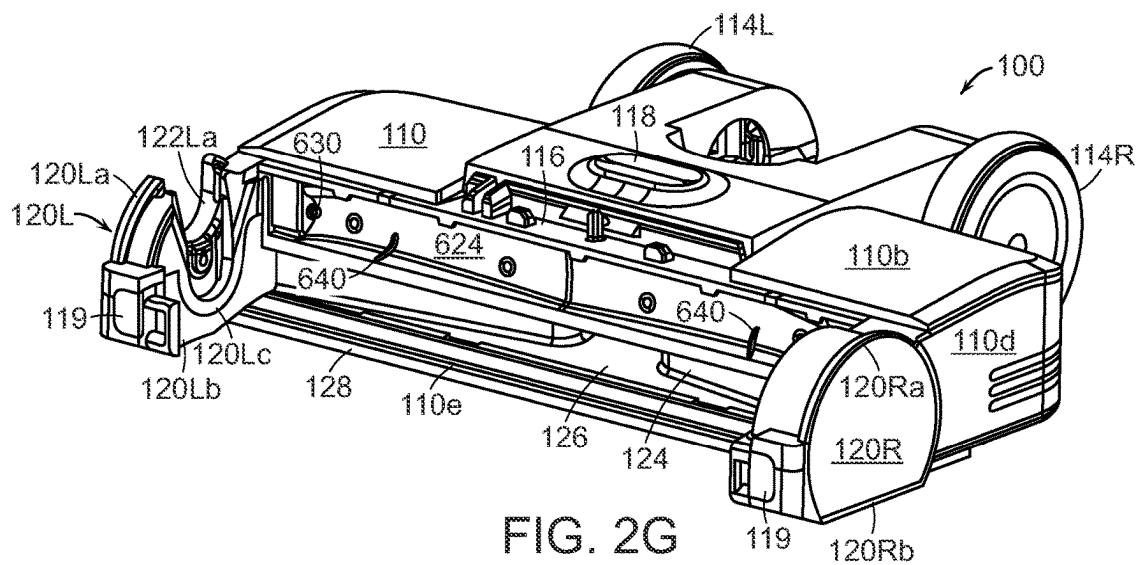


FIG. 2F



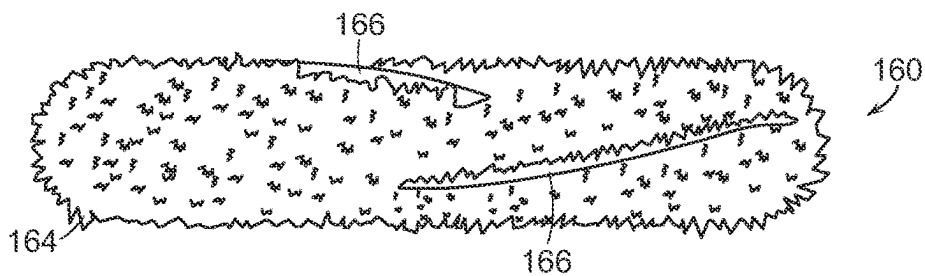
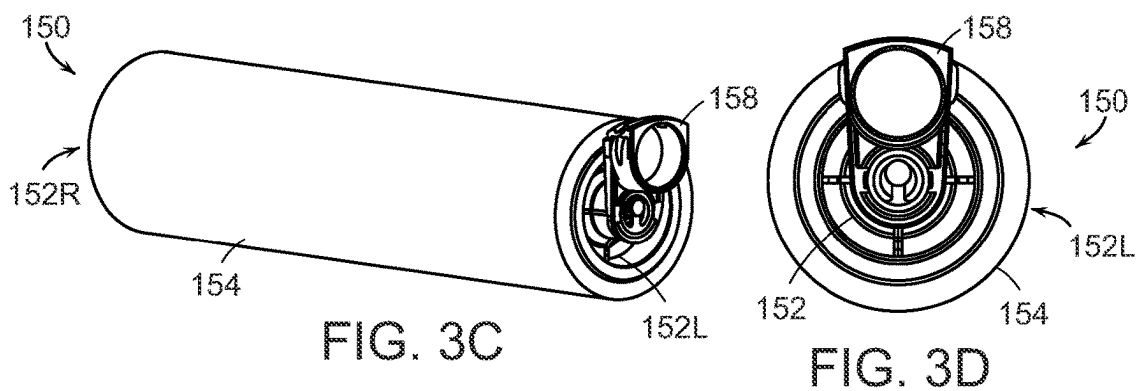
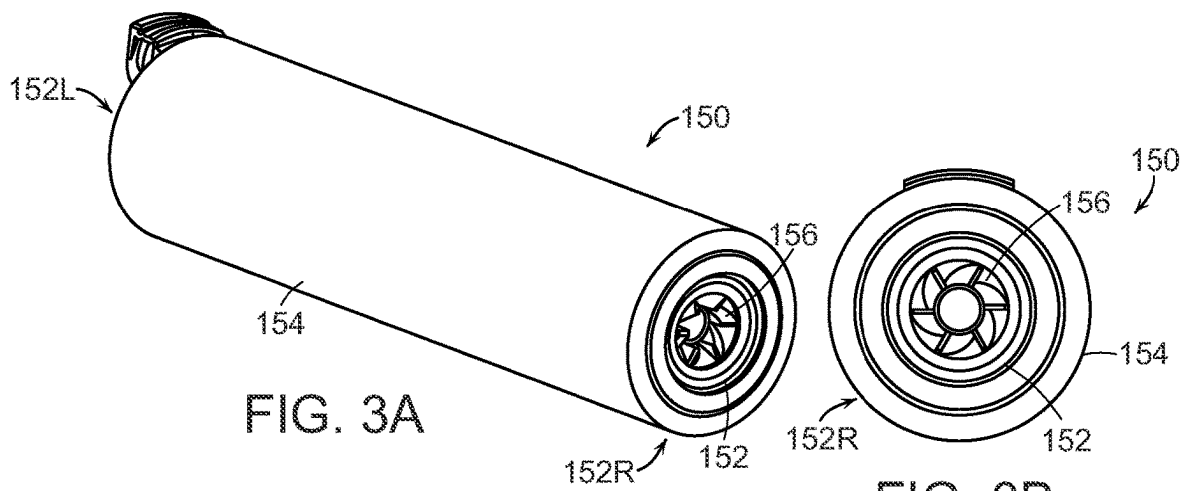


FIG. 4A

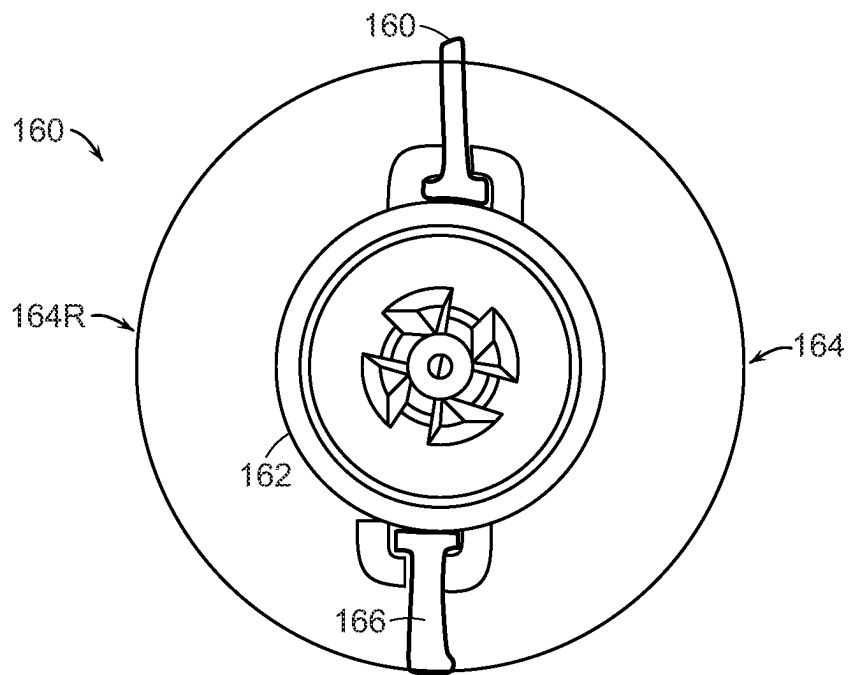


FIG. 4B

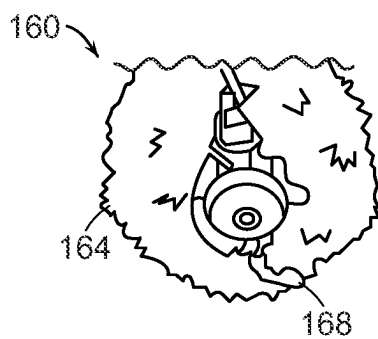


FIG. 4C

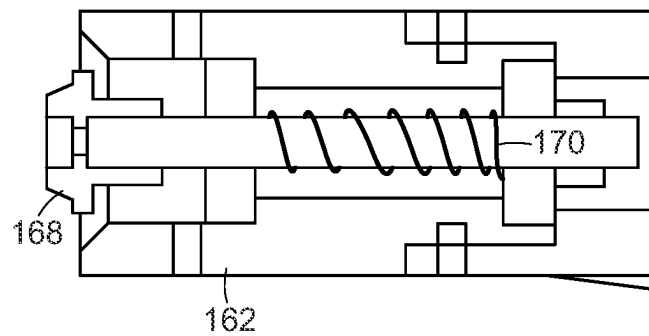


FIG. 4D

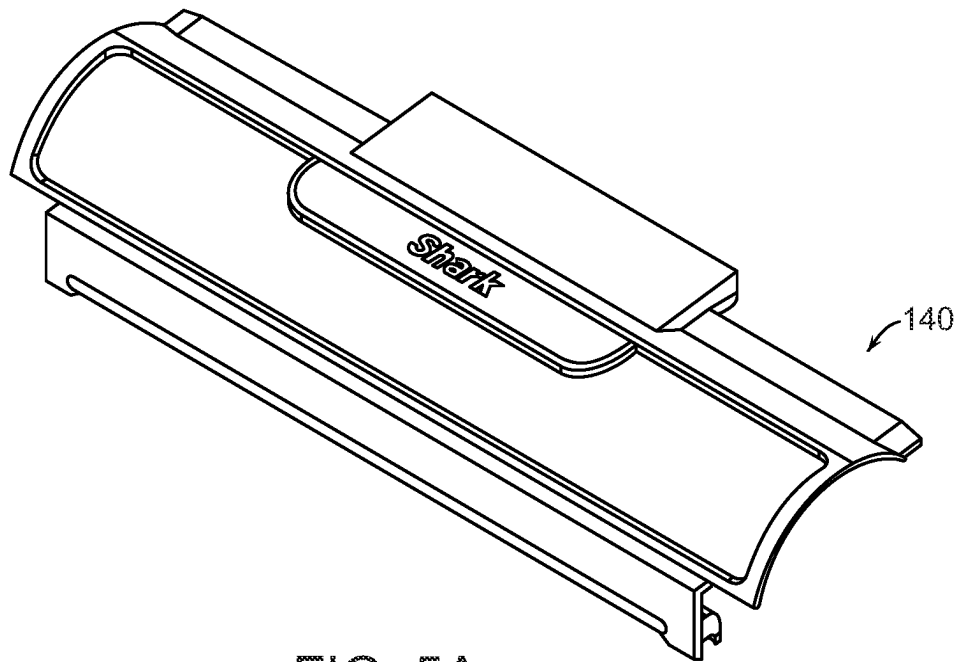


FIG. 5A

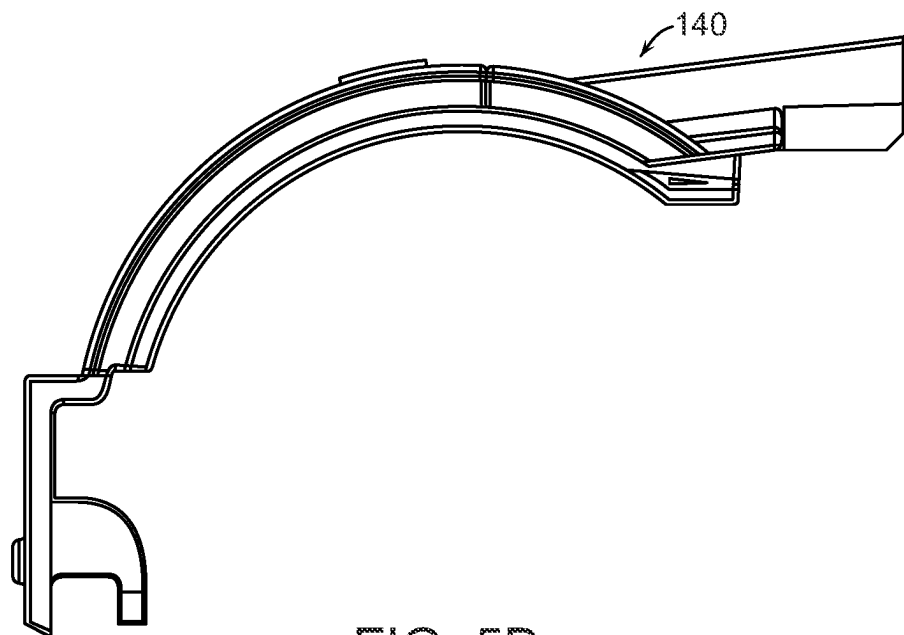
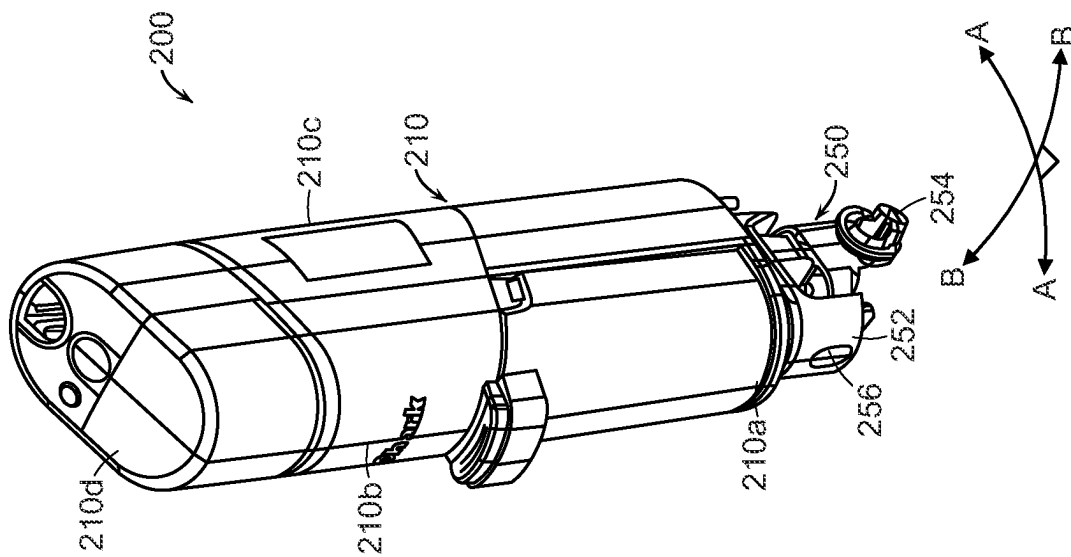
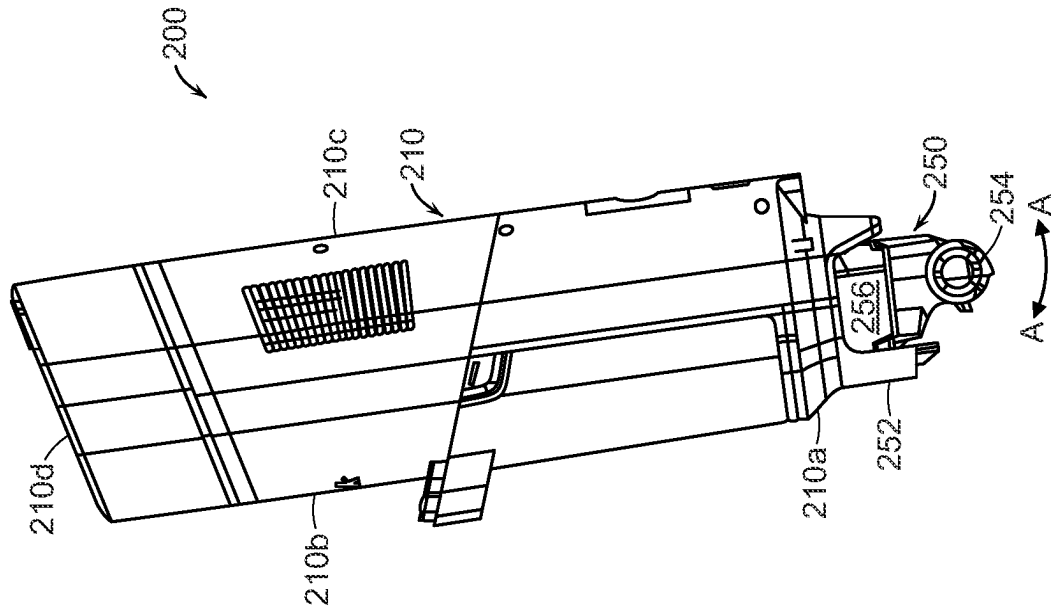


FIG. 5B



GA
GLE




 ҚАЗАҚСТАН РЕСПУБЛИКАСЫНЫҢ
 БІЛІМ ЖӘНЕ ҒЫЛЫМ МИНИСТРЛІГІ

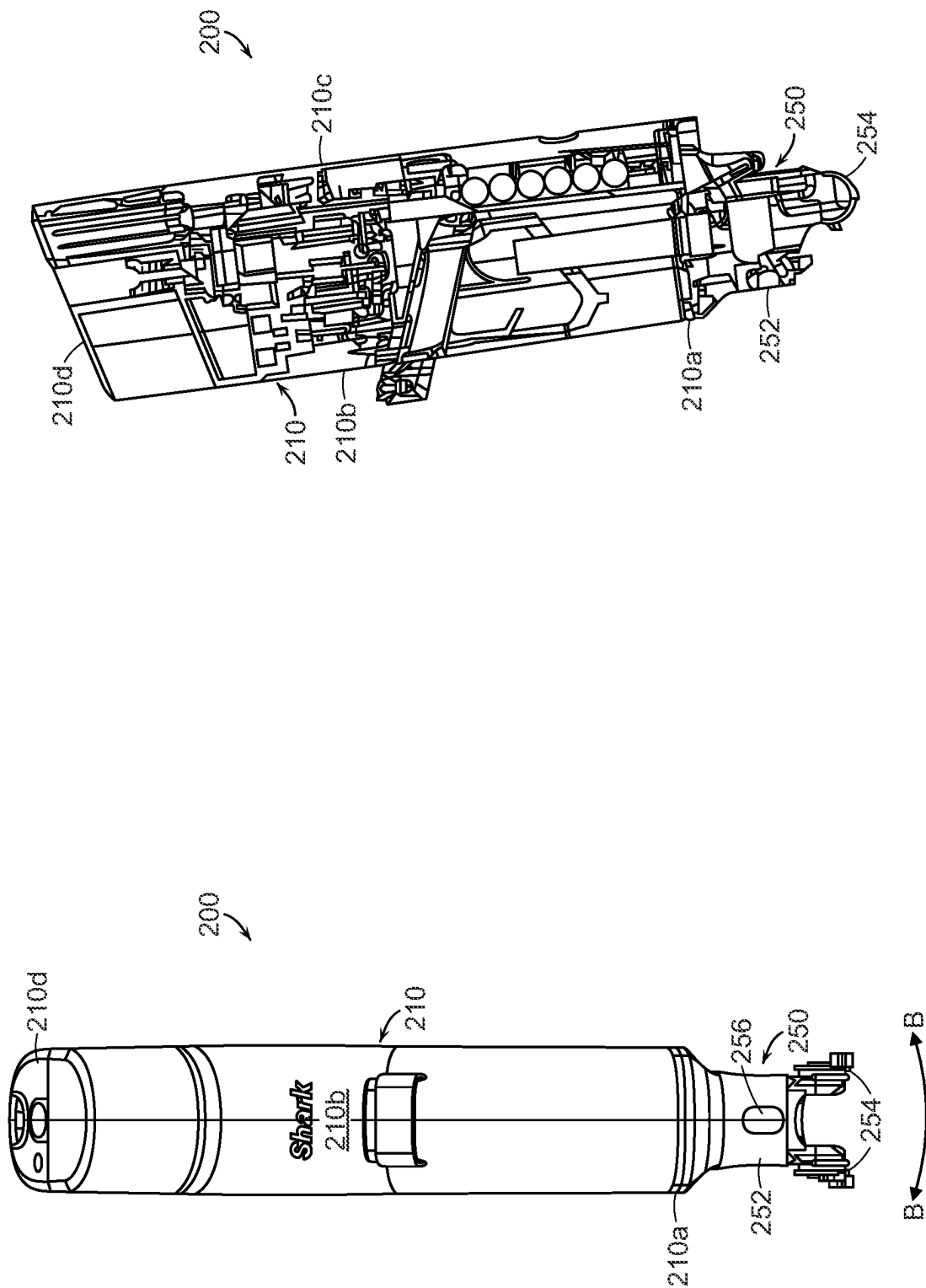


FIG. 6D

FIG. 6C

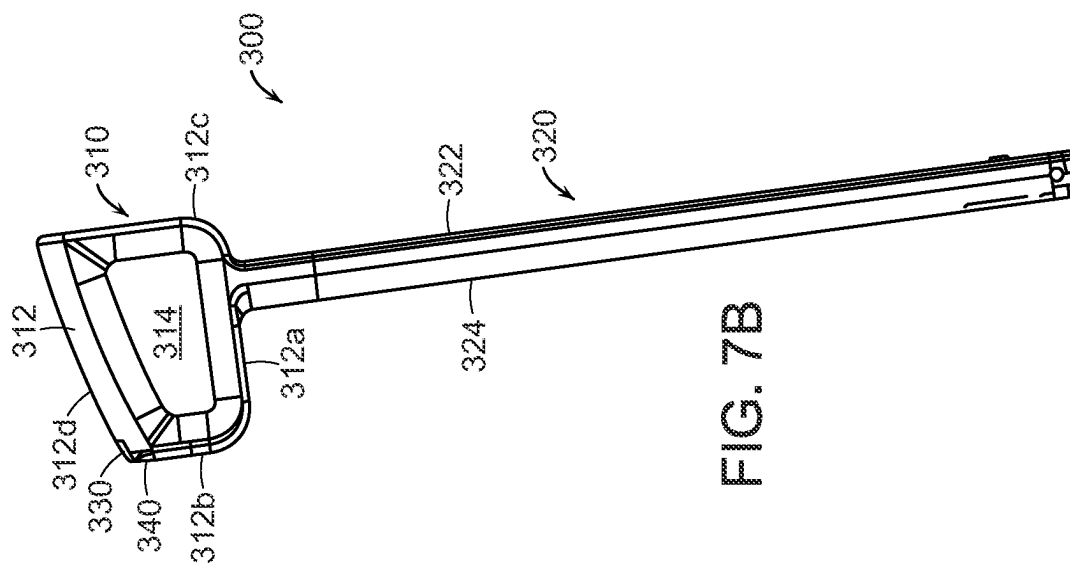


FIG. 7B

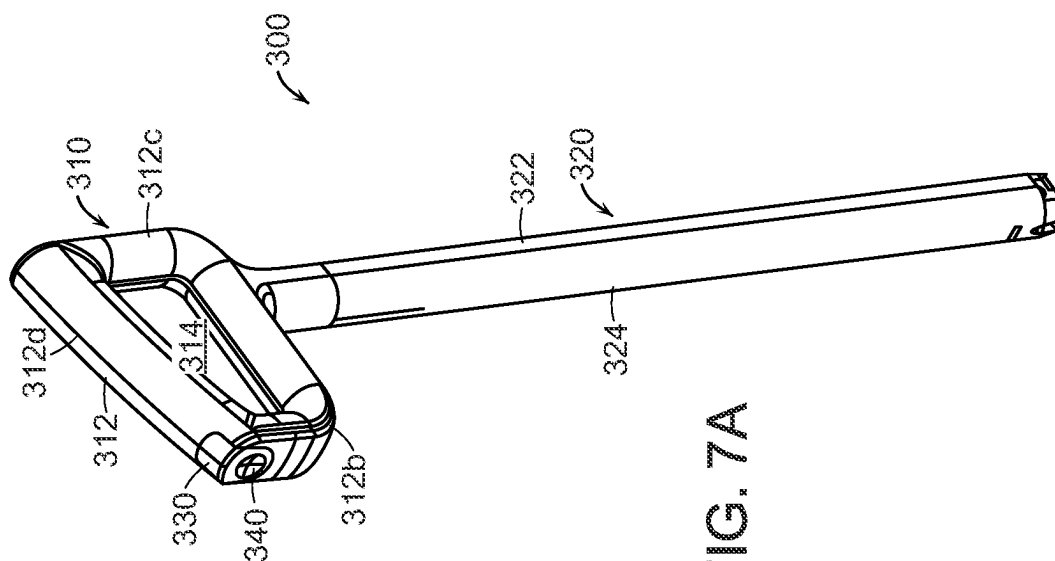


FIG. 7A

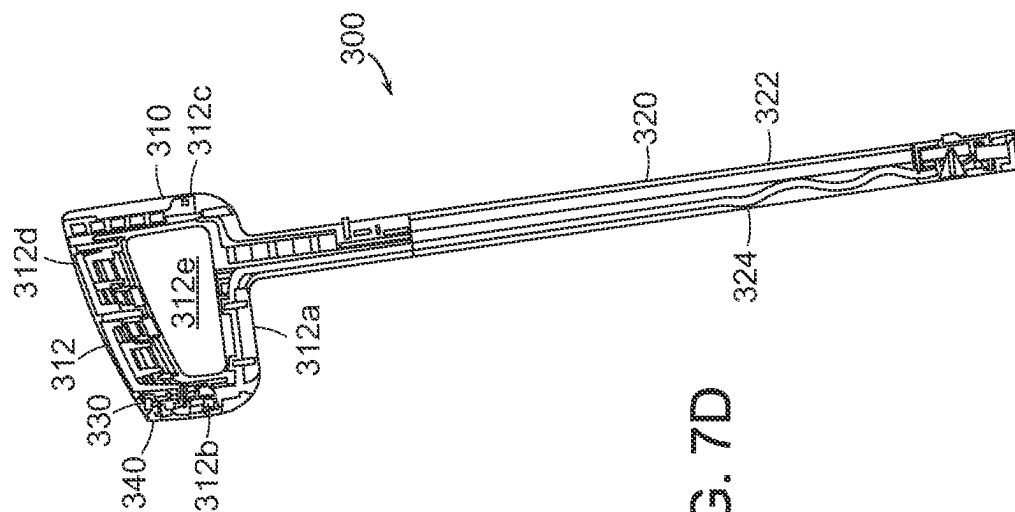


FIG. 7D

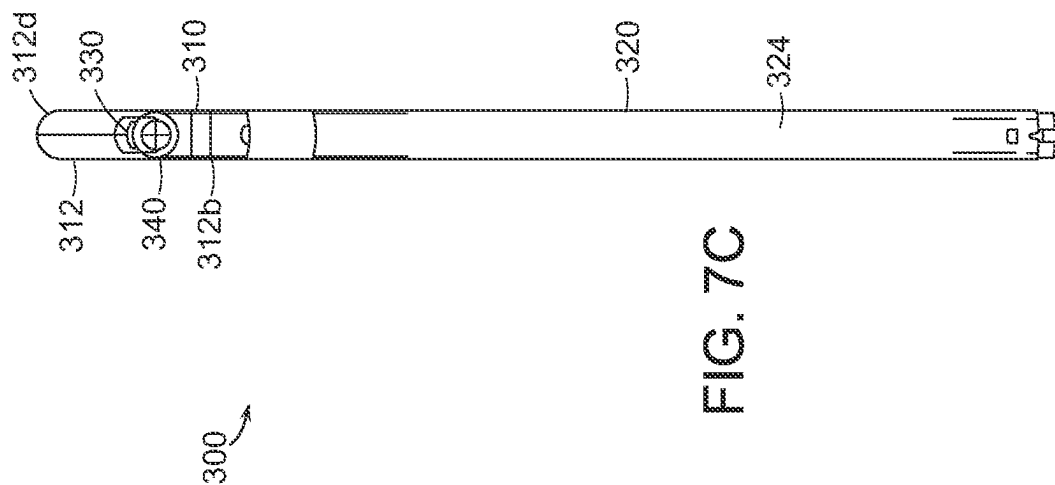


FIG. 7C

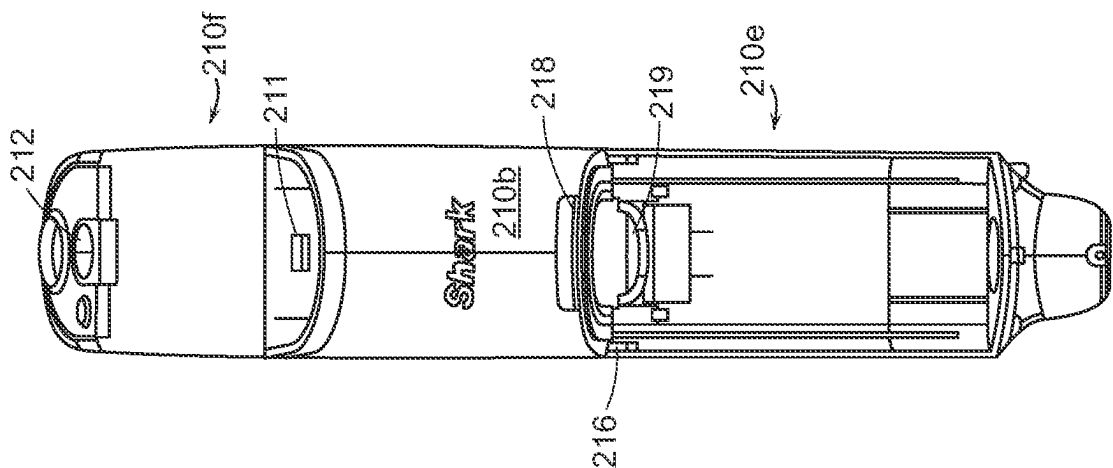


FIG. 8B

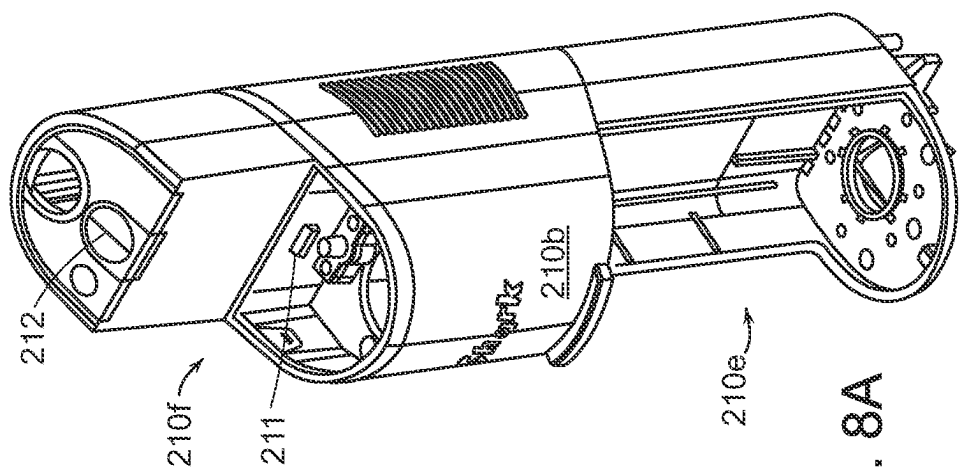


FIG. 8A

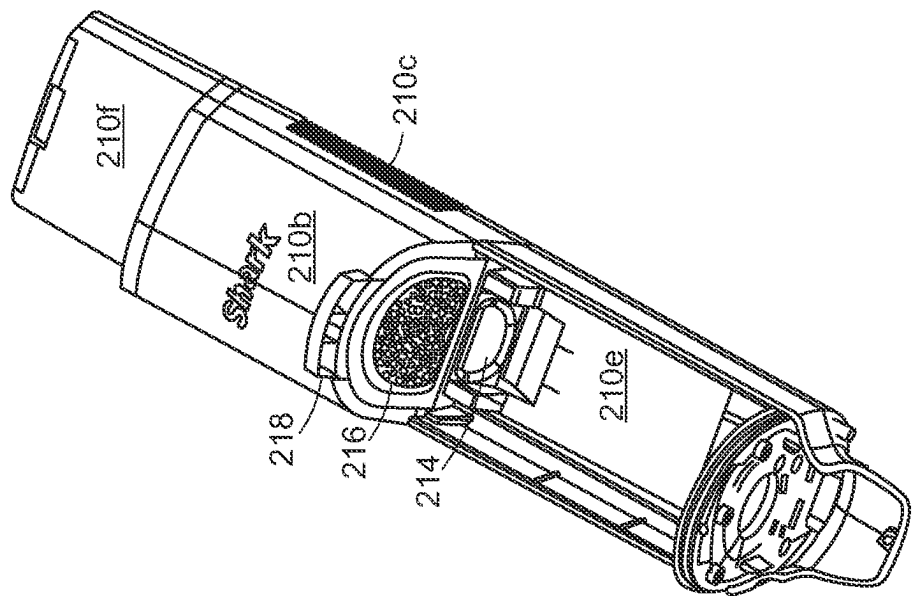


FIG. 8D

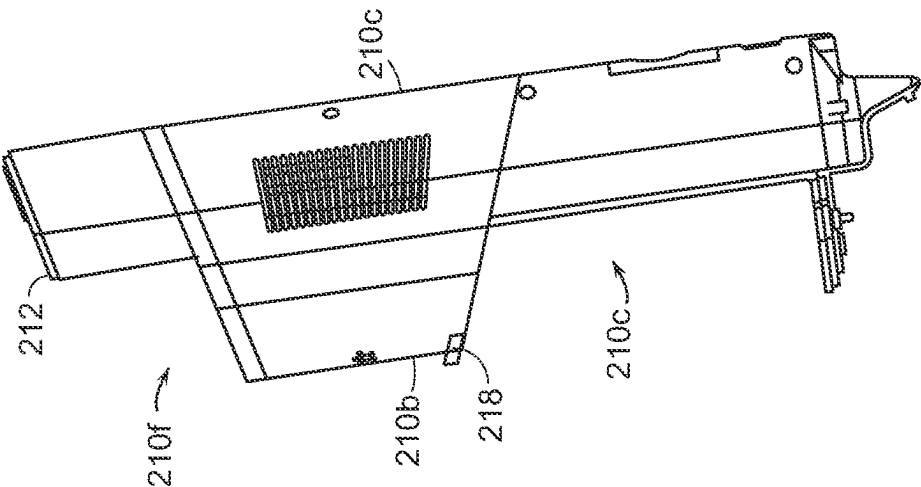
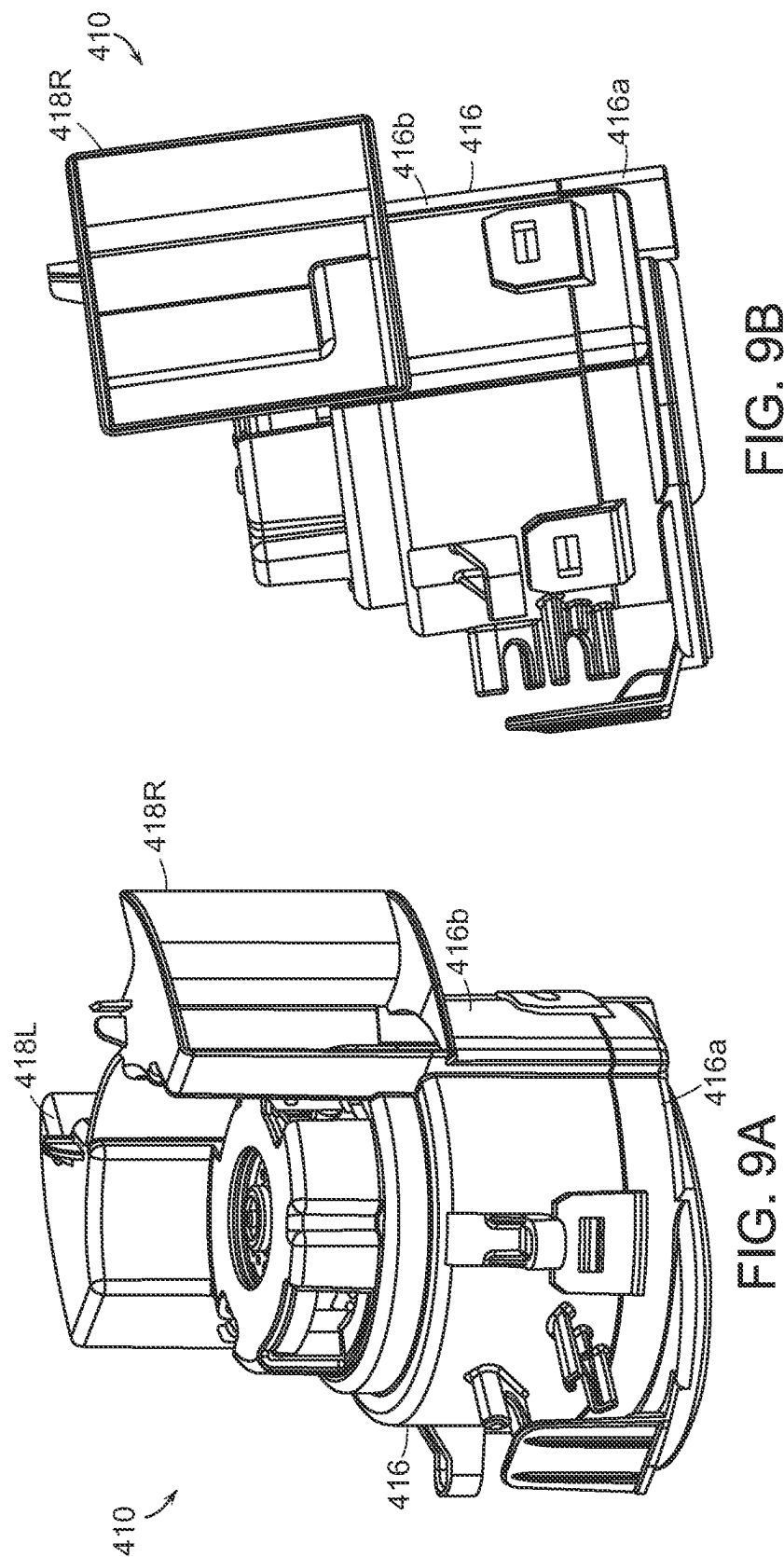


FIG. 8C



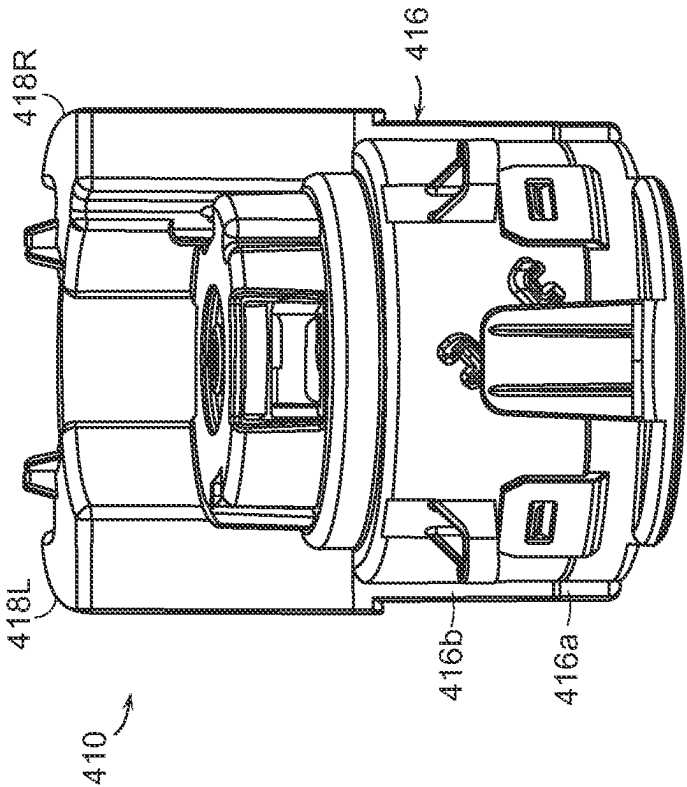


FIG. 9C

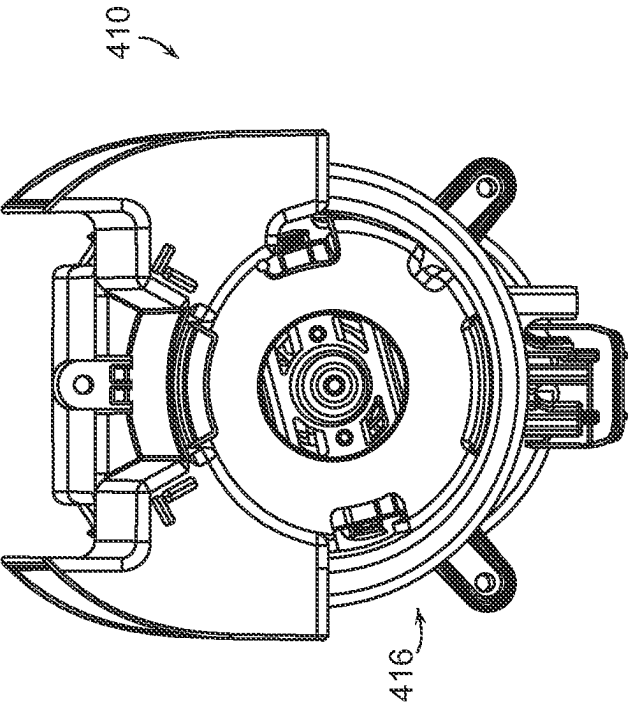


FIG. 9D

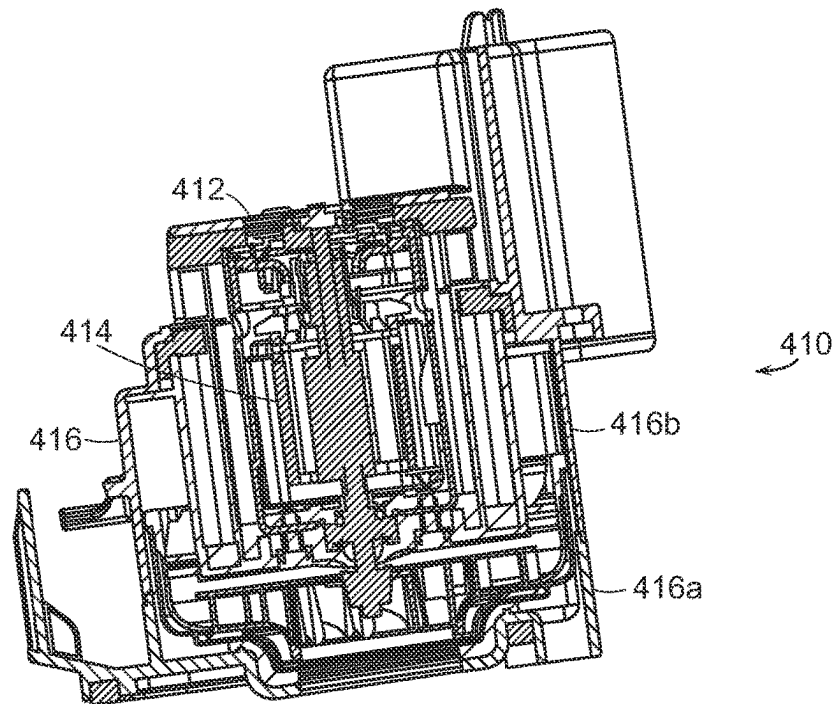


FIG. 9E

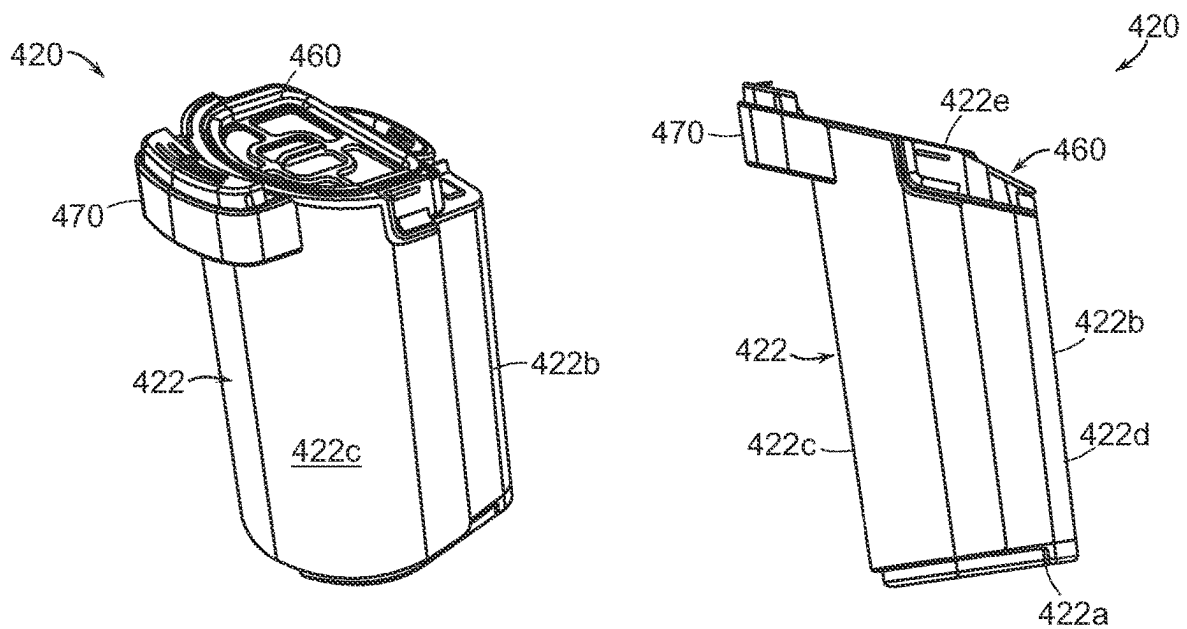


FIG. 10A

FIG. 10B

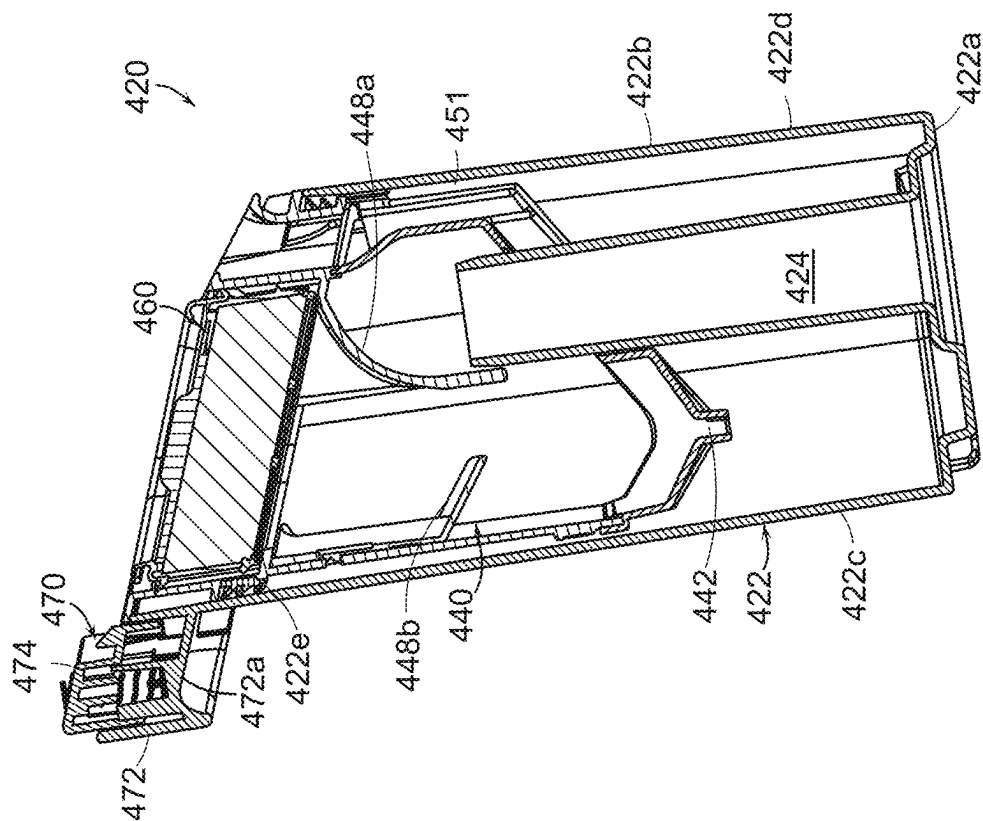


FIG. 10D

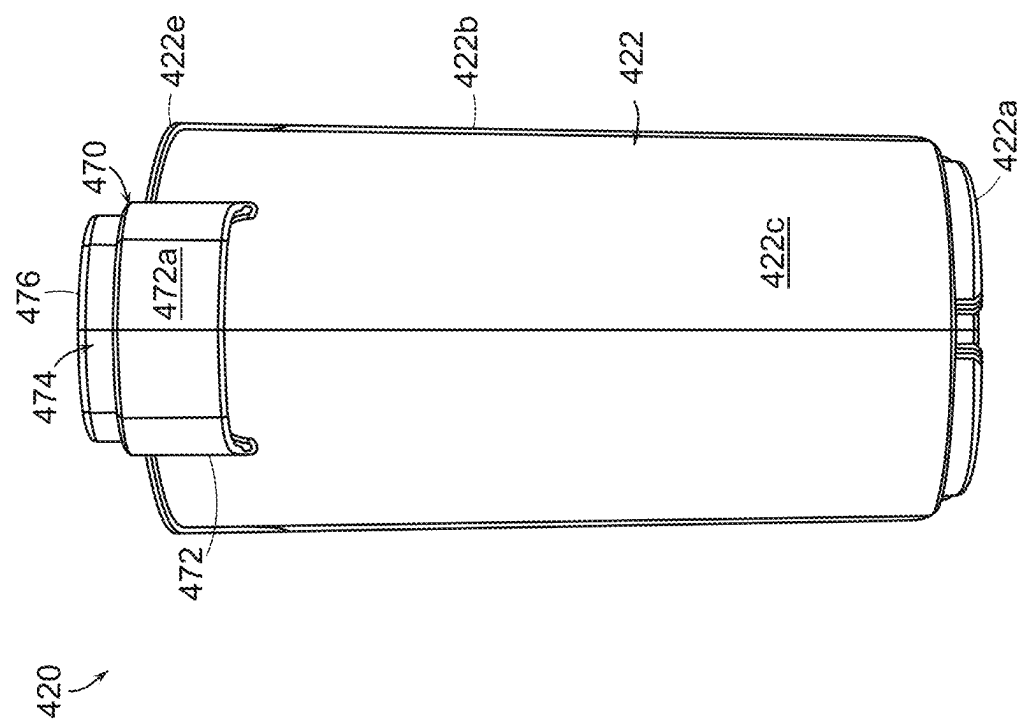


FIG. 10C

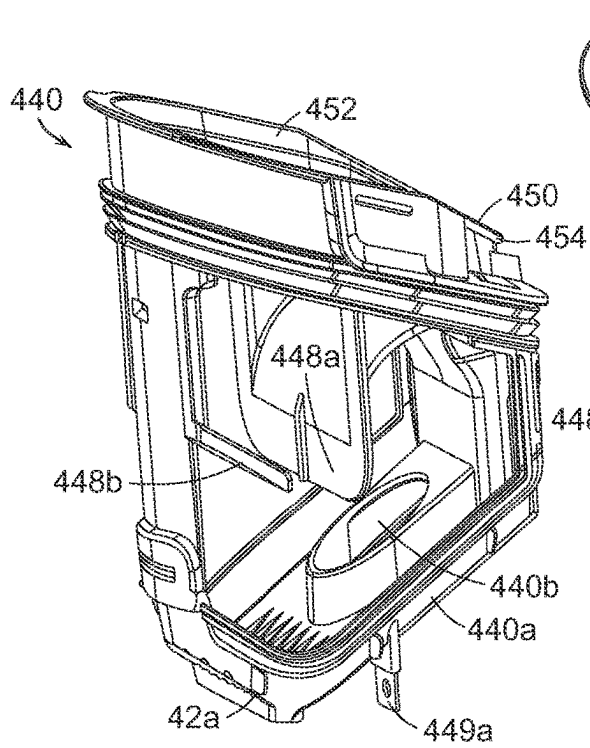


FIG. 10E

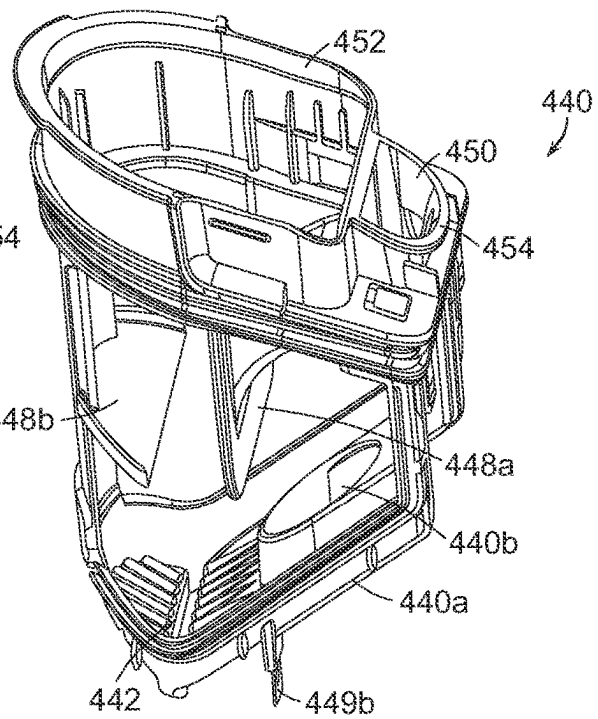


FIG. 10F

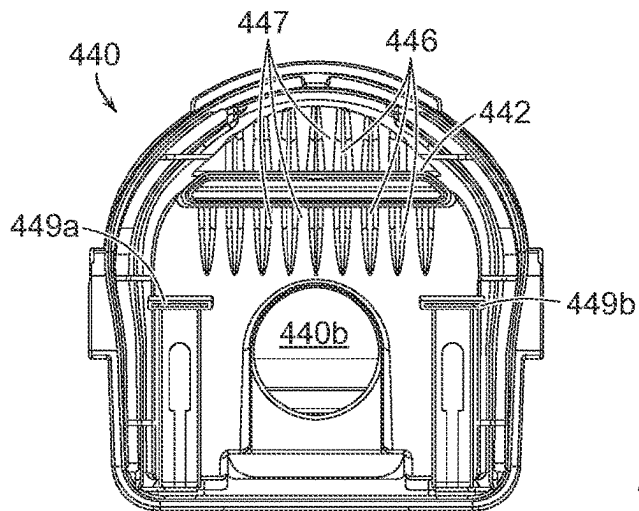


FIG. 10G

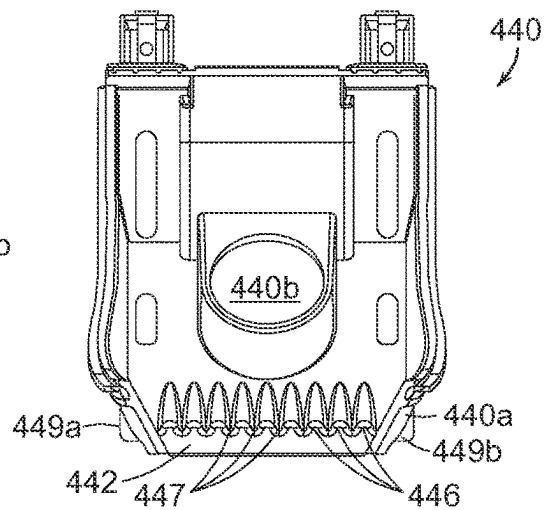


FIG. 10H

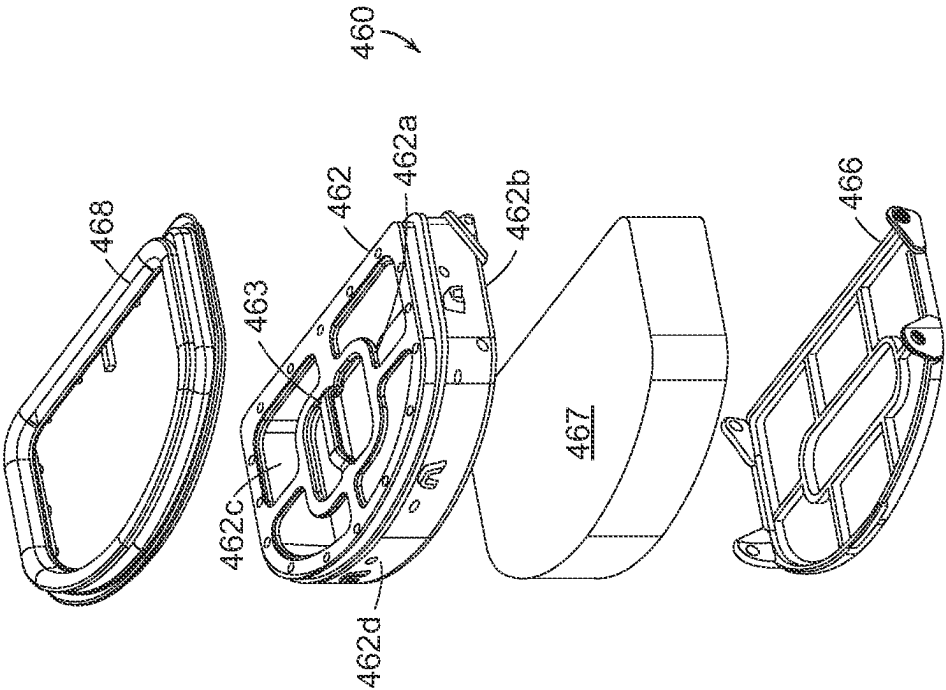


FIG. 10J

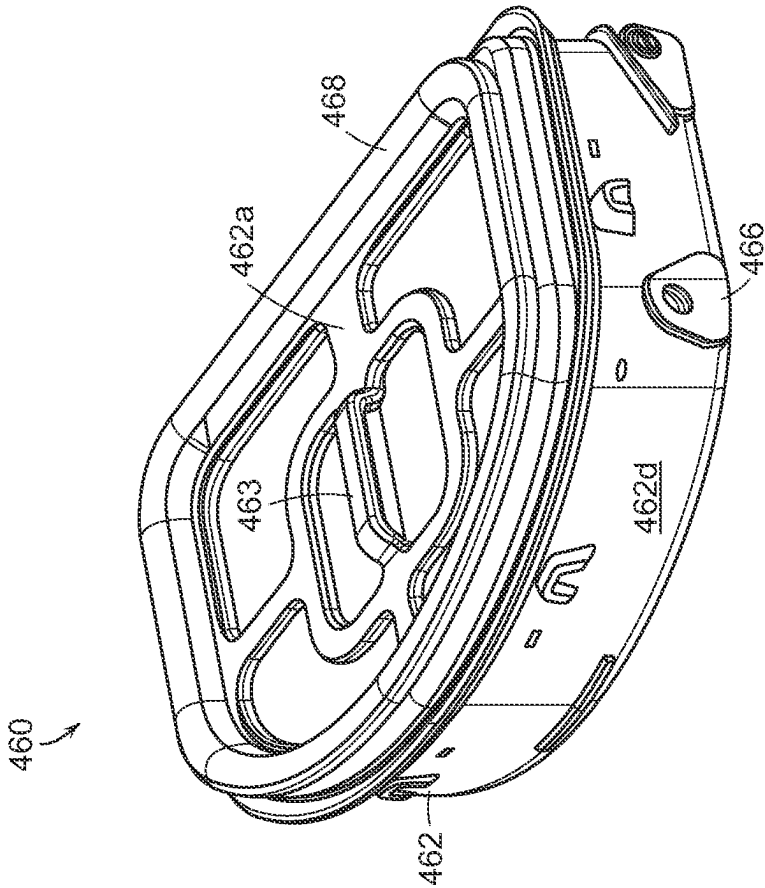


FIG. 10I

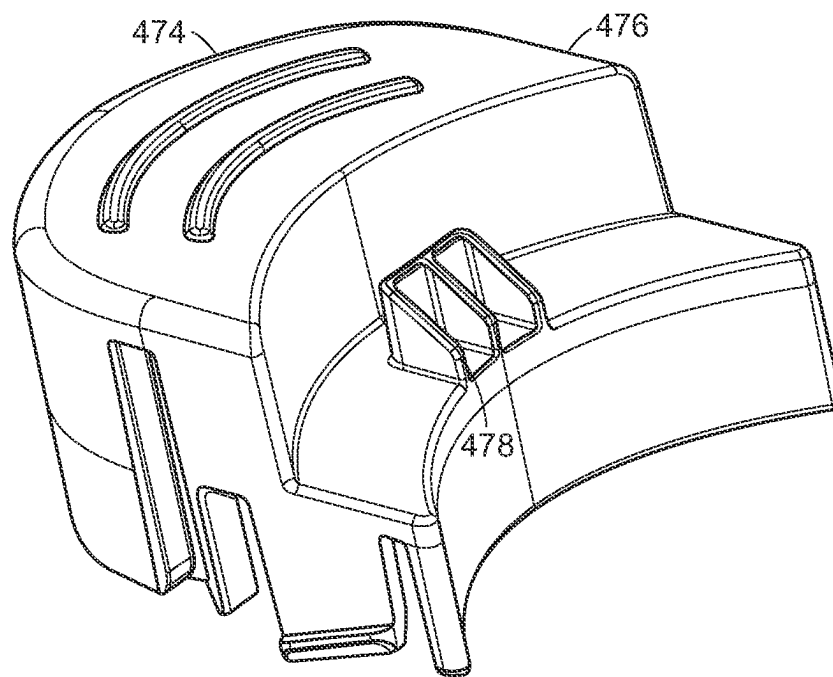


FIG. 10K

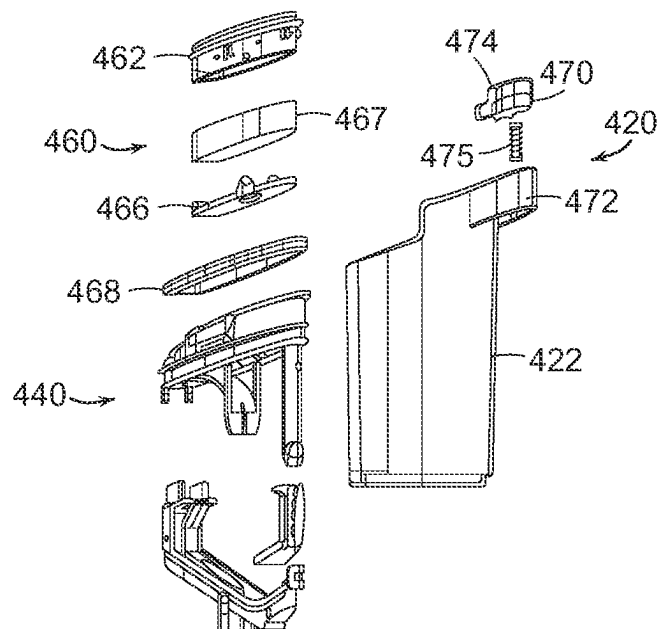


FIG. 10L

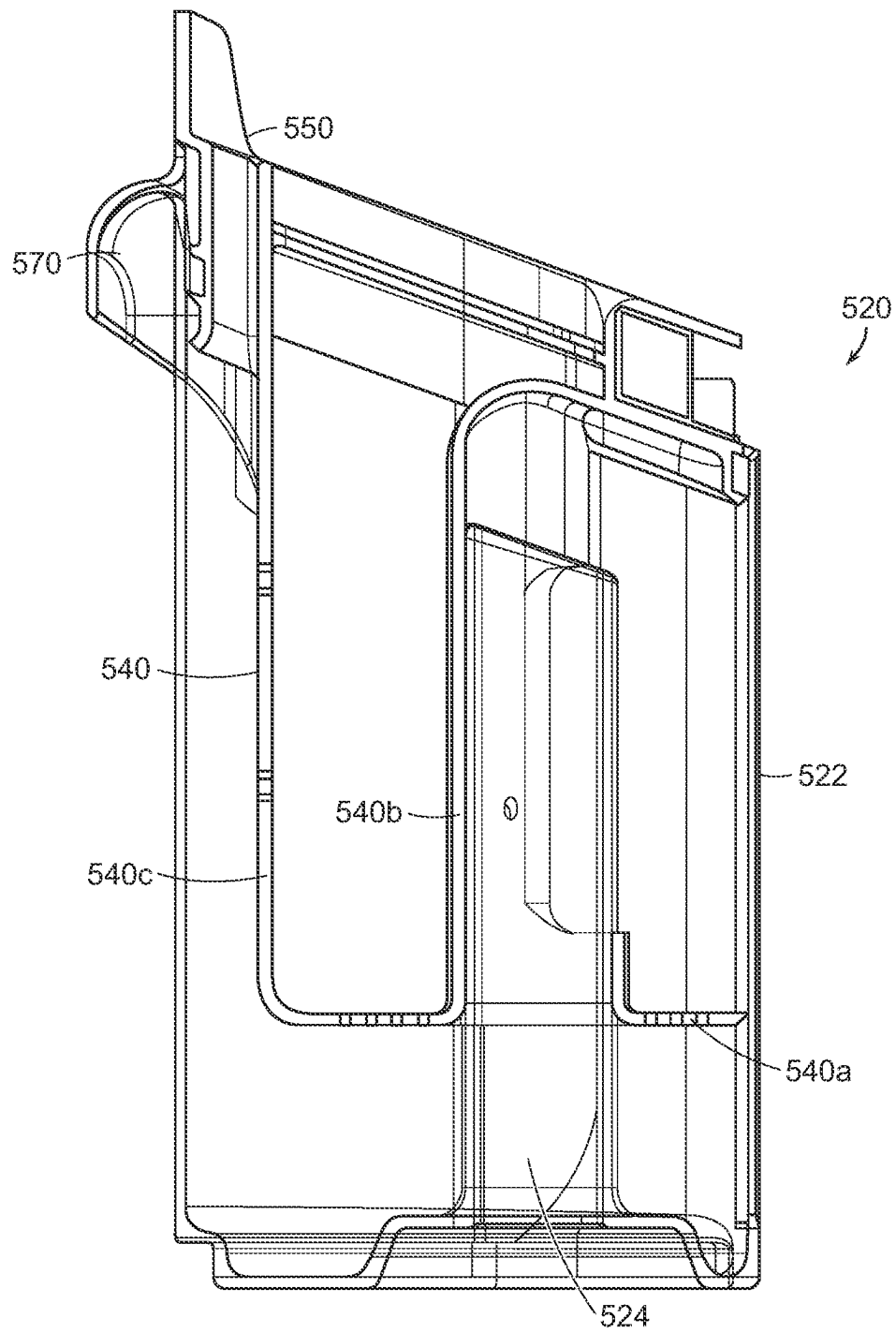
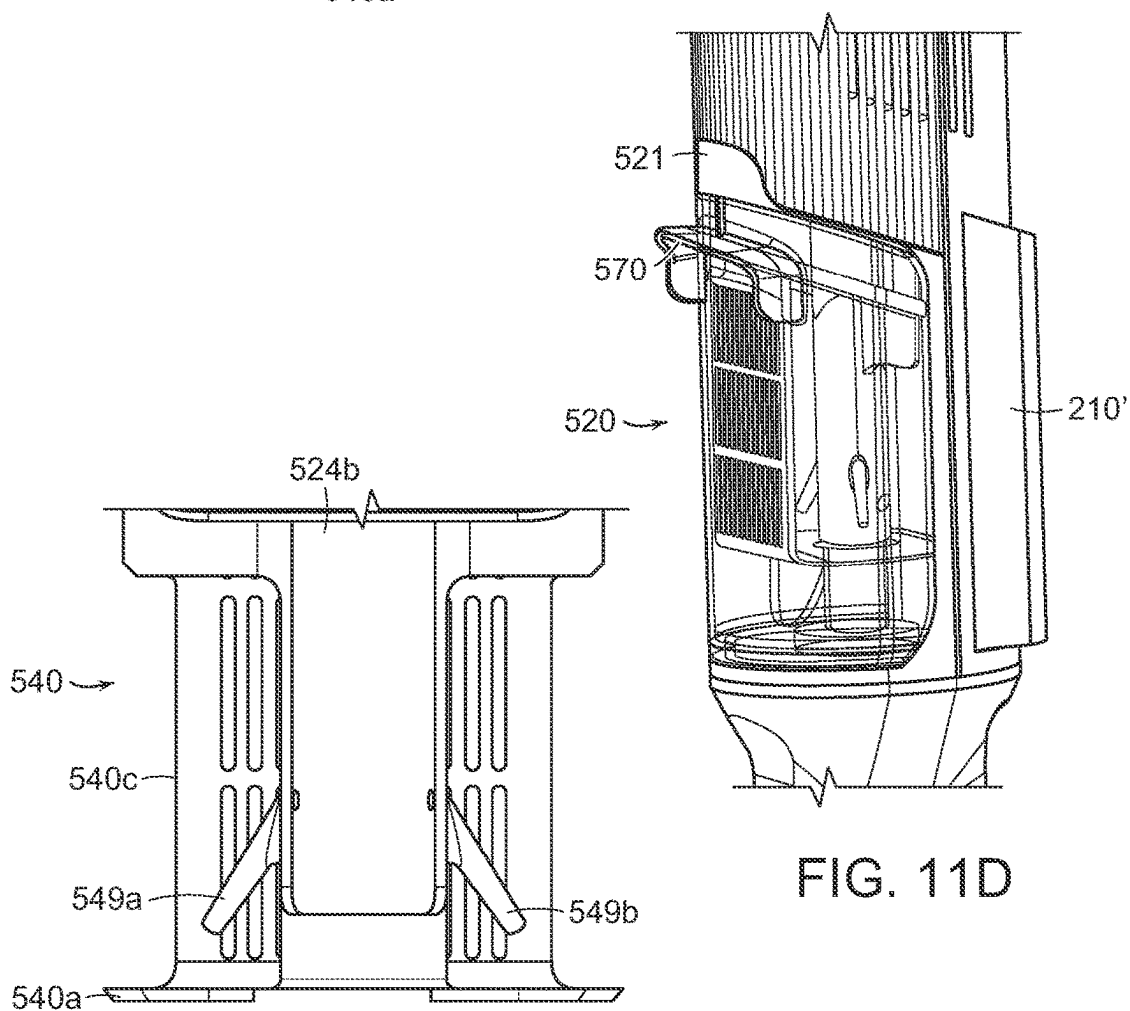
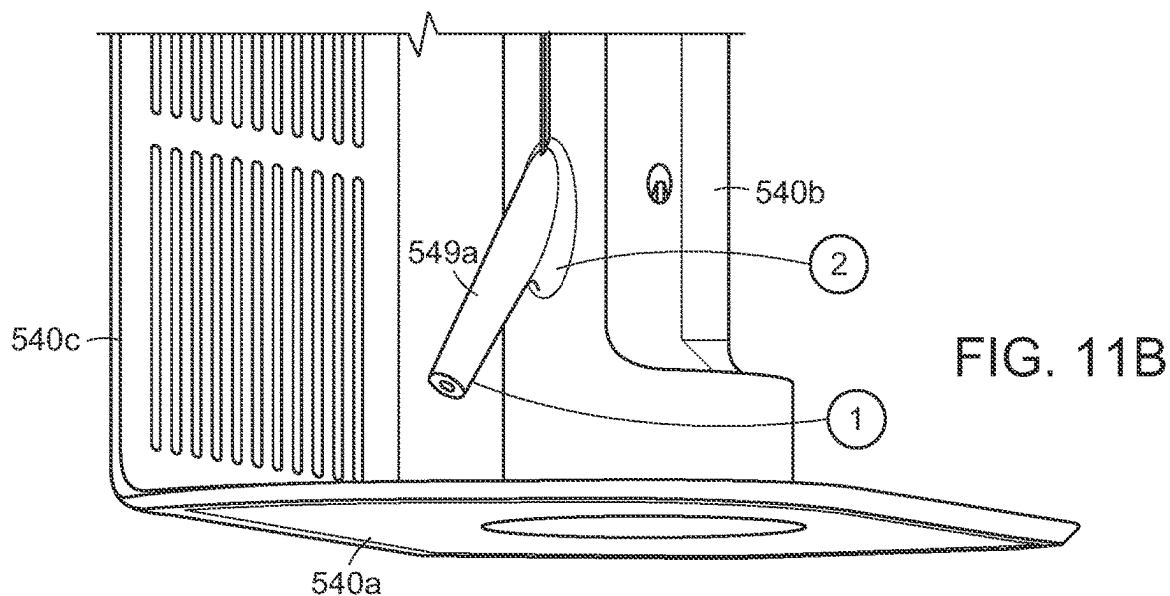


FIG. 11A



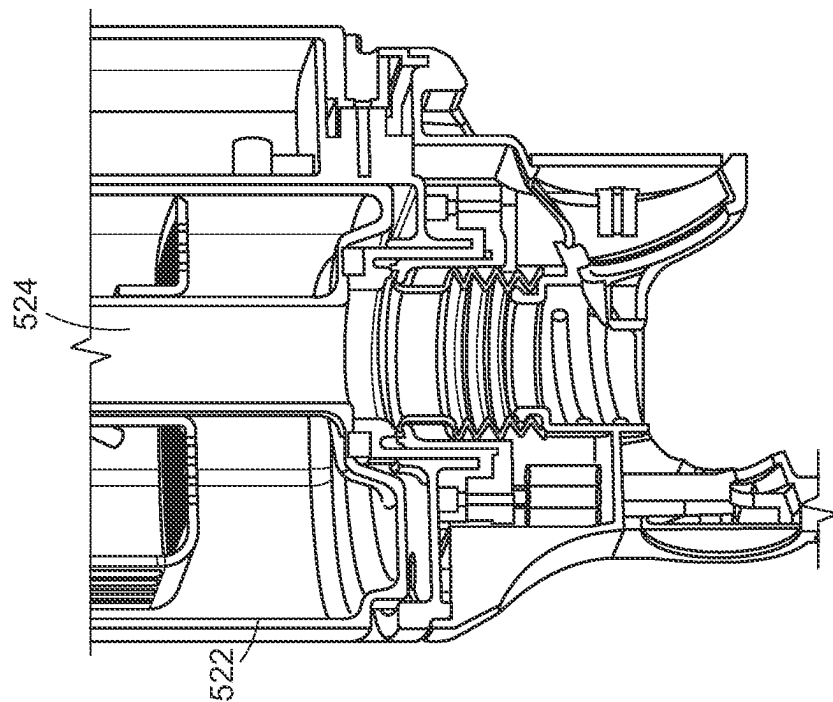


FIG. 11F

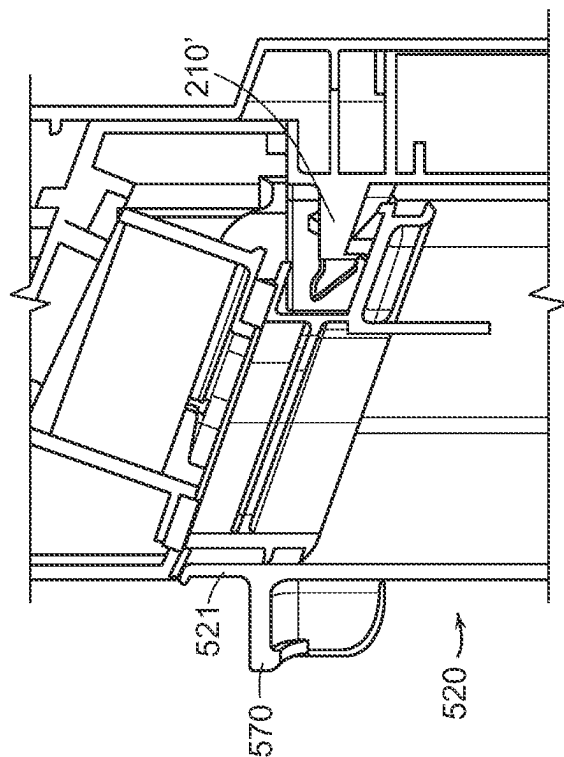


FIG. 11E

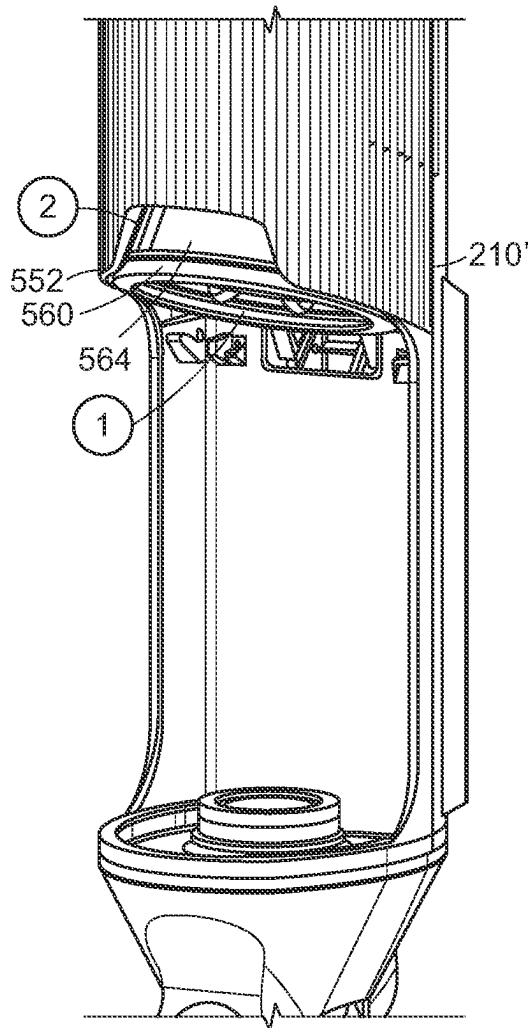


FIG. 11G

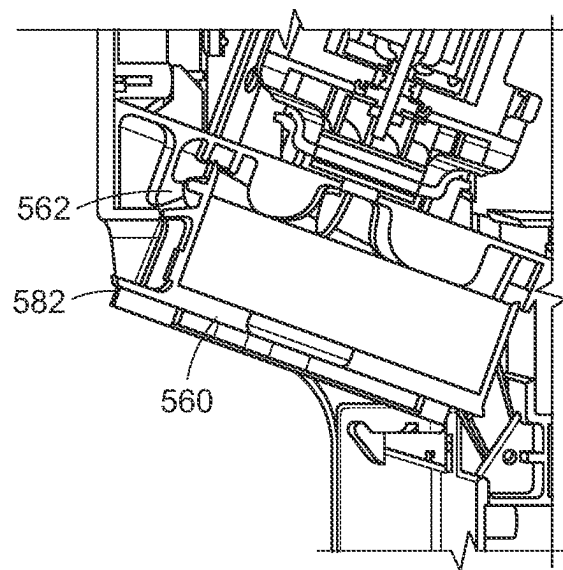


FIG. 11H

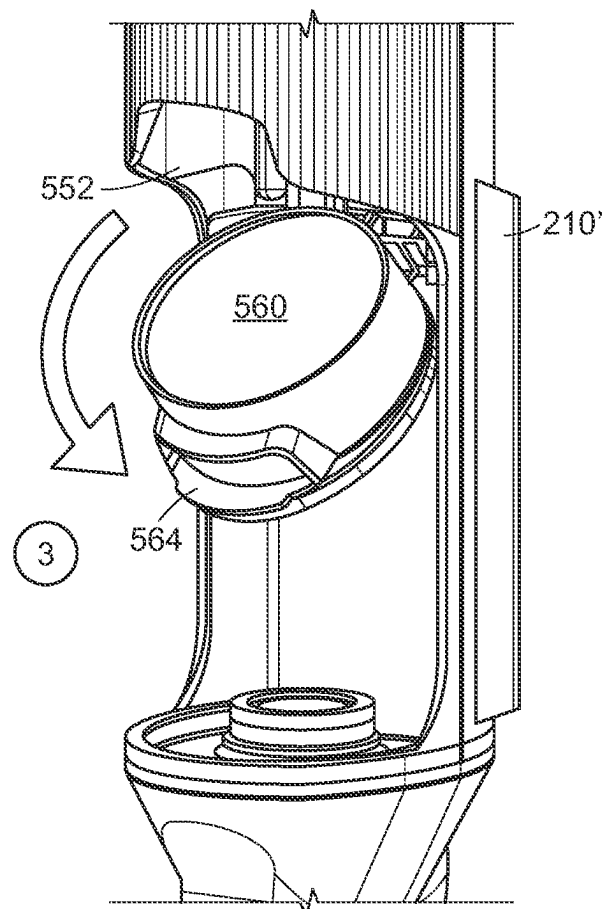


FIG. 11I

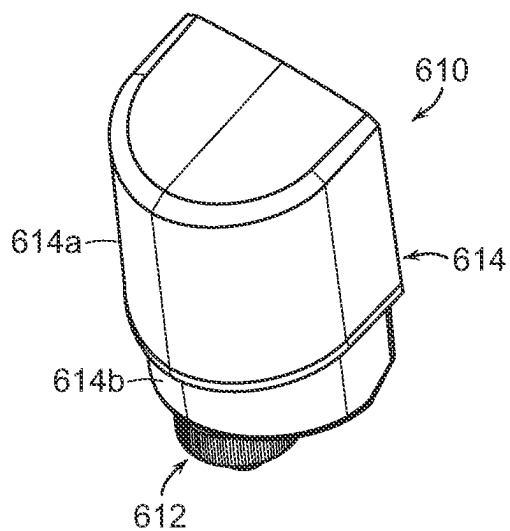


FIG. 12A

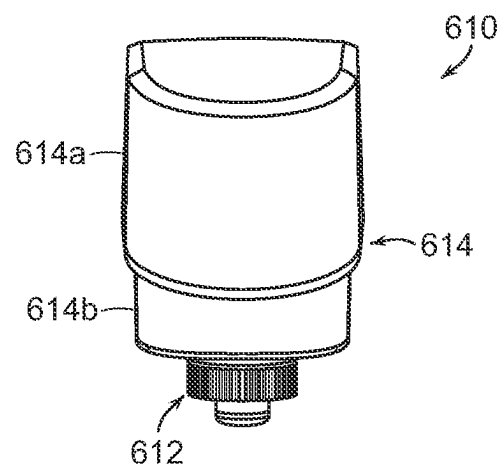


FIG. 12B

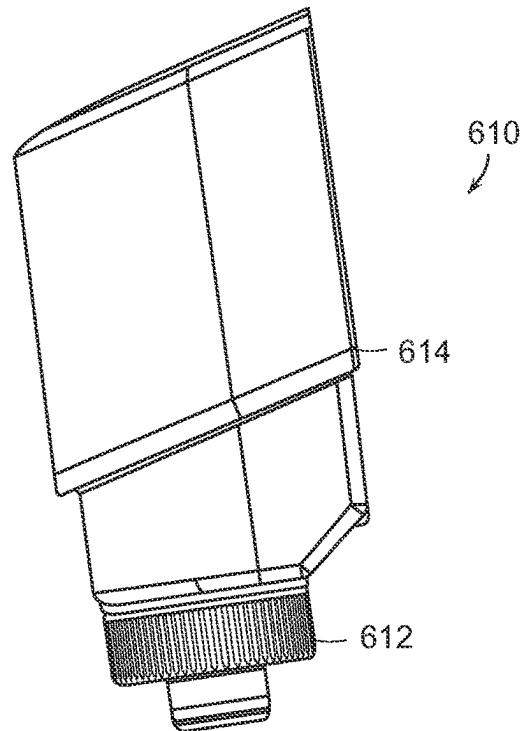


FIG. 12C

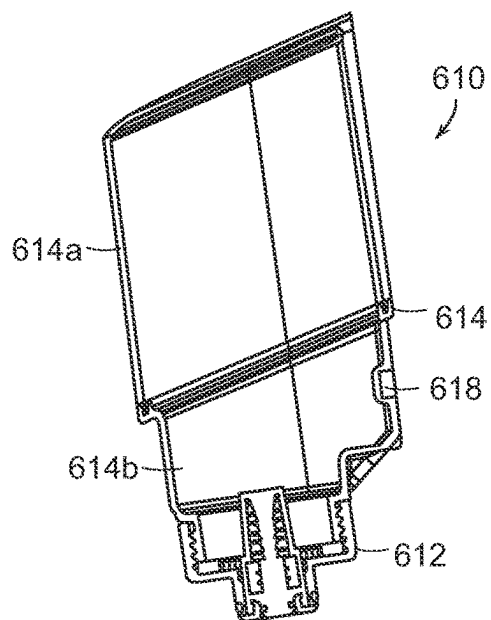


FIG. 12D

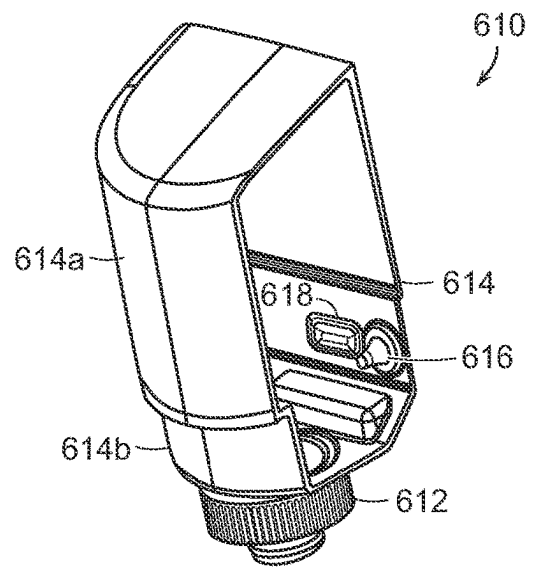


FIG. 12E

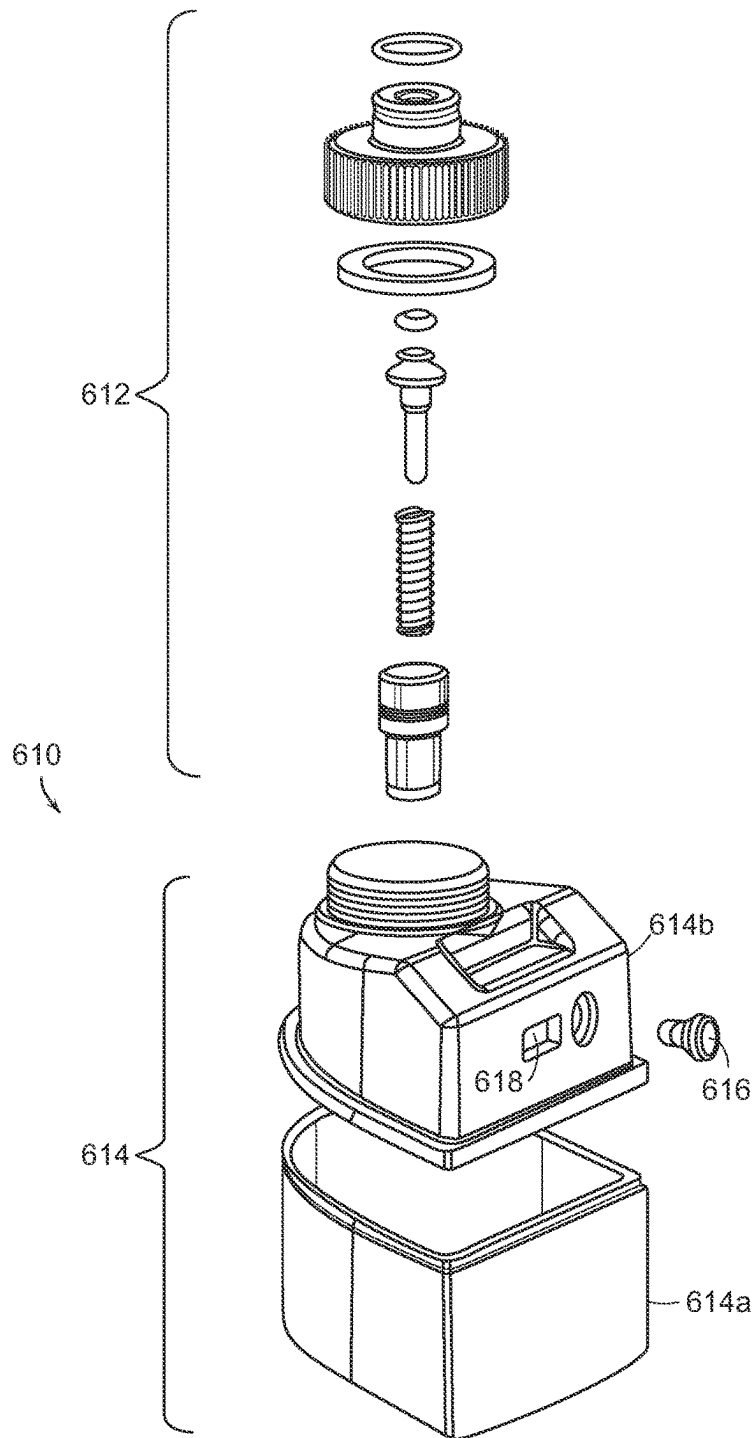
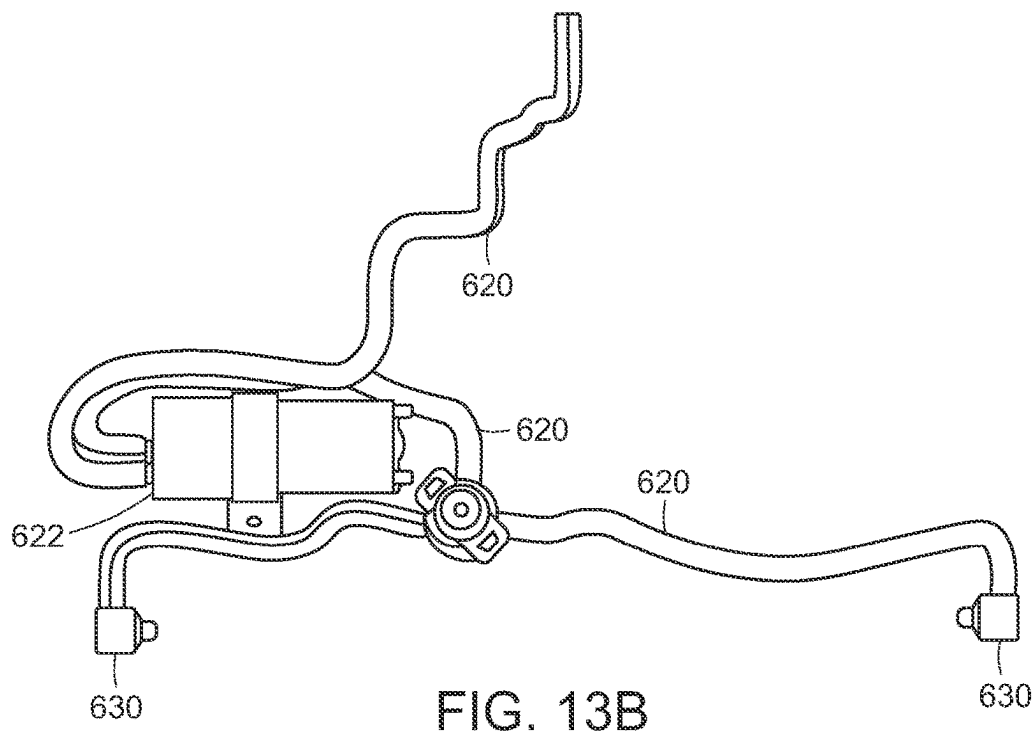
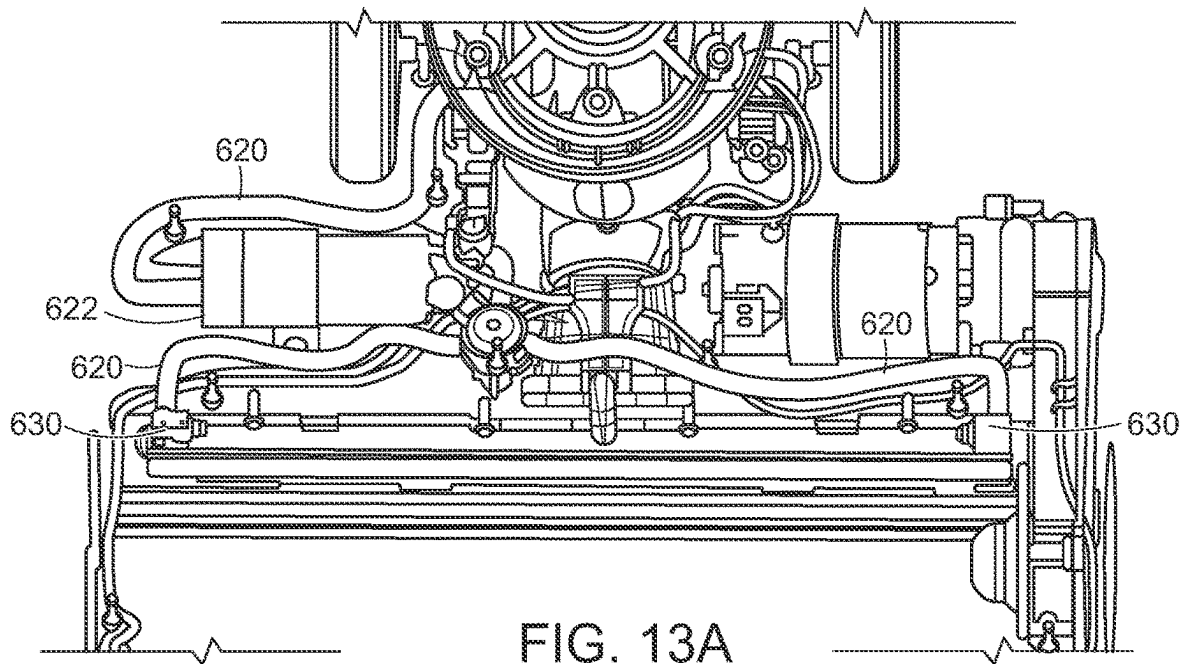


FIG. 12F



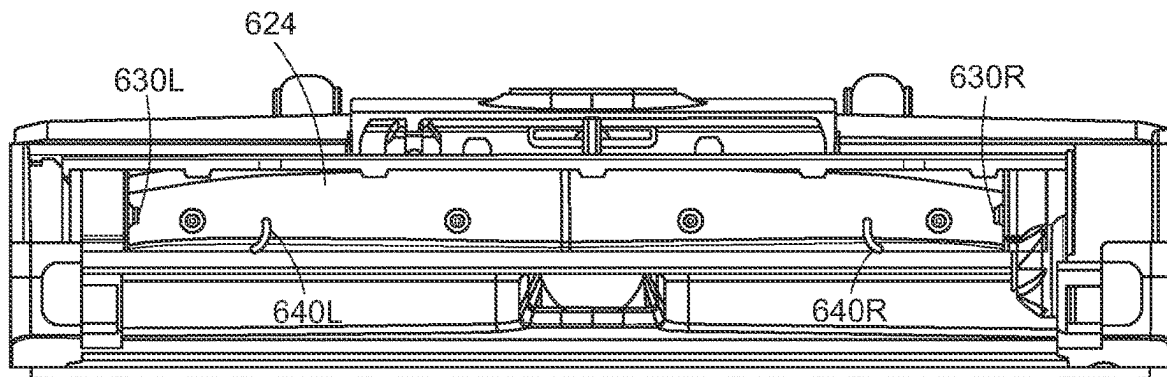


FIG. 13C

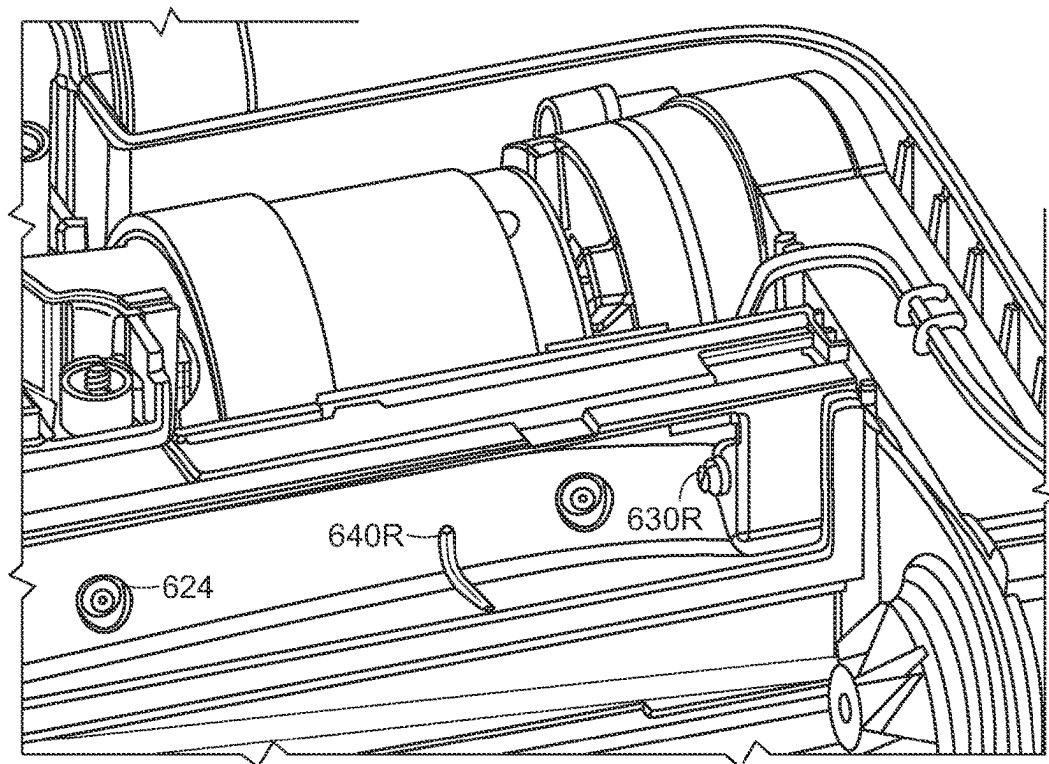


FIG. 13D

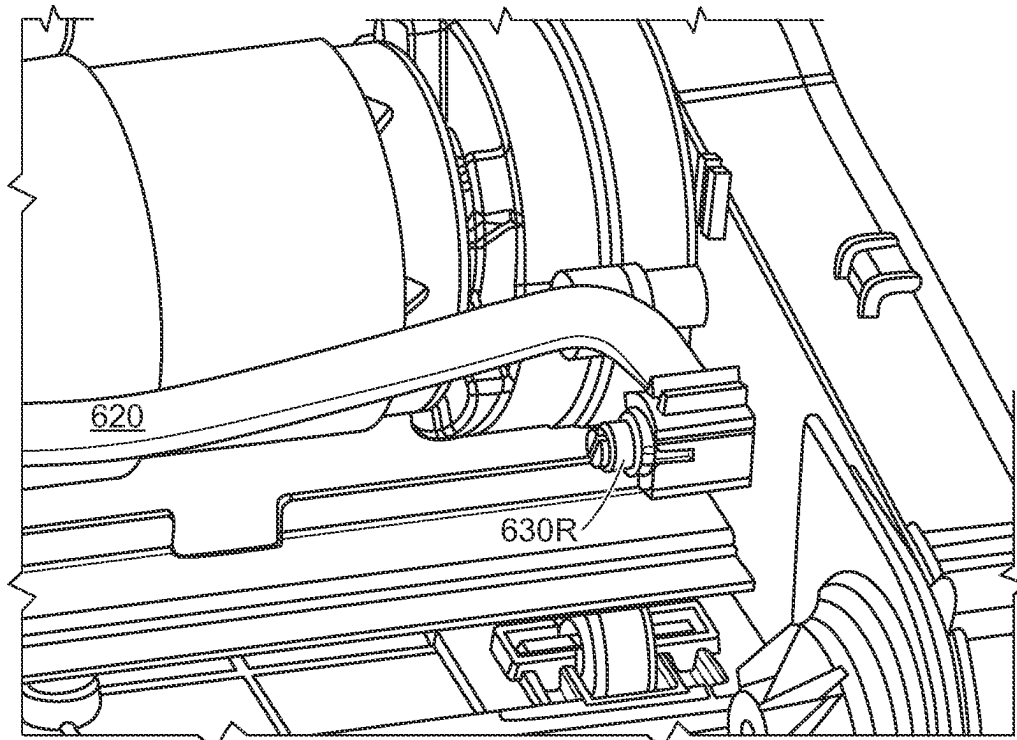


FIG. 13E

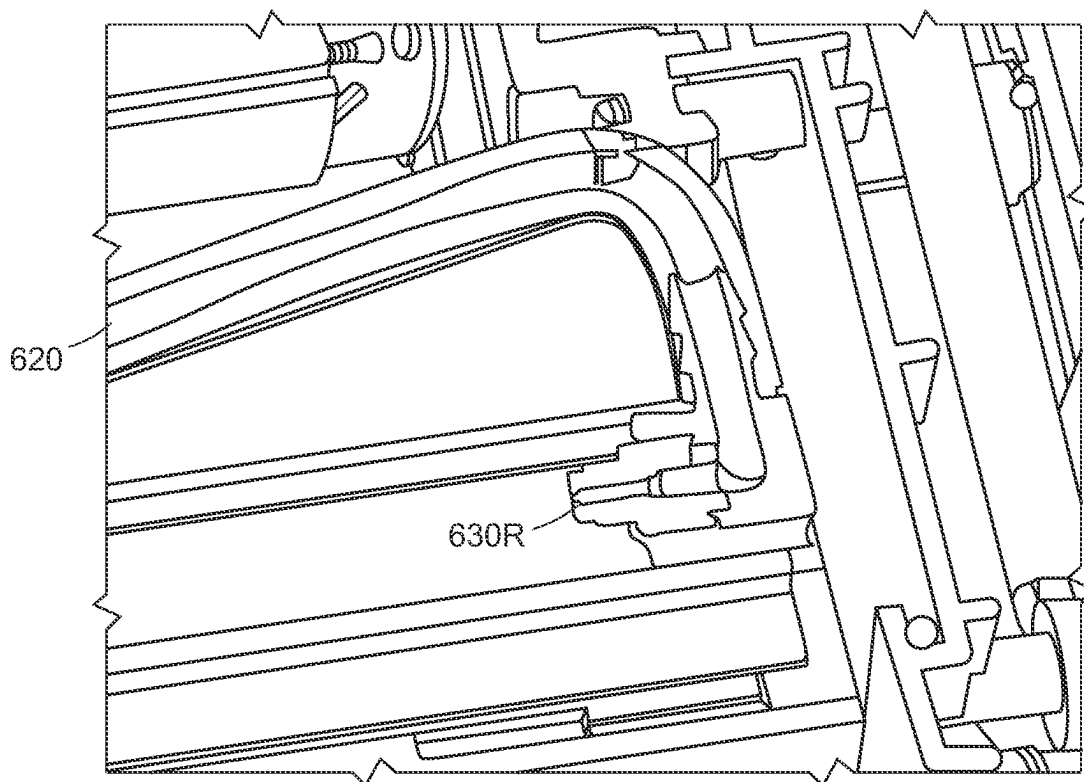


FIG. 13F

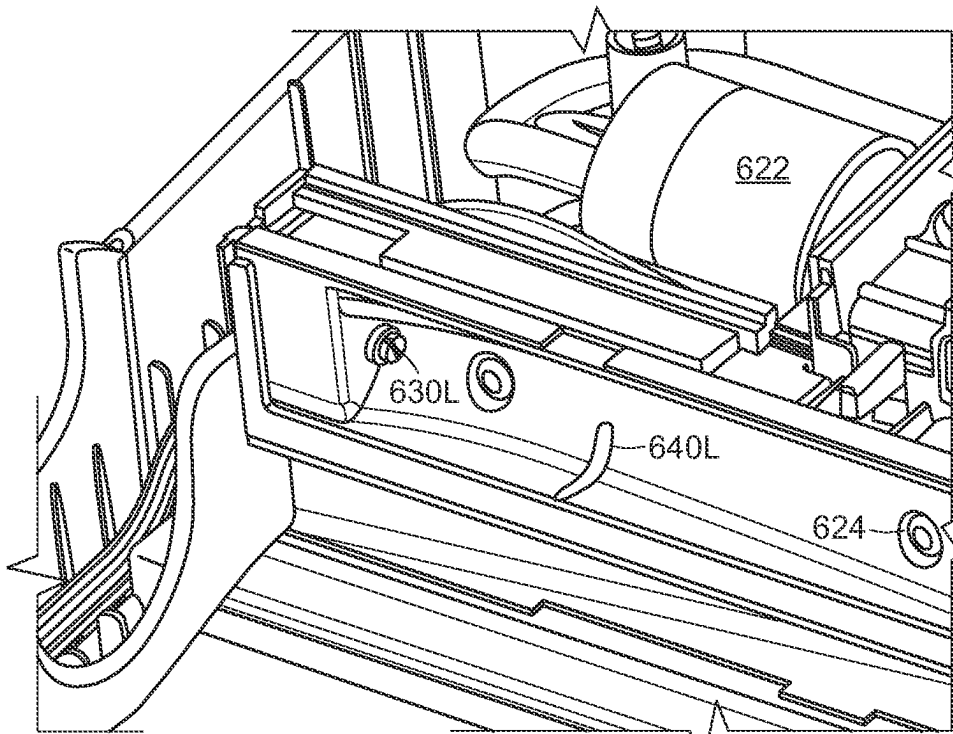


FIG. 13G

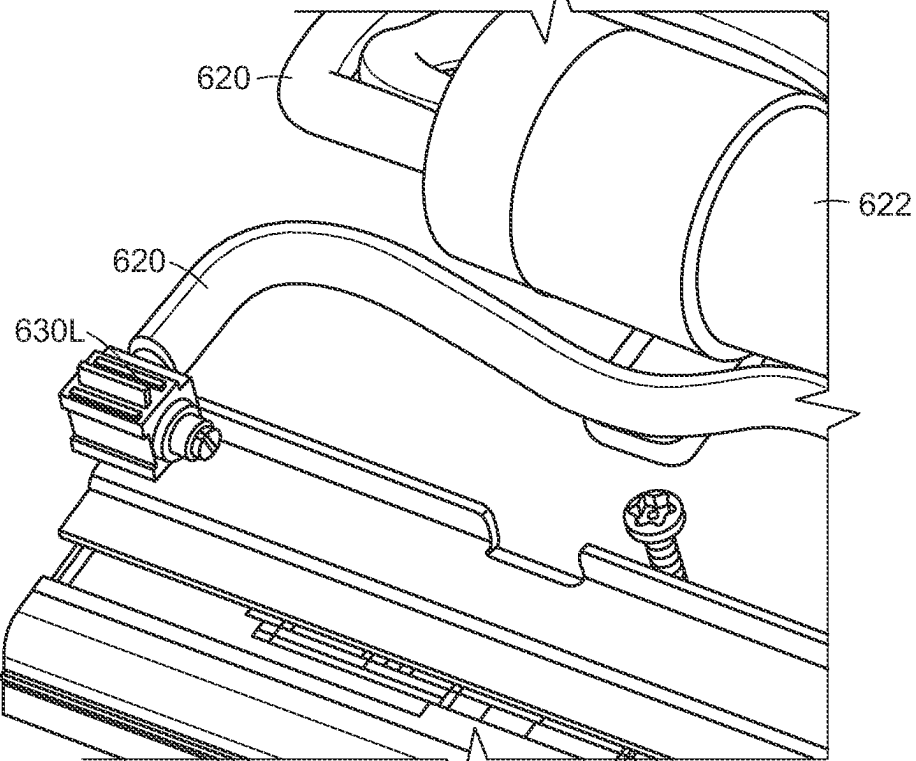


FIG. 13H

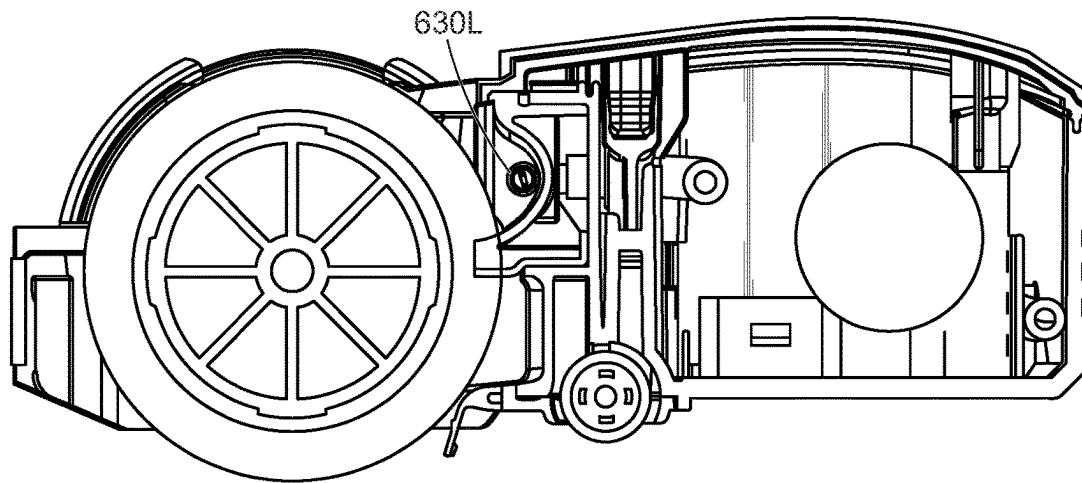


FIG. 13I

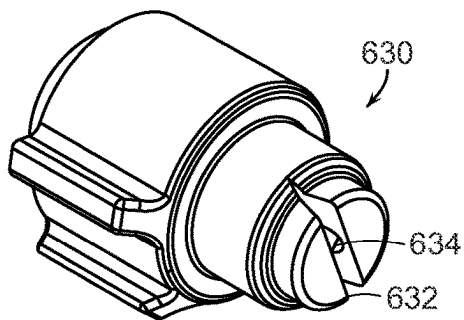


FIG. 14A

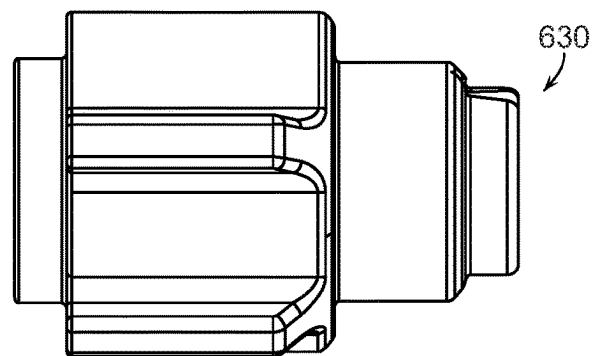


FIG. 14B

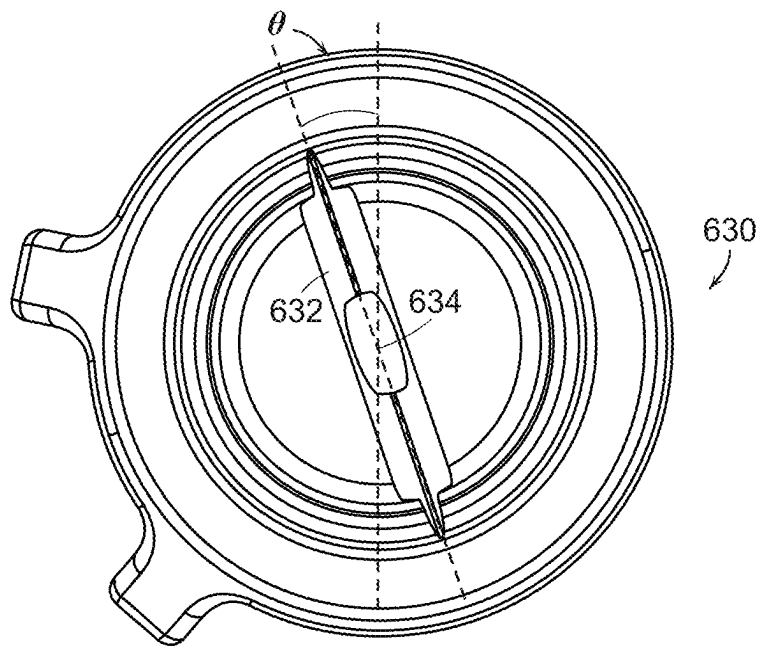


FIG. 14C

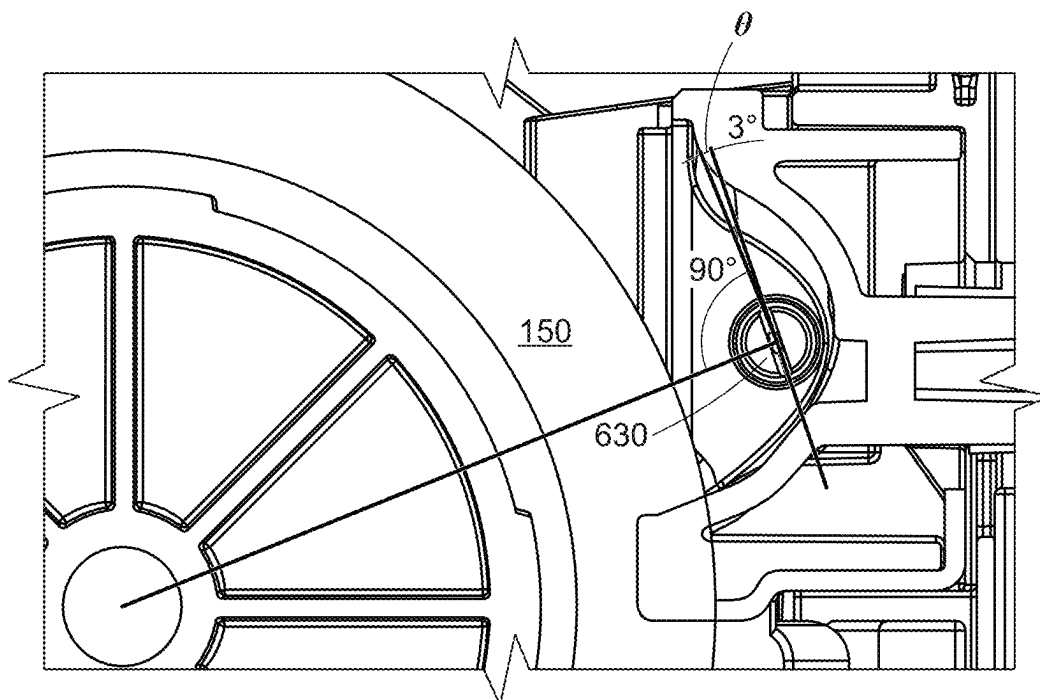


FIG. 14D

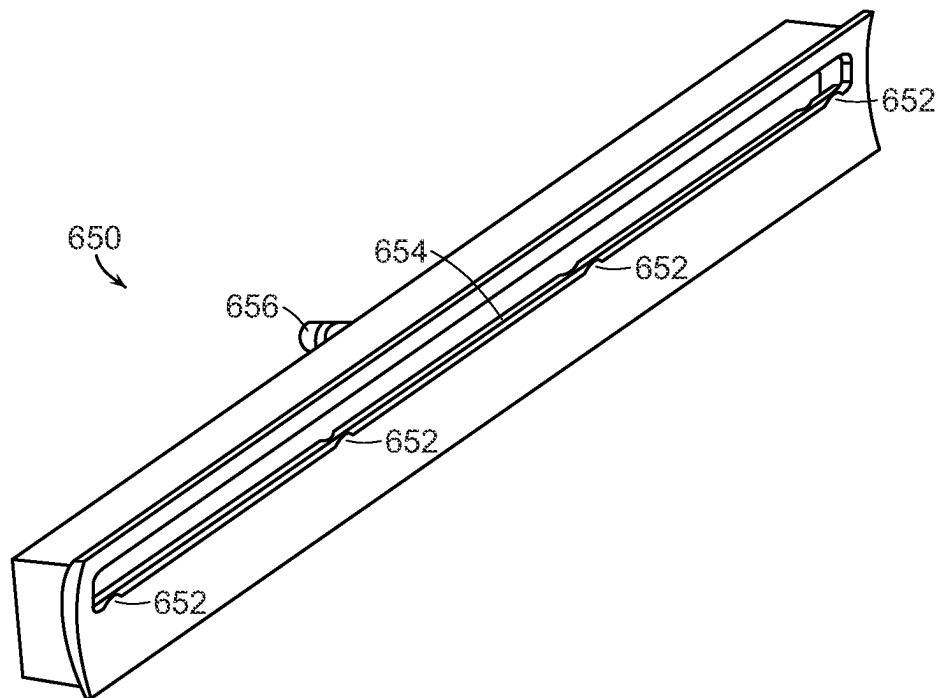


FIG. 15A

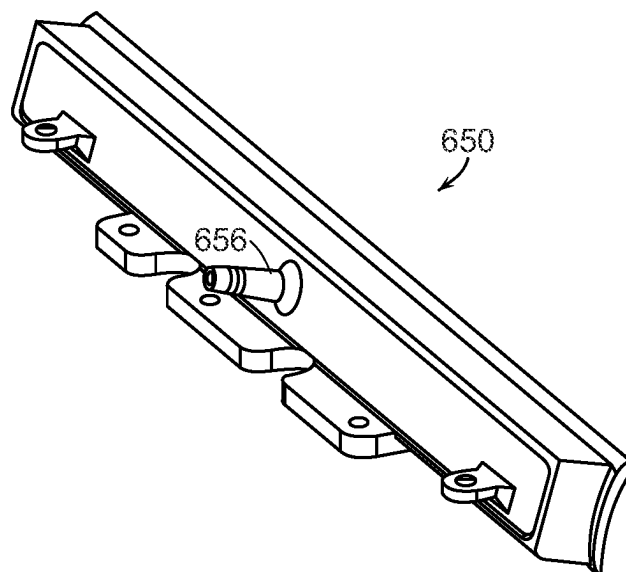


FIG. 15B

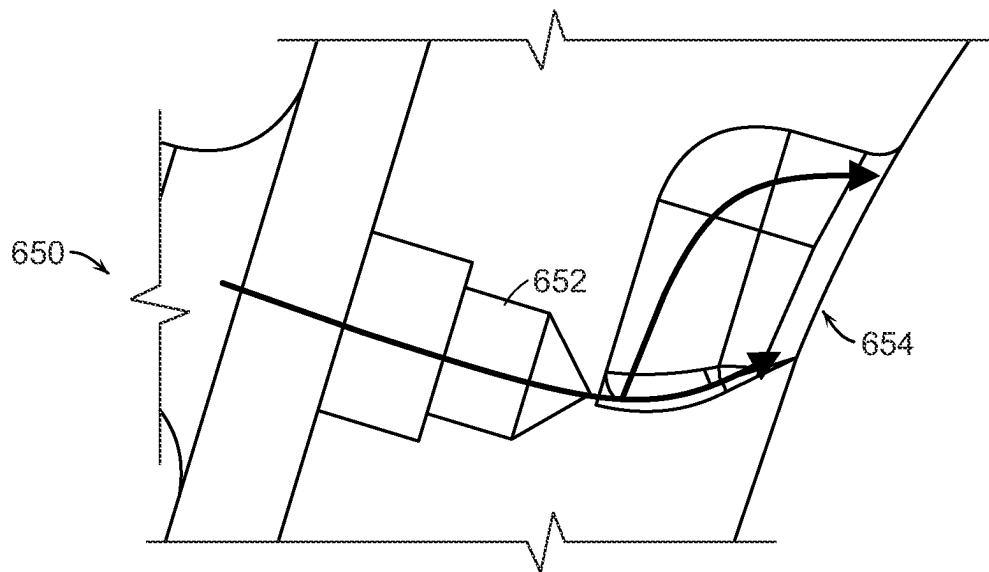


FIG. 15C

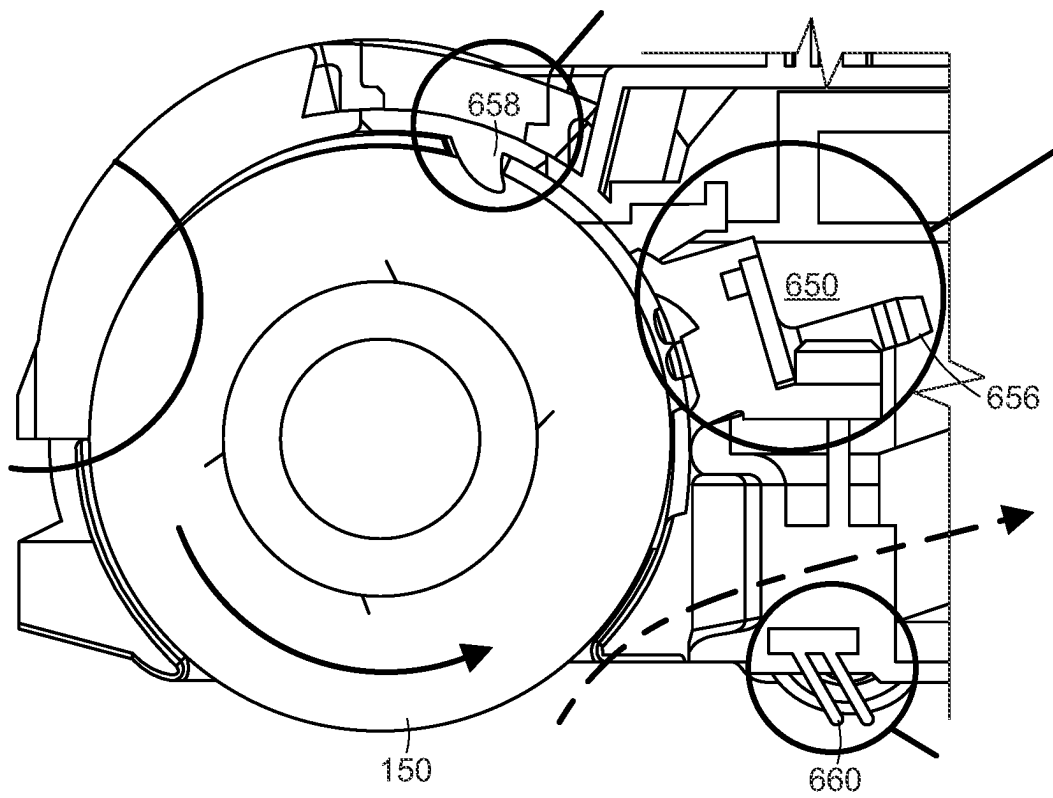


FIG. 15D

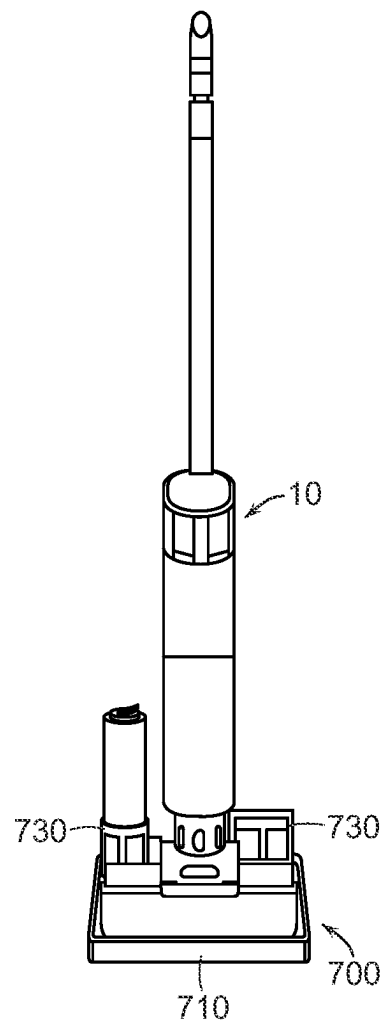


FIG. 16A

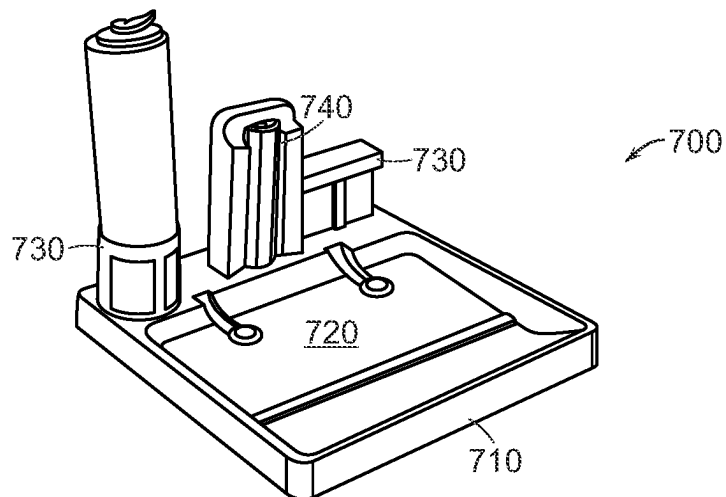


FIG. 16B

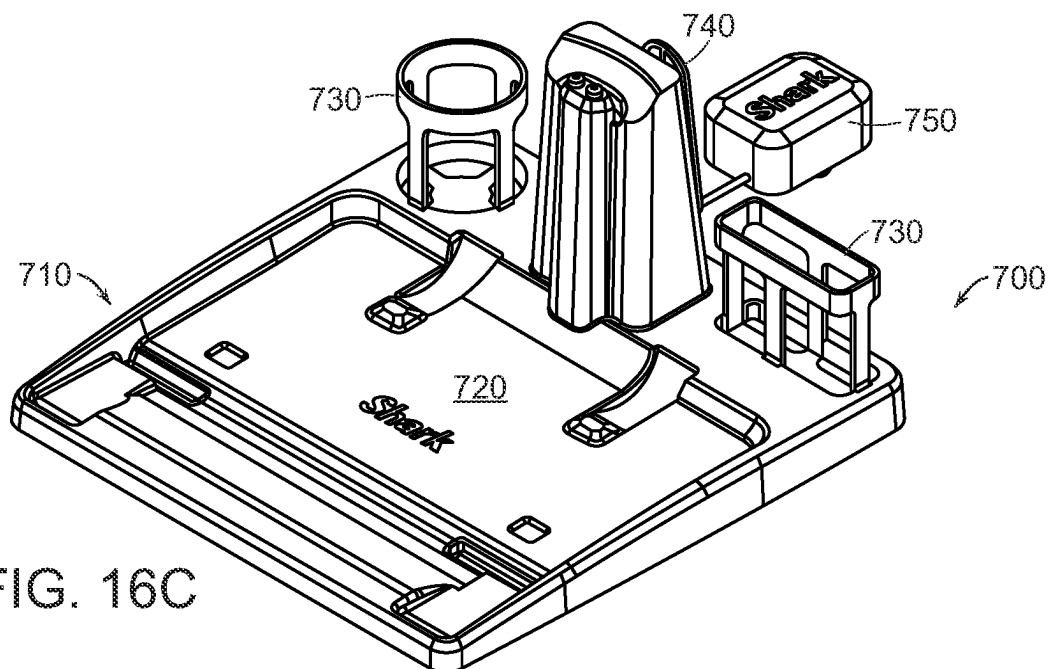


FIG. 16C

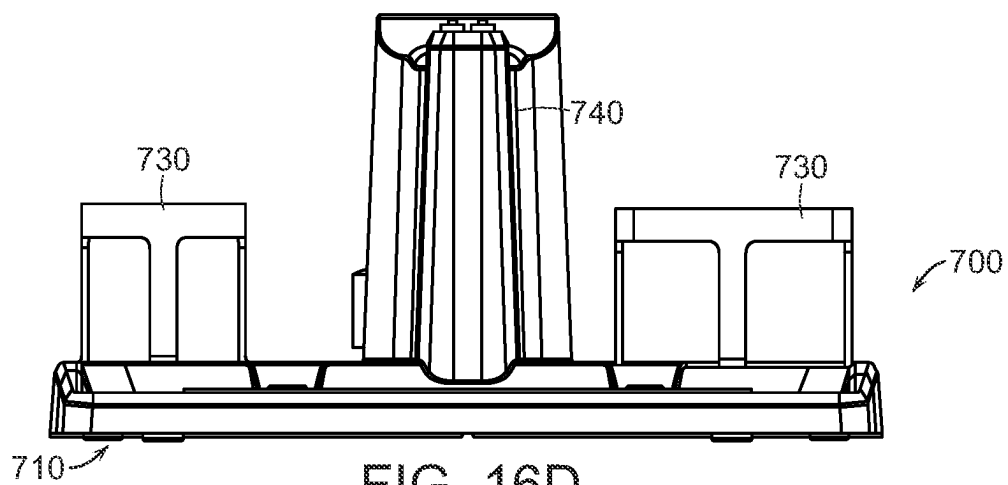


FIG. 16D

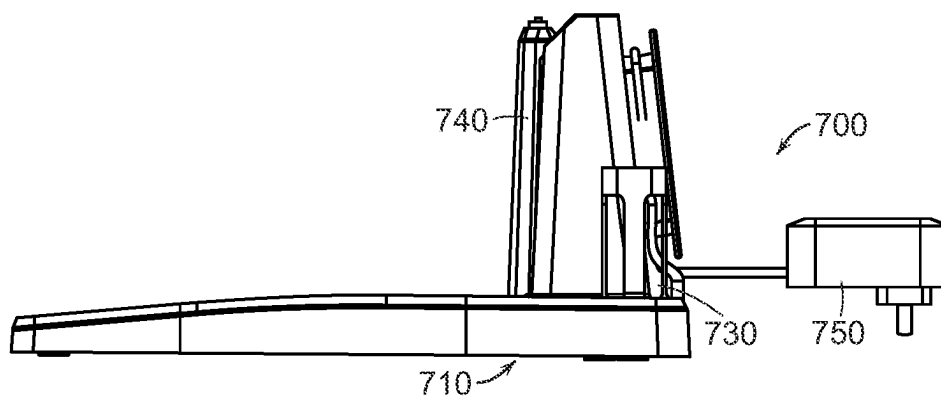


FIG. 16E

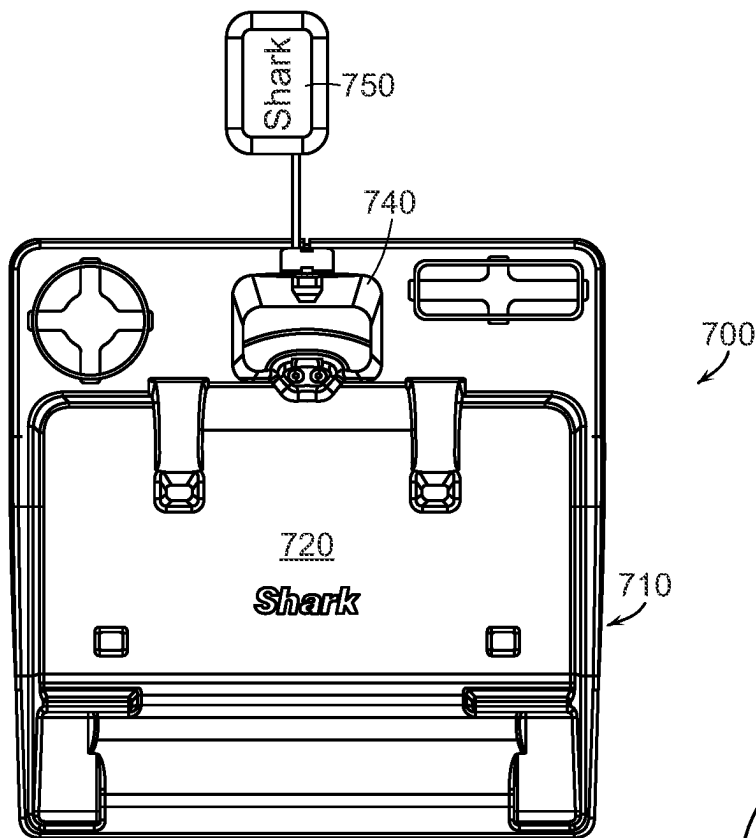


FIG. 16F

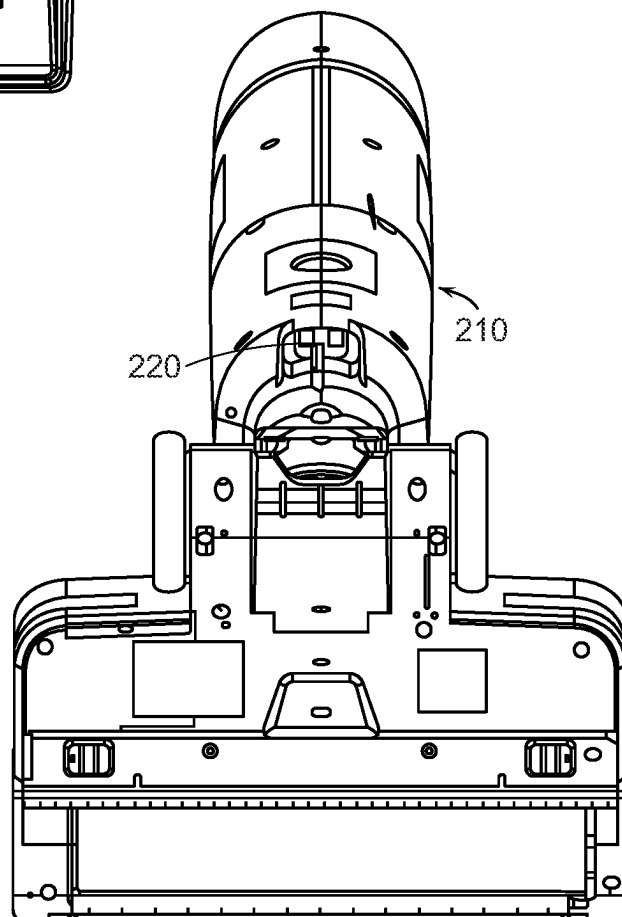


FIG. 16G

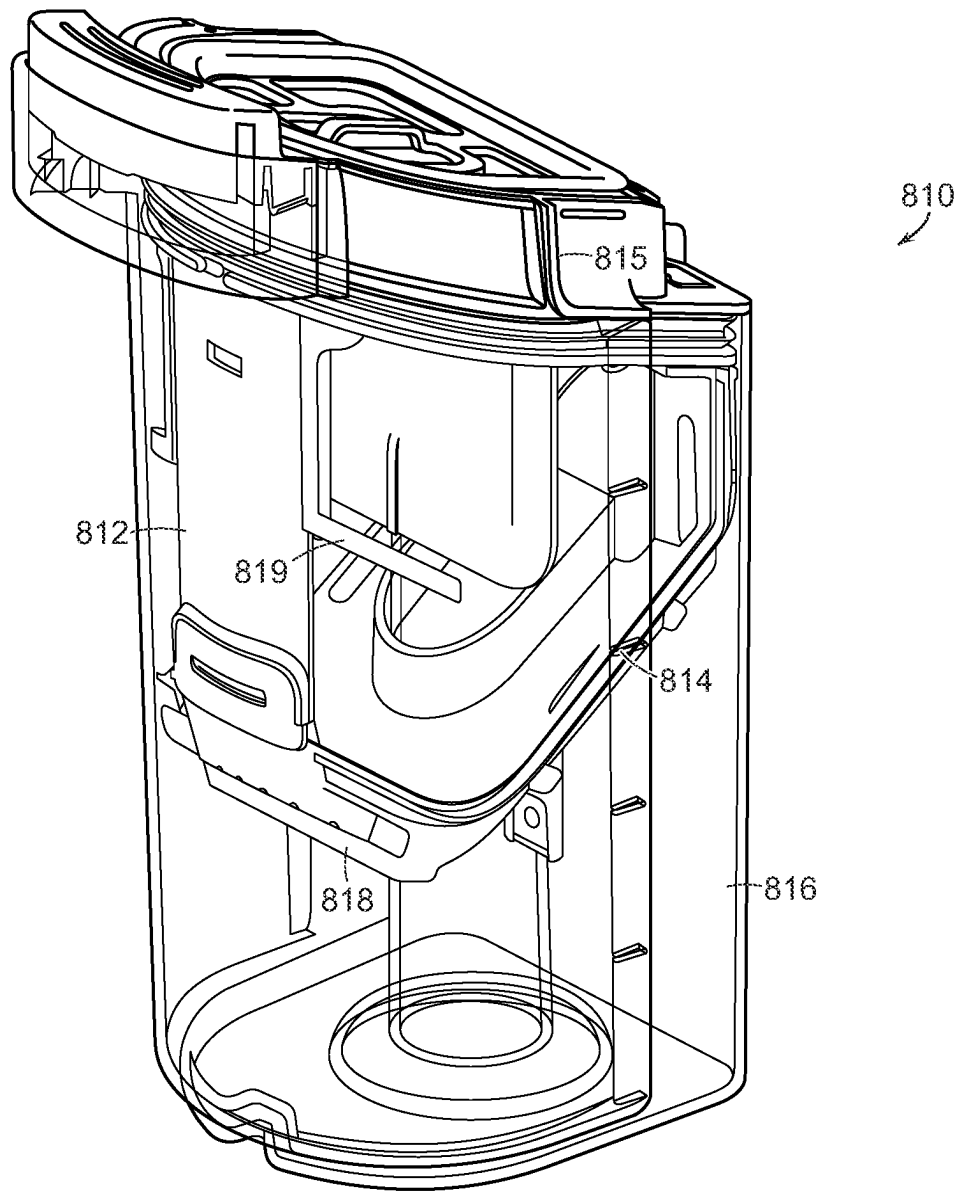


FIG. 17A

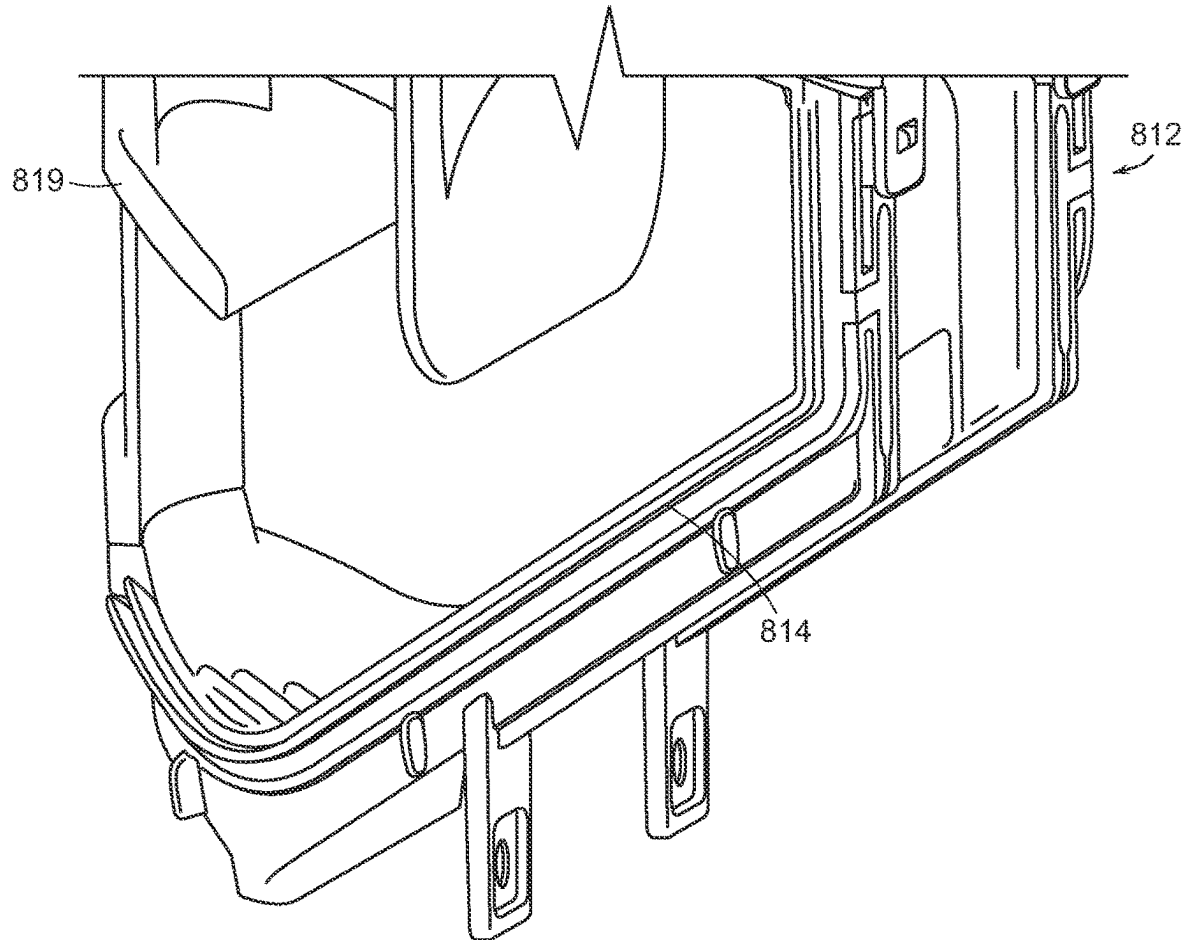


FIG. 17B

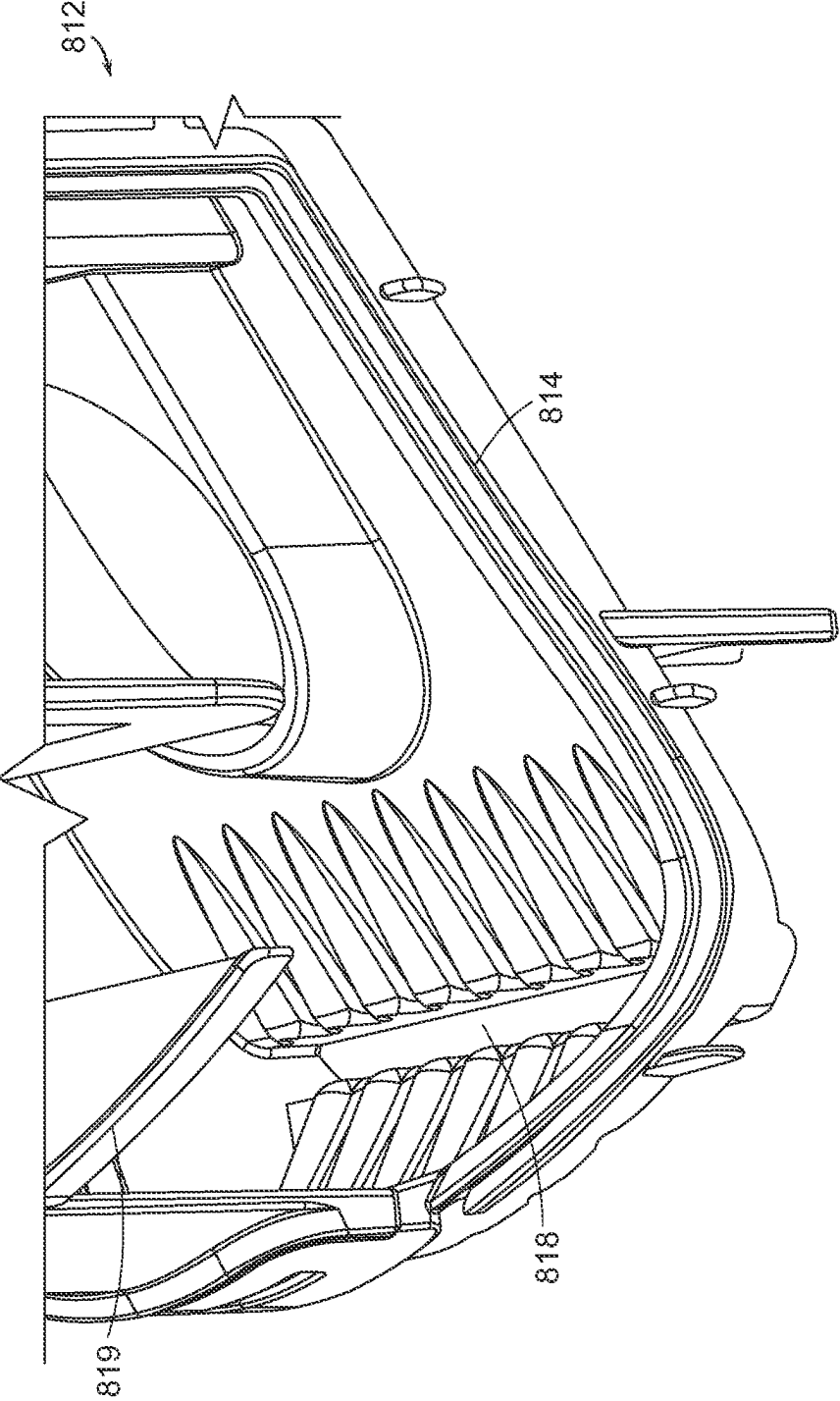


FIG. 17C

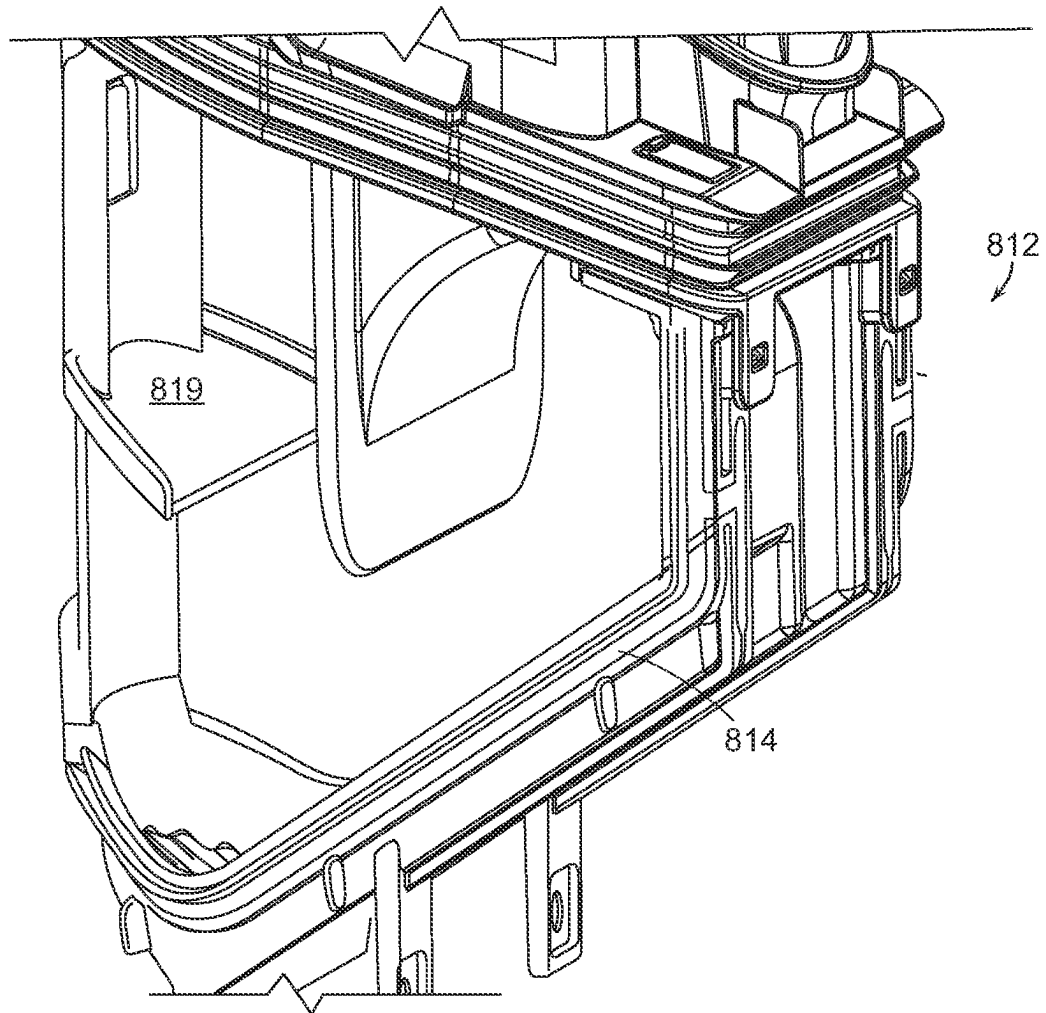


FIG. 17D

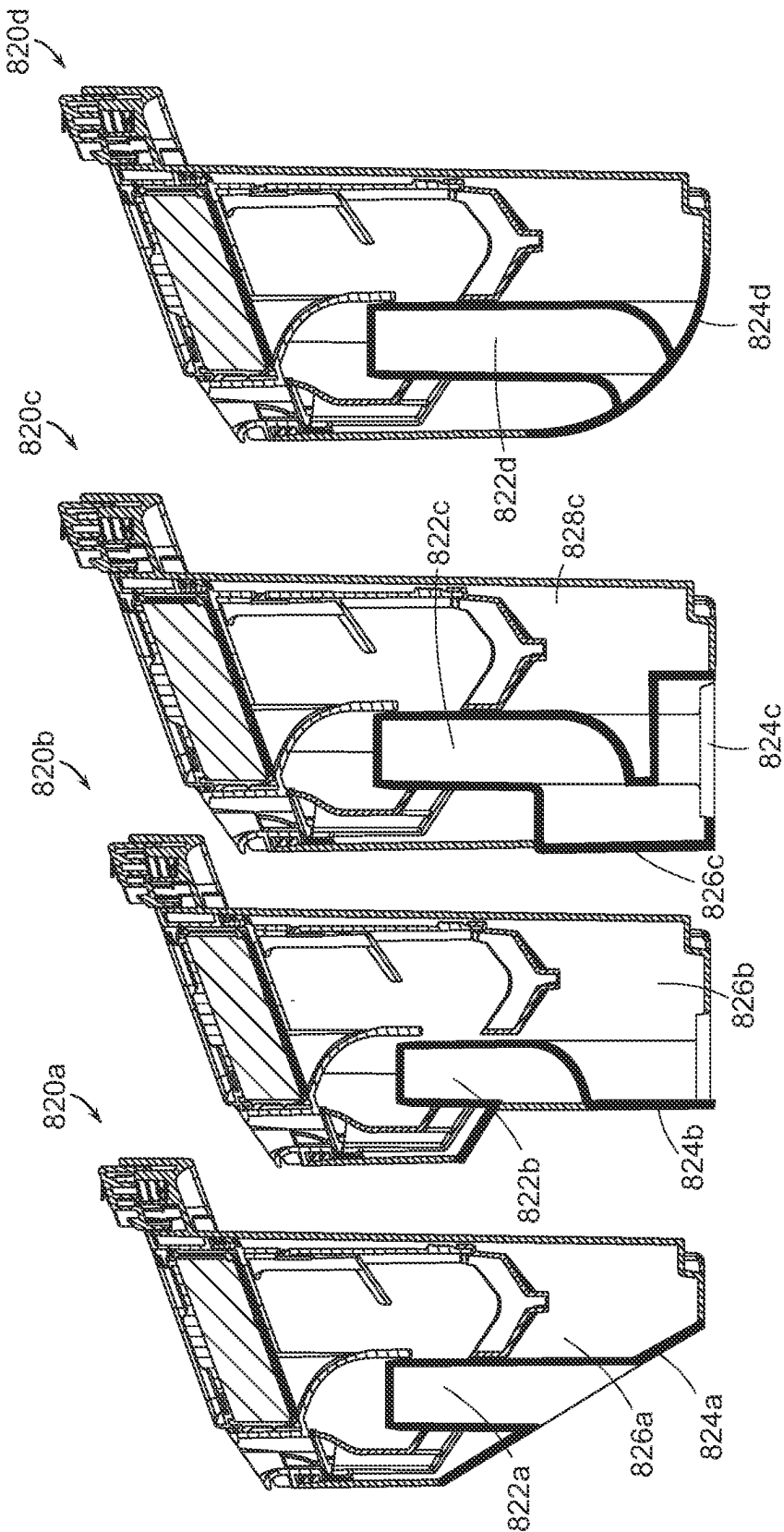


FIG. 18D

FIG. 18C

FIG. 18B

FIG. 18A

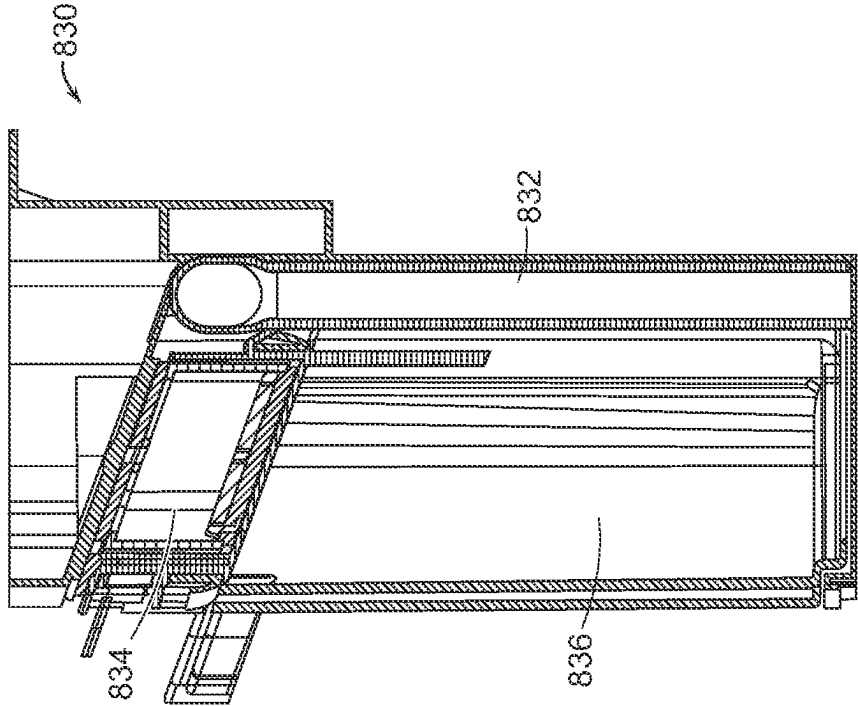


FIG. 19B

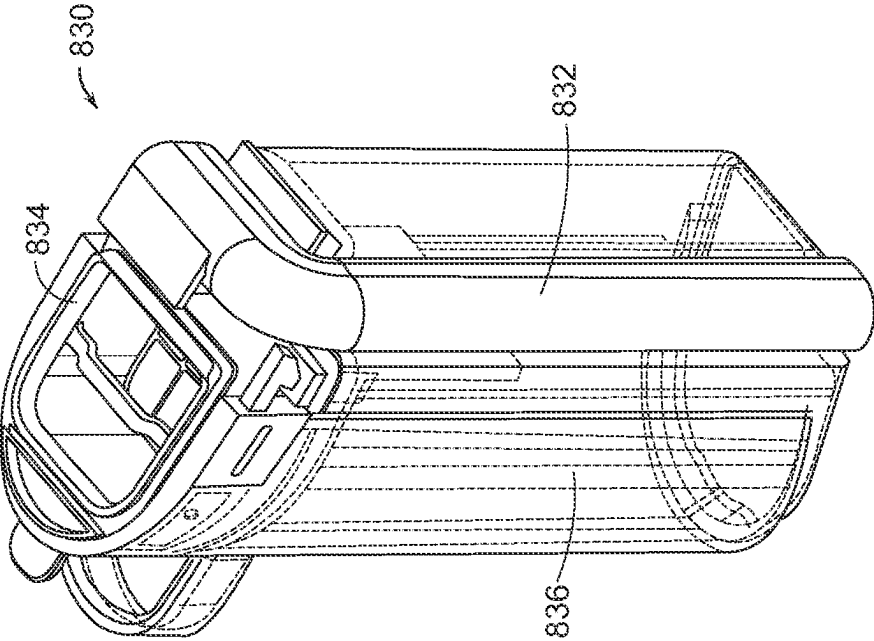


FIG. 19A

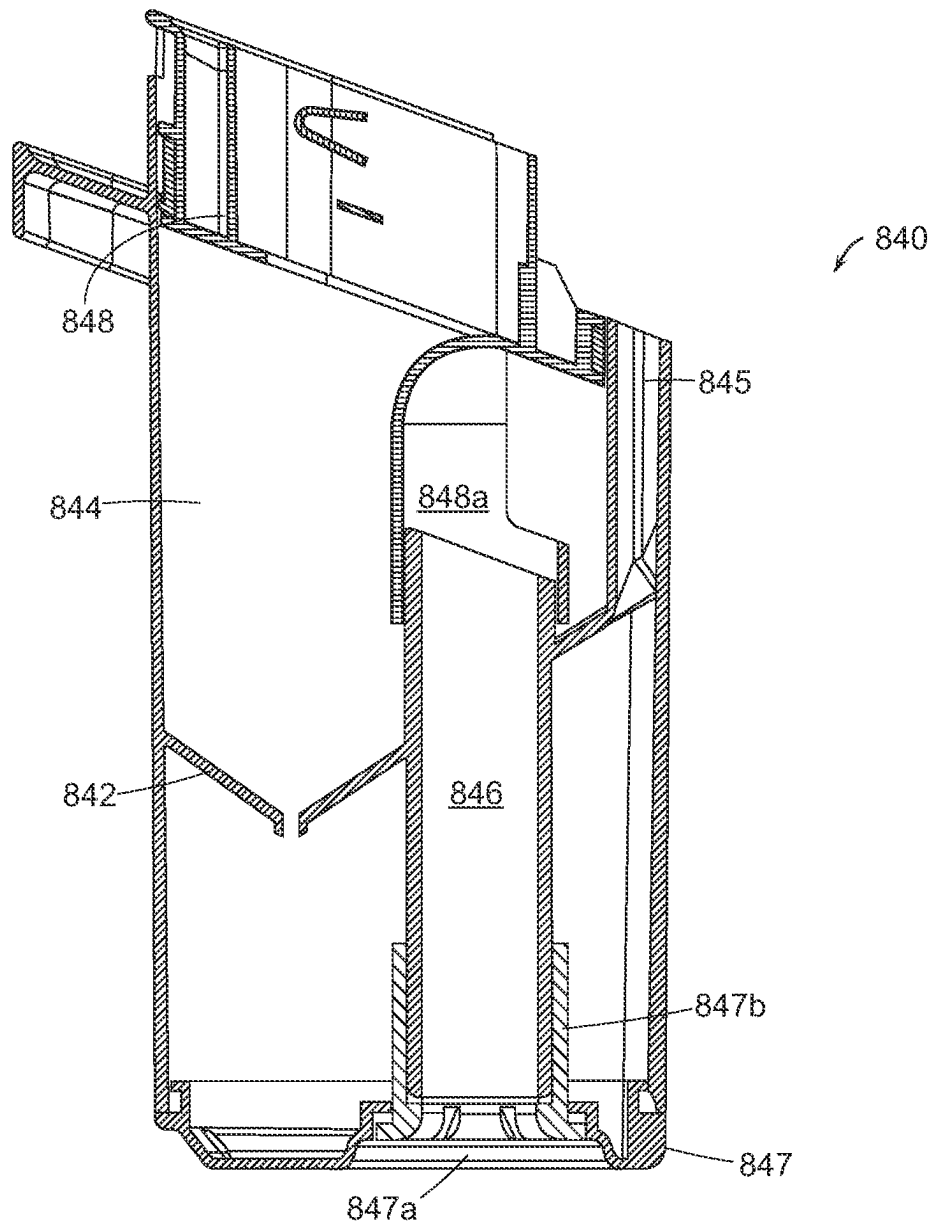


FIG. 20

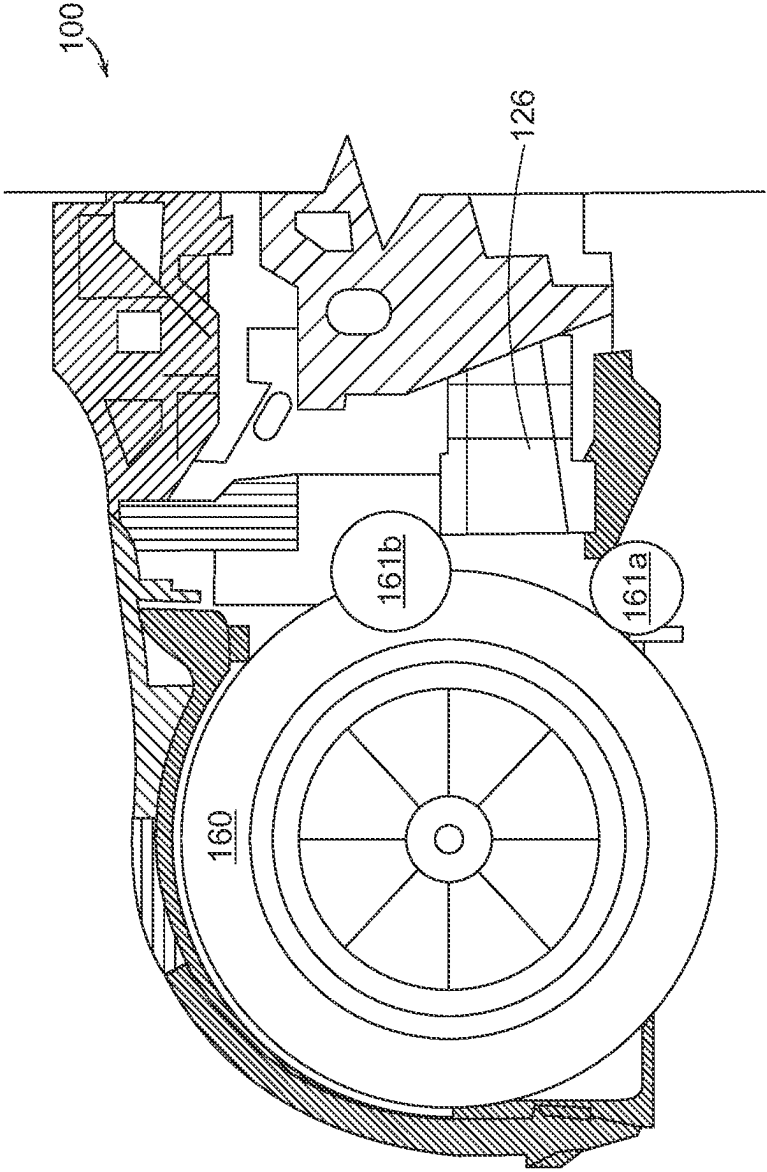


FIG. 21

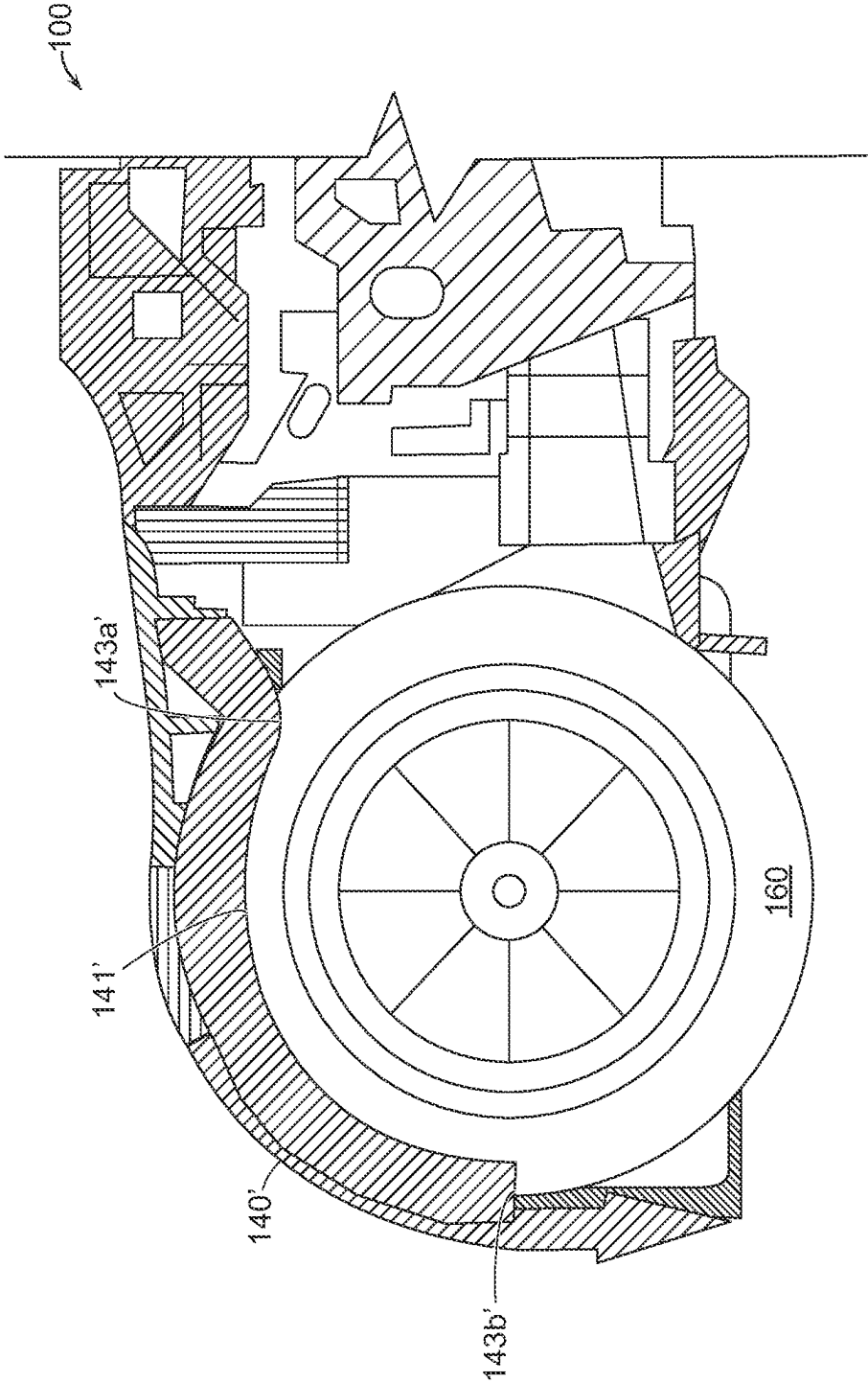


FIG. 22

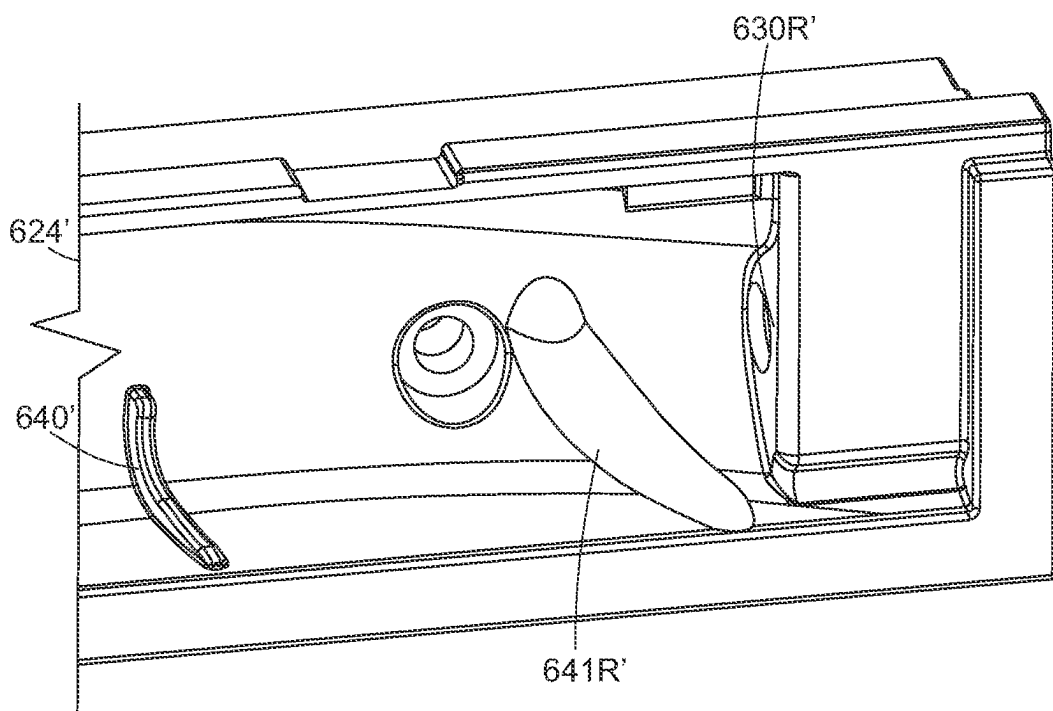


FIG. 23

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WET DRY APPLIANCE**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims priority to U.S. Provisional Patent Application No. 63/285,867 filed on Dec. 3, 2021 and entitled "EXTRACTION CLEANER," the entire contents of which are hereby expressly incorporated by reference herein.

FIELD

A cleaning device operable in wet and dry modes is provided.

BACKGROUND

Conventional cleaning devices, such as dry vacuums and wet vacuums, perform cleaning operations using suction to take in debris and waste. Dry vacuums operate through the use of suction and may employ a brushroll or agitator to assist in freeing the debris and waste from a surface. Wet vacuums operate through the use of suction and a brushroll or pad, but they also supply fluid to the to-be-cleaned surface in order to assist in removal of debris and waste. The supply of fluid can occur directly, wherein fluid is sprayed onto a surface, or indirectly, wherein fluid is sprayed onto an applicator such as a brushroll. When fluid is sprayed onto a brushroll, application of the fluid can be uneven, resulting in inconsistent cleaning of the surface.

When fluid is applied to the surface, the fluid will, incidentally, mix with the debris and waste, and the resulting slurry is drawn into the device using suction. Once inside the cleaning device, the fluid may need to be separated from the debris in order to ease in disposal. Disposal may require total removal of a fluid recovery tank, as well as disassembly of the tank.

Accordingly, there remains a need to provide a better fluid application method to improve consistent fluid application, as well as to enable ease of waste disposal after being drawn into a vacuum cleaning device.

SUMMARY

A cleaning device operable in dry and wet cleaning modes is provided. Related apparatuses and techniques are also provided.

In one embodiment, a cleaning device is provided having a housing body with an upright handle assembly and a head assembly coupled thereto. A fluid recovery tank can be removably coupled to the housing body, and it can include a container having an opening in a top portion thereof extending into an inner chamber configured to hold a liquid therein. The inner chamber can have an inlet formed in a bottom wall and configured to receive fluid from the head assembly. The fluid recovery tank can also include a separator removably disposed within the opening in the container and configured to separate solid debris from fluid drawn into the inner chamber. The fluid recovery tank can also include a lid removably coupled to the separator. The lid can have an outlet fluidly coupled to the suction source such that a suction force can be applied through the outlet, the chamber, and the inlet to the head assembly to draw fluid and debris into the inner chamber. The fluid recovery tank can further include a pour spout formed in the separator and having an opening formed there through and in fluid communication

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with the inner chamber, wherein the separator is configured to allow fluid to be poured from the pour spout while retaining solid debris in the inner chamber.

One or more of the following features can be included in any feasible combination. For example, the separator can include a lower surface that is spaced from the lid and from the bottom wall of the container. The lower surface of the separator can have an opening formed therein to allow fluid to pass therethrough while substantially preventing solid debris from passing therethrough. In certain embodiments, the lower surface of the separator can be substantially V-shaped with opposed walls oriented at an angle relative to one another and mated along a lower-most end thereof. The opening can be formed between the walls at the lower-most end.

In another example, the separator can include a sidewall having a channel formed therein and aligned with the pour spout to allow fluid to flow from the container out the pour spot.

In another example, the lid can include a removable filter disposed across the outlet. In certain aspects, the removable filter can include a first filter material having a first porosity and a second filter material having a second porosity that is different than the first porosity.

In another example, the cleaning device can include a latch on the container and configured to engage the housing body to retain the fluid recovery tank on the housing body. In certain aspects, the latch can be positioned on a first side of the container and the pour spout can be positioned on a second side of the container opposite the first side.

In another example, the inlet in the bottom wall can include a hollow standpipe extending from the bottom wall toward the lid and having an inner lumen there through for receiving fluid and debris from the head assembly. In certain aspects, the hollow standpipe can extend through an opening in the separator to deliver fluid and debris into the separator.

In another example, the separator can include at least one deflector configured to direct fluid away from the lid. In certain aspects, the at least one deflector can be located proximate to the inlet. In further aspects, the at least one deflector can include a first deflector located proximate the inlet and a second deflector located below the inlet.

In another embodiment, a fluid recovery tank for use on a cleaning device is provided. The fluid recovery tank can include a container having a bottom wall and sidewalls defining an inner chamber therein. The top of the container can be open, and the bottom wall can include a fluid inlet therein. The cleaning device can include a separator removably disposed within the open top of the container and extending into the chamber. A removable lid can be disposed in the separator. The separator can have a pour spout formed therein. The separator can separate the chamber into an upper portion and a lower portion. The separator can be configured to retain solid debris within the upper portion while allowing liquid in the bottom portion to be poured out of the pour spout.

One or more of the following features can be included in any feasible combination. For example, the separator can include a lower surface that is spaced from the lid and from the bottom wall of the container. The lower surface of the separator can have an opening formed therein to allow fluid to pass there through while substantially preventing solid debris from passing there through.

In another example, the separator can include a sidewall having a channel formed therein and aligned with the pour spout to allow fluid to flow from the container out the pour spot.

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In another example, the lid can include a removable filter disposed therein and configured to allow a suction force to be applied there through. In other aspects, the removable filter can include a first filter material having a first porosity and a second filter material having a second porosity that is

different than the first porosity.

In another example, the fluid recovery device can include a spring-biased latch movably mounted on an exterior surface of the container. In other aspects, the latch can be positioned on a first side of the container and the pour spout is positioned on a second side of the container opposite the first side.

In another example, the inlet in the bottom wall can include a hollow standpipe extending from the bottom wall toward the lid and having an inner lumen there through for receiving fluid and debris from the head assembly. The hollow standpipe can extend through an opening in the separator.

In another example, the pour spout can be configured to be in an open position when the lid is in an open position and a closed position.

In another example, the separator can include a seal configured to frictionally engage the container to create a water-tight seal therewith.

In another example, the separator can include at least one deflector configured to direct fluid away from the lid. In certain aspects, the at least one deflector can be located proximate to the fluid inlet. In further aspects, the at least one deflector can include a first deflector located proximate the inlet and a second deflector located below the fluid inlet.

In another embodiment, a cleaning device is provided and can include a head assembly containing a brushroll and a housing body coupled to the head assembly and having an upright handle extending therefrom. The housing body can include a suction source in fluid communication with the head assembly, a fluid supply assembly configured to retain fluid and to deliver fluid to the head assembly, and a fluid recovery tank configured to receive fluid from the head assembly using the suction source. The head assembly can include at least one spray nozzle in fluid communication with the fluid supply assembly and configured to emit fluid onto a first portion of the brushroll. The head assembly can further include at least one deflector disposed therein and configured to redirect fluid emitted by the at least one spray nozzle onto a second portion of the brushroll.

One or more of the following features can be included in any feasible combination. For example, the at least one spray nozzle can include a left spray nozzle disposed on the left side of the head assembly and a right spray nozzle disposed on the right side of the head assembly. In other aspects, the at least one deflector includes a left deflector configured to redirect fluid emitted by the left spray nozzle and a right deflector configured to redirect fluid emitted by the right spray nozzle. In other aspects, the second portion of the brushroll can include a left end portion of the brushroll and a right end portion of the brushroll. The first portion of the brushroll can be a mid-position located between the left end portion and the right end portion. In still further aspects, the left spray nozzle and the right spray nozzle can be longitudinally aligned with one another.

In another embodiment, the at least one deflector can be curved toward the at least one spray nozzle.

In another embodiment, the at least one spray nozzle is configured to spray fluid in a substantially planar orientation. In other aspects, the at least one nozzle is configured to spray fluid in a fan pattern at a spray angle between about 10 and 60 degrees. In another example, the at least one spray

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nozzle is configured to spray fluid in a flat fan pattern along a plane substantially tangential to the brushroll.

In another embodiment, a cleaning device is provided and can include a base housing having a brushroll chamber with a brushroll disposed therein, and a suction outlet disposed therein and in fluid communication with the brushroll chamber. The cleaning device can include at least one fluid delivery spray nozzle disposed in the brushroll chamber and configured to emit fluid onto a first portion of the brushroll. The cleaning device can include at least one deflector disposed in the brushroll chamber and configured to redirect fluid emitted by the fluid assembly onto a second portion of the brushroll.

One or more of the following features can be included in any feasible combination. For example, the cleaning device can include a fluid recovery tank in fluid communication with the suction inlet and configured to receive fluid and debris from the suction inlet.

In another example, the at least one fluid delivery spray nozzle can include first and second fluid delivery spray nozzles positioned on opposite ends of the brushroll and configured to deliver fluid along a length of the brushroll. The at least one deflector can be positioned within the brushroll chamber between the first and second fluid delivery spray nozzles. In certain embodiments, the at least one deflector can include a first deflector positioned adjacent to the first fluid delivery spray nozzle for deflecting fluid sprayed from the first fluid delivery nozzle, and a second deflector positioned adjacent the second fluid delivery spray nozzle for deflecting fluid sprayed from the second fluid delivery spray nozzle.

In another embodiment, the at least one deflector can include an elongate projection formed on an inner surface of the brushroll chamber and positioned within a flow path of fluid delivered by the at least one fluid delivery nozzle.

In other aspects, the first portion of the brushroll can include a mid-portion of the brushroll and the second portion of the brushroll comprises first and second end portions of the brushroll.

In another example, the at least one deflector can be molded into a housing at least partially defining the brushroll chamber.

In another example, the at least one deflector can be positioned on an inner surface of the brushroll chamber above an opening within the brushroll chamber defining the suction inlet.

In another example, the at least one nozzle can be configured to spray fluid in a substantially planar fan-shaped pattern.

In another example, the at least one nozzle can include a first nozzle disposed in a right side of the brushroll chamber and a second nozzle disposed in the left side of the brushroll chamber. The at least one deflector can be a first pair of deflectors configured to redirect fluid emitted by the first nozzle and a second pair of deflectors configured to redirect fluid emitted by the second nozzle.

In another embodiment, a fluid recovery tank for use on a cleaning device is provided. The fluid recovery tank can include a container having a bottom wall and sidewalls defining an inner chamber therein, a top of the container being open, the bottom wall including a fluid inlet therein, and the container having a longitudinal axis extending from the top to the bottom wall. The fluid recovery tank can include a latch movably coupled to an outer sidewall of the container and configured to move along an axis substantially parallel to the longitudinal axis of the chamber. The latch can include an engagement feature configured to engage a

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portion of a cleaning device to mate the container to a cleaning device. The fluid recovery tank can include a separator removably disposed within the open top of the container and extending into the chamber, the separator being configured to separate solid debris from fluid within the inner chamber.

One or more of the following features can be included in any feasible combination. For example, the latch can be movably disposed within a housing formed on the outer sidewall of the container. The latch can extend above an upper surface of the container. In certain embodiments, the latch can be spring-biased to a locking position. In another example, the latch can be disposed entirely outside the container.

In another embodiment, the separator can have a bottom wall that is spaced from the bottom wall of the container. The bottom wall of the separator can have an opening formed therein to allow fluid to pass there through while substantially preventing solid debris from passing there through such that solid debris is retained within the separator.

In another example, the fluid recovery tank can include a lid removably disposed within an opening formed in a top of the separator. In other aspects, the lid can include a filtration device.

In another embodiment, a fluid recovery tank for use on a cleaning device is provided. The fluid recovery tank can include a container defining an inner chamber therein. The container can have a suction outlet configured to couple to a suction source for allowing a suction force to be applied to the inner chamber. The container can also have a fluid inlet for allowing fluid to be drawn into the chamber by the suction force. The fluid recovery tank can include a separator removably disposed within the container and having an opening formed therein and configured to allow fluid to pass there through while substantially preventing solid debris from passing there through such that solid debris is retained within the separator. The fluid recovery tank can include a button movably disposed on an exterior of the container and configured to move between a locked configuration in which the button is configured to engage a cleaning device to retain the container on the cleaning device, and an unlocked configuration in which the button is configured to disengage from a cleaning device to allow the container to be removed from the cleaning device.

One or more of the following features can be included in any feasible combination. For example, a top of the button can be substantially parallel to a top of the container when the button is in the locked configuration and the unlocked configuration. The button can be biased to the locked configuration.

In another embodiment, a top of the button can extend above the top of the container when the button is in the locked configuration, and the top of the button can extend above the top of the container when the button is in the unlocked configuration.

In other aspects, the button can be disposed within a housing formed on an external surface of the container. The housing can include a biasing element disposed therein and can bias the button to the locked configuration. The button can include a protrusion configured to be received by a complementary depression on a cleaning device. In other aspects, the button can be curved to align with a contour of the cleaning device.

In another embodiment, the fluid recovery tank can include a lid removably disposed within an opening in the separator, the lid defining the suction outlet. The lid can

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include at least one filter disposed therein and extending across the suction outlet. The at least one filter can include a first filter material having a first porosity and a second filter material having a second porosity greater than the first porosity.

In another example, the fluid inlet can include a hollow standpipe extending into the inner chamber.

The details of one or more variations of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features and advantages of the subject matter described herein will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF DRAWINGS

These and other features will be more readily understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

FIG. 1A is a front perspective view of one embodiment of a cleaning device;

FIG. 1B is a front view of the cleaning device of FIG. 1A;

FIG. 1C is a right side view of the cleaning device of FIG. 1A;

FIG. 1D is cross-sectional right side view of the cleaning device of FIG. 1A;

FIG. 2A is a front perspective view of a head assembly of the cleaning device of FIG. 1A;

FIG. 2B is a right side view of the head assembly of FIG. 2A;

FIG. 2C is a front view of the head assembly of FIG. 2A;

FIG. 2D is a bottom view of the head assembly of FIG. 2A;

FIG. 2E is a cross-sectional side view of the head assembly of FIG. 2A;

FIG. 2F is a front perspective view of the head assembly of FIG. 2A, having a linking assembly removed;

FIG. 2G is a front perspective view of the head assembly of FIG. 2F, having a brushroll removed and showing a right support structure;

FIG. 2H is a front perspective view of the head assembly of FIG. 2F, having a brushroll removed and showing a left support structure;

FIG. 3A is a right front perspective view of a brushroll used in the cleaning device of FIG. 1A;

FIG. 3B is a left rear perspective view of the brushroll of FIG. 3A;

FIG. 3C is a right side view of the brushroll of FIG. 3A;

FIG. 3D is a left side view of the brushroll of FIG. 3A;

FIG. 4A is a top perspective of another embodiment of a brushroll;

FIG. 4B is a right side view of the brushroll of FIG. 4A;

FIG. 4C is a partial left perspective view of the brushroll of FIG. 4A;

FIG. 4D is a partial cross-sectional view of the brushroll of FIG. 4A;

FIG. 5A is a front perspective view of a brushroll cover used with the head assembly of FIG. 2A;

FIG. 5B is a right side view of the brushroll cover of FIG. 5A;

FIG. 6A is a front perspective view of a body assembly of the cleaning device of FIG. 1A;

FIG. 6B is a right side view of the body assembly of FIG. 6A;

FIG. 6C is front view of the body assembly of FIG. 6A;

FIG. 6D is a cross-sectional side view of the body assembly of FIG. 6A;

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FIG. 7A is a front perspective view of a handle assembly of the cleaning device of FIG. 1A;

FIG. 7B is a right side view of the handle assembly of FIG. 7A;

FIG. 7C is a front view of the handle assembly of FIG. 7A;

FIG. 7D is a cross-sectional side view of the handle assembly of FIG. 7A

FIG. 8A is a front perspective view of the body assembly of FIG. 6A, having a fluid supply tank and a recovery tank removed from their respective retaining areas;

FIG. 8B is a front view of the body assembly of FIG. 8A;

FIG. 8C is a right side view of the body assembly of FIG. 8A;

FIG. 8D is a bottom perspective view of the body assembly of FIG. 8A;

FIG. 9A is a front perspective view of a motor assembly of the cleaning device of FIG. 1A;

FIG. 9B is a right side view of the motor assembly of FIG. 9A;

FIG. 9C is a front view of the motor assembly of FIG. 9A;

FIG. 9D is a top view of the motor assembly of FIG. 9A;

FIG. 9E is cross-sectional side view of the motor assembly of FIG. 9A;

FIG. 10A is a front perspective view of a fluid recovery tank of the cleaning device of FIG. 1A;

FIG. 10B is a right side view of the fluid recovery tank of FIG. 10A;

FIG. 10C is a front view of the fluid recovery tank of FIG. 10A;

FIG. 10D is a cross-sectional side view of the fluid recovery tank of FIG. 10A;

FIG. 10E is a front perspective view of a separator of the fluid recovery tank of FIG. 10A;

FIG. 10F is a rear side perspective view of the separator of FIG. 10E;

FIG. 10G is a bottom view of the separator of FIG. 10E;

FIG. 10H is a partial cross-sectional view of the separator of FIG. 10E;

FIG. 10I is a perspective view of a lid of the fluid recovery tank of FIG. 10A;

FIG. 10J is an exploded view of the lid of FIG. 10I;

FIG. 10K is a rear perspective view of a latch of the fluid recovery tank of FIG. 10A;

FIG. 10L is an exploded view of the fluid recovery tank of FIG. 10A;

FIG. 11A is a right side view of another embodiment of a fluid recovery tank;

FIG. 11B is a partial left perspective view of a separator of the fluid recovery tank of FIG. 11A;

FIG. 11C is a partial front view of the separator of FIG. 11B;

FIG. 11D is a front perspective view of the fluid recovery tank of FIG. 11A, having been installed in a body assembly of a cleaning device;

FIG. 11E is a partial cross-sectional view of the fluid recovery tank of FIG. 11D, showing an upper end engaged in the body assembly;

FIG. 11F is a partial cross-sectional view of the fluid recovery tank of FIG. 11D, showing a lower end engaged in the body assembly;

FIG. 11G is a front perspective view of a filter engaged in a body assembly according to the embodiment of FIG. 11A;

FIG. 11H is a partial cross-sectional view of the filter of FIG. 11G engaged in the body assembly;

FIG. 11I is a front perspective view of the filter of FIG. 11G being disengaged from the body assembly;

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FIG. 12A is a front perspective view of a fluid supply tank of the cleaning device of FIG. 1A;

FIG. 12B is a front view of the fluid supply tank of FIG. 12A;

FIG. 12C is a right side view of the fluid supply tank of FIG. 12A;

FIG. 12D is a cross-sectional side view of the fluid supply tank of FIG. 12A;

FIG. 12E is a front perspective cross-sectional view of the fluid supply tank of FIG. 12A;

FIG. 12F is an exploded view of the fluid supply tank of FIG. 12A;

FIG. 13A is a top view of a head assembly having no upper housing and showing components used in wet vacuum modes;

FIG. 13B is a top view of the tubing, fluid pump, and nozzles shown FIG. 13A;

FIG. 13C is a front view of the application face of the head assembly of FIG. 13A;

FIG. 13D is a partial front perspective view of a right side of the head assembly of FIG. 13A, having a brushroll removed;

FIG. 13E is a partial front perspective view of the right side of the head assembly of FIG. 13A, having an application face removed;

FIG. 13F is a partial cross-sectional view of the right side of the head assembly of FIG. 13A;

FIG. 13G is a partial front perspective view of the left side of the head assembly of FIG. 13A

FIG. 13H is a partial front perspective view of the left side of the head assembly of FIG. 13A, having the application face removed;

FIG. 13I is a front view of the application face of the head assembly of FIG. 13A;

FIG. 14A is a front perspective view of a spray nozzle of the cleaning device of FIG. 1A;

FIG. 14B is a right side view of the spray nozzle of FIG. 14A;

FIG. 14C is a front view of the spray nozzle of FIG. 14A;

FIG. 14D is a partial cross-sectional view of a head assembly including the spray nozzle of FIG. 14A;

FIG. 15A is a front perspective view of an alternative embodiment of an application face;

FIG. 15B is a rear perspective view of the application face of FIG. 15A;

FIG. 15C is a partial cross-sectional view of the application face of FIG. 15A;

FIG. 15D is a partial cross-sectional view of a head assembly including the application face of FIG. 15A;

FIG. 16A is a front view of the cleaning device of FIG. 1A placed on a charging mat;

FIG. 16B is a front perspective view of the charging mat of FIG. 16A;

FIG. 16C is a front perspective view of the charging mat of FIG. 16A;

FIG. 16D is a front view of the charging mat of FIG. 16A;

FIG. 16E is a right side view of the charging mat of FIG. 16A;

FIG. 16F is a top view of the charging mat of FIG. 16A; and

FIG. 16G is a rear perspective view of the cleaning device of FIG. 1A, without a charging mat;

FIG. 17A is a perspective view of another embodiment of a fluid supply tank;

FIG. 17B is a partial perspective view of a separator of the fluid supply tank of FIG. 17A;

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FIG. 17C is another partial perspective view of the separator of FIG. 17B;

FIG. 17D is another partial perspective view of the separator of FIG. 17B;

FIG. 18A is a left side view of the fluid supply tank of FIG. 12A, having a hollow standpipe according to another embodiment;

FIG. 18B is a left side view of the fluid supply tank of FIG. 12A, having a hollow standpipe according to another embodiment;

FIG. 18C is a left side view of the fluid supply tank of FIG. 12A, having a hollow standpipe according to another embodiment;

FIG. 18D is a left side view of a the fluid supply tank of FIG. 12A, having a hollow standpipe according to another embodiment;

FIG. 19A is a right rear perspective view of another embodiment of a fluid supply tank having an external hollow standpipe;

FIG. 19B is a right side view of the fluid supply tank of FIG. 19A;

FIG. 20 is a right side view of another embodiment of a fluid supply tank having a fixed separator and an pivoting bottom;

FIG. 21 is a partial cross-sectional view of a head assembly having a passive roller according to another embodiment; and

FIG. 22 is a partial cross-sectional view of a head assembly having a brushroll cover with an arcuate inner extension;

FIG. 23 is a partial perspective view of a fluid application face according to another embodiment.

It is noted that the drawings are not necessarily to scale. The drawings are intended to depict only typical aspects of the subject matter disclosed herein, and therefore should not be considered as limiting the scope of the disclosure.

DETAILED DESCRIPTION

Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present invention.

Further, in the present disclosure, like-named components of the embodiments generally have similar features, and thus within a particular embodiment each feature of each like-named component is not necessarily fully elaborated upon. Additionally, to the extent that linear or circular dimensions are used in the description of the disclosed systems, devices, and methods, such dimensions are not intended to limit the types of shapes that can be used in conjunction with such systems, devices, and methods. A person skilled in the art will recognize that an equivalent to such linear and circular dimensions can easily be determined for any geometric shape.

A cleaning device is provided that includes fluid delivery and recovery systems that can be operated in combination with, or in place of, traditional vacuum modes in order to

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improve cleaning capabilities of the device. In certain exemplary embodiments, the cleaning device includes features to aid in fluid delivery in a more uniform manner to improve cleaning, as well as to aid in the recovery and disposal of fluid and debris (waste). For example, the cleaning device can include features located within the cleaning heading assembly that evenly distribute fluid onto a brushroll, thus allowing fluid to be evenly applied to a surface to be cleaned. The cleaning device can also include fluid recovery tank for collecting waste from the surface to be cleaned, and it can include features to aid in separating the debris from the fluid, as well as features to facilitate disposal of the waste. For example, the fluid recovery tank can include a pour spout to facilitate disposal of fluid, while retaining debris within the tank. The fluid recovery tank can also include a unique latch mechanism that aids in removal of the fluid recovery tank from the device without interfering with suction delivery to the tank and waste collection within the tank.

With reference now to FIGS. 1A-1D, an exemplary embodiment of a cleaning device 10 is shown. The illustrated cleaning device 10 generally includes a head assembly 100, a body assembly 200, a handle assembly 300, and a vacuum assembly 400 (not shown). The cleaning device is shown disposed atop a charging mat 700, which will be discussed below. As will be explained in detail below, the device 10 also includes fluid delivery and fluid recovery assemblies. In the illustrated embodiment, the handle assembly 300 includes a handle 310 and a stem 320, and the body assembly 200 includes a body housing 210 coupled to the stem 320. The head assembly 100 can be coupled to the body housing 210 opposite the stem 320. The head assembly 100 can include a head housing 110, and small wheels 112L, 112R (not shown) rotatably and large wheels 114L, 114R rotatably coupled to the head housing 110 and configured to allow the cleaning device 10 to roll along a surface, and a brushroll, not shown in FIGS. 1A-1D, disposed in the head assembly 100 and configured to rotate during operation of the cleaning device 10.

The vacuum assembly 400 (not shown), is disposed within the head and body assemblies 100, 200 and is capable of taking in fluid, dirt, debris, and other waste through suction and storing it within the cleaning device 10. In certain embodiments, the vacuum assembly 400 can include a motor and a motor fan, as will be discussed in more detail below. The motor and motor fan can be entirely contained in a motor housing disposed within the body assembly 200. As will be discussed in more detail below, hosing (not shown) can be coupled to the motor fan and it can be disposed to run through the body assembly 200 to the head assembly 100 to allow the motor to generate a suction force to draw waste into the device 10. Waste taken in by the vacuum assembly 400 through the hosing (not shown) can be deposited into a recovery tank removably disposed within the body assembly, as will also be discussed in more detail below.

The cleaning device 10 can further include a fluid supply tank 610 capable of supplying fluid to an area to be cleaned in order to aid in a cleaning process. The fluid can be mixed with dirt and debris, and the waste can be drawn back into the cleaning device 10 with suction generated by the motor and deposited in the recovery tank 420.

FIGS. 2A-2H depict the head assembly 100 in more detail. More specifically, FIGS. 2A-2E depict the head assembly 100 including an articulator 250, to be described in more detail below, and FIGS. 2F-2H depict the head assembly 100 without the articulator 250. As shown, the head assembly 100 includes a head housing 110, which is substantially T-shaped when viewed from the bottom as

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shown in FIG. 2D. The head assembly 100 can include wheels disposed on the head housing 110, which aid in movement of the cleaning device 10. In the illustrated embodiment, a set of left and right small wheels 112L, 112R is disposed on the bottom side 110a of the head housing 110 beneath left and right portions respectively, and a set of left and right large wheels 114L, 114R is disposed on left and right rear sides of the head housing 110, respectively. Both the small wheels 112L, 112R and the large wheels 114L, 114R rotate to allow the cleaning device 10 to be maneuvered on surfaces.

FIG. 2E shows a cross-section of the head assembly 100 with hosing 230 extending from the front of the head assembly and up through the articulator. The hosing 230 will be discussed in more detail below, with respect to the vacuum assembly and the dry and wet cleaning operations.

FIGS. 2G and 2H show the head assembly 100 with a brushroll cover 140 and a brushroll 150 removed, thereby providing a more-detailed view of the inner form of the head housing 110. On an upper portion of the front side 110d of the head housing 110 is a cover support 116, which can removably receive a brushroll cover 140. Rearward of the cover support 116, located on the top side 110b of the head housing 110, is a cover button 118, which can be actuated to release the brush cover 140 mounted to the head housing 110 to enable removal of the brush cover 140. Extending from the front side 110e of the head housing 110, especially as seen in FIGS. 2G and 2H, are the brushroll supports 120L, 120R, which are capable of holding a brushroll 150 for use during cleaning operations. The left and right brushroll supports 120L, 120R extend from the front side 120e of the head housing 110. The left and right supports 120L, 120R each have a rounded top edge 120La, 120Ra and a substantially flat bottom edge 120Lb, 120Rb. Brushroll support structures 122L, 122R are located on interior faces 120Lc, 120Rc of both the left and right supports 120L, 120R. The left support structure 122L, seen in FIG. 2G, includes a rounded cutout portion 122La capable of receiving a complementary structure on the brushroll 150, and the right support structure 120R includes a rotatably coupled extension 122Ra having a geometric interface capable of meshing with a corresponding geometric indentation of the brushroll 150.

The head assembly can further include light sources 119 disposed on the front of the left and right support structures 120L, 120R, which can act to illuminate a surface to be cleaned in order to aid in a cleaning process. The light sources 119 can be any kind of light source known, including a light-emitting diode, and the like.

Also depicted in FIGS. 2G and 2H, a fluid application face 624 is located on the front side 110e of the head housing 110. The fluid application face 624 has a generally hemi-cylindrical, concave shape so that, when a brushroll 150 is installed between the left and right supports 120L, 120R, the fluid application face 624 at least partially surrounds the outer face of the brushroll 150. The fluid application face 624 occupies a top half of the front side 110e of the head housing 110. One or more spray nozzles 630 and deflectors 640 can be disposed on the fluid application face 624. The spray nozzles 630 and deflectors 640 will be discussed in more detail below with respect to FIGS. 13A-14C.

Beneath the fluid application face 624 and occupying a lower half of the front side 110e of the head housing 110 is an intake face 124. The intake face includes a recessed façade leading to a central intake 126 therein. The left and right sides 124L, 124R of the intake face 124 can angle inward and lead to the central intake 126 itself. The central

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intake 126 forms a suction inlet that generally allows for dirt, debris, and waste to be taken into the cleaning device 10, as will be discussed in more detail below.

At the lower edge of the front side 110e of the head housing 110 below the intake face 124 is a flexible guide 128. The flexible guide 128 can be gradually sloped upward from the front side 110e toward the rear, and it can extend the entire width of the front side 110e between the left and right supports 120L, 120R. The flexible guide 128 is formed such that, in a neutral position, the guide 128 extends below the bottom side 110a of the head housing 110, which can be seen especially in FIGS. 2B and 2E. As a result, when the cleaning device 10 is placed upon a surface, the flexible guide 128 is biased against the surface, thereby allowing for waste to be fed toward the intake face 124 without leaving a gap for waste to avoid the cleaning device 10.

FIGS. 3A-3D shown an embodiment of a brushroll 150, which can be received between the left and right support structures 120L, 120R depicted in FIGS. 2A-H. In various cleaning operations, the brushroll 150 is configured to rotate to loosen waste deposited on a surface to be cleaned. The brushroll 150 is also able to direct waste into the cleaning device. While the configuration of the brushroll 150 can vary, in one embodiment the brushroll 150 can have a substantially elongate cylindrical shape and can include a central dowel 152 and a cleaning material 154 surrounding the dowel 152. The dowel 152 can be cylindrical, having right and left ends 152R, 152L, and it can be made of a rigid material, such as a hard plastic, metal, rubber, or a combination thereof, to provide the brushroll 150 with some support. The cleaning material 154 is attached to the outer surface of the dowel 152 along the entire length thereof and can be made of various cleaning materials, such as micro-fiber, bristles, or other materials known in the art, alone or in some combination. Further, the cleaning material 154 can be disposed in a variety of configurations on the outer surface of the dowel 152, and it can be formed from one or more materials which can be intermixed or separated into specific regions.

As mentioned previously, the brushroll 150 includes structures which can be retained by the left and right support structures 120L, 120R of the head assembly 110. These structures can be located on or extending from the right and left ends 152R, 152L of the dowel 152. FIGS. 3A and 3B show the right end 152R of the dowel 152 in more detail. The right end 152 of the dowel 152 includes an indentation 156 having a geometric pattern that corresponds to a geometric interface of the protrusion 122Ra on the right support structure 120R. The geometric design is capable of providing a friction fit between the protrusion 122Ra and the brushroll 150, such that during a cleaning operation when the protrusion 122Ra is rotatably driven, the brushroll 150 will rotate as well. FIGS. 3C and 3D show the left end 152L of the dowel 152 in more detail. The left end 152L of the dowel 152 includes an extension piece or tab 158 rotatably coupled thereon and configured to be received within the rounded cutout portion 122La of the left support structure 120L of the head assembly 110. When not in use, the tab 158 can be grasped by a user and the brushroll 150 can be removed from the left and right support structures 120L, 120R such as for inspection and maintenance or replacement.

FIGS. 4A-4D depict another embodiment of a brushroll. The brushroll 160, as shown in FIG. 4A, has a similar structure to brushroll 150 depicted in FIGS. 3A-3D, and so elements with comparable structure and operation will not be described in detail. Generally, the brushroll 160 is sub-

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stantially cylindrical in shape, and it includes a central dowel **162** surrounded by a cleaning material **164**. The brushroll **160** also includes a helical paddle **166** secured to the central dowel **162** and winding around the dowel **162** about an outer surface thereof. The paddle **166** extends radially beyond the cleaning material **164**, as can be seen in FIGS. **4A** and **4B**. The paddle **164** can be made of any suitable material, and may include, for example, rubber, plastic, or another similar material, and with this structure, in operation, the paddle **164** can flex and bend when impacting a cleaning surface in order to drive dirt, debris, and waste into the cleaning device **10**.

On an outer surface of the brushroll **160**, similar to the brushroll **150**, are structures which can interface with corresponding structures on a head assembly **100'** (not shown). For example, FIG. **4B** depicts a right side **162R** having a geometric interface which can operate similarly to the structure seen above in FIGS. **3A** and **3B**. On a left side **162L**, the brushroll **160** includes a rounded protrusion **168**, which can be received by a corresponding structure located on a brushroll support (not shown). FIG. **4D** depicts a partial-cross-sectional view of the protrusion **168**, which is biased to an extended position by a spring **170**. The end of the protrusion **168** is shaped to mate with a brush support and secure the brushroll **160** so that, during cleaning operations, the brushroll **160** can rotate to drive dirt, debris, and waste to the cleaning device **10**.

As mentioned above, the head assembly **100** further includes a brushroll cover **140** removably attached to the top side **110b** of the head housing **110**. FIGS. **5A-5D** show an embodiment of a brushroll cover **140**, which can be secured to the head assembly **110** to cover a brushroll **150**, **160** in order to prevent splash and spray from the brushroll **150**, **160** while it is in operation.

The brushroll cover **140** is shaped to extend the entire width of the head housing **110** between the left and right sides **110c**, **110d**, and it extends from the top side **110b** of the head housing **110**, over the rounded top edges **120La**, **120Ra** of the left and right support structures **120L**, **120R**, and ends just above the surface upon which the cleaning device **10** rests. This arrangement of the brushroll cover **140** can be seen in FIGS. **2A**, **2C**, and **2F**. The brushroll cover **140** extends to be nearly even with the bottom side **110a** of the head housing **110**, thereby creating a small gap to allow for the introduction of waste into the cleaning device **10** during a cleaning process. The brushroll cover **140** can be removably attached to the head housing **110** at the top side **110b** via the cover support **116**, as explained above. The brushroll cover **140** can also be attached to the head housing **110** via a hinge (not shown), such that the cover **140** can provide easy access to the brushroll **160**.

With reference now to **6A-8D**, various views and components of an embodiment of the body assembly **200** of the cleaning device **10** are shown.

The body assembly **200** can be operatively coupled to the head assembly **100** via an articulator **250**. The articulator **250**, as introduced with respect to FIGS. **2A-2D** and as shown again coupled to the body assembly **200** in FIGS. **6A-6D**, is coupled to the bottom of the body assembly **200** and can be at least partially disposed within the head assembly **100**. The illustrated articulator **250** is configured to articulate about two degrees of freedom. The articulator **250** has an outer housing **252** having a substantially elliptical cross-section that tapers in an upward direction to become larger, eventually coinciding with the size of the body assembly **200**. A first point of articulation **254**, allowing for articulation about a first degree of freedom, is mounted

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within the head assembly **100**. The first point of articulation **254** allows for the body assembly **200** to pivot between a forward direction and a backward direction, as indicated by the arrows A-A in FIGS. **6A-6D**. A second point of articulation **256**, located above the first point of articulation **254**, allows for the body assembly **200** to pivot between a left direction and a right direction, as indicated by the arrows B-B in FIG. **2A**. One or both points of articulation **254**, **256** can be articulated at a given time. Further, in other embodiments, the body assembly **200** can articulate in any number of degrees of freedom about any number of points of articulation.

The body assembly **200** includes a body housing **210** having a substantially cylindrical form with an elliptical cross-section. The body housing **210** includes a housing base **210a** coupled to the articulator **250**, a rounded front side **210b** and a rounded rear side **210c** extending upward from the housing base **210a**, and a top side **210d**. The top side **210d** of the body housing **210** in the illustrated embodiment is substantially flat and slopes downward at an angle from the rear side **210c** to the front side **210b**. The top side **210d** of the body housing **210** is coupled to the handle assembly **300**, which extends from the body assembly **200** in a direction opposite the head assembly **100**.

FIGS. **7A-7D** show the handle assembly in more detail, including the handle **310** coupled to the stem **320**. The illustrated handle **310** has a substantially trapezoidal handle frame **312** surrounding an interior handle aperture **314**. The illustrated handle frame **312** has a substantially flat bottom section **312a**, and a front section **312b** and a back section **312c** extending upward from the bottom section **312a** at substantially right angles relative to the bottom section **312a**. The front section **312b** is shorter than the back section **312c**, and the top of each of the front section **312b** and the back section **312c** are connected by a top section **312d**. The top section **312d** is angled downward toward the front section **312b** by virtue of the height discrepancy between the front section **312b** and the back section **312c**. The handle **312** further includes a power button **330** disposed on an upper exterior of the front section **312b**, and an area rug button **340** disposed on a front exterior of the top section **312d**. The functions of these buttons will be described in more detail below.

The illustrated handle assembly **300** further includes a stem **320** disposed between an underside of the bottom section **312a** of the handle **312** and the top side **210d** of the body housing **210**. The stem **320** is substantially linear and has a nearly flat rear face **322** and a rounded front face **324**, such that the stem **320** has a substantially semicircular cross-section. A person skilled in the art will appreciate that the handle assembly can have a variety of other configurations.

Referring again to the body assembly **200** the body housing **210** includes first and second cavities **210e**, **210f** for receiving components of the cleaning device **10**. The first and second cavities **210e**, **210f** are sized to receive a recovery tank **420** and a fluid supply tank **610** respectively, as shown in FIGS. **6A-6D**, such that when retained in their respective cavities, the recovery tank **420** and fluid supply tank **610** are shaped to conform to the overall cylindrical shape of the body assembly **200**.

FIGS. **8A-8D** show the body assembly **200** with the recovery tank **420** and fluid supply tank **610** removed from the first and second cavities **210e**, **210f**, respectively. The first cavity **210e**, located in the lower front side **210b** of the body housing **210**, is sized to removably receive the recovery tank **420** such that, when retained in the first cavity **210e**,

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the recovery tank **420** occupies the entirety of a lower region of the front side **210b** of the body housing **210**. The first cavity **210e** can include a seal **214** disposed on an upper side thereof and configured to seal against an upper portion of the retained recovery tank **420**. Next to the seal **214** is a divider **216**. The divider **216** can be porous to allow air to flow through the system during dry and wet cleaning operations, as will be described below in more detail. The recovery tank **420** is removable from the body housing **210** after actuation of a latch assembly **460** (not shown) extending outward from an upper extent of the recovery tank **420**, which releases the recovery tank **420** from engagement with a retaining slot **218**, located toward the front of the first cavity **210e**. The second cavity **210f**, located in an upper front portion **210b** of the body housing, and occupying a substantial portion of the top side **210d**, receives the fluid supply tank **610** for use in wet cleaning processes. A fluid tank switch **212** is disposed in the top side **210d** of the body housing **210** between the second cavity **210f**. When the fluid tank switch **212** is actuated, a tank engagement feature **211** recedes into the body housing **210**, and the fluid supply tank **610** can be removed from the second cavity **210f**. The recovery tank **420** and the fluid supply tank **610** will be described in greater detail below with respect to the cleaning processes that the cleaning device **10** may perform.

As previously indicated, the cleaning device **10** can operate in both wet and dry cleaning modes. Dry cleaning modes generally include modes related to traditional vacuuming operations, such as vacuuming on hard surfaces or on softer surfaces, such as carpet. Dry cleaning modes rely on suction to take dirt and debris into the cleaning device for convenient disposal. In some dry cleaning modes, a brushroll can rotate to agitate debris and waste on a cleaning surface. The brushroll can loosen the dirt and debris while simultaneously directing it toward a suction intake of a cleaning device. In other dry cleaning modes, a brushroll does not rotate, and instead, suction is relied on alone to force dirt and debris into a cleaning device. Wet cleaning modes can generally include a cleaning device supplying fluid either directly or indirectly to a surface to aid in cleaning. The supplied fluid can act to loosen dirt and debris stuck to the surface, and the dirtied fluid can be taken into the cleaning device through suction or other means. In some wet cleaning modes, like some dry cleaning modes described above, a brushroll can further assist in loosening dirt and debris off the surface and directing it toward a suction intake. In these wet cleaning modes, the fluid can be supplied directly to the brushroll in order to simultaneously apply the fluid to the surface while agitating the dirt and debris found on the surface. In other wet cleaning modes, fluid can be supplied directly to the surface and the brushroll can agitate the wetted surface. In still other modes, fluid can be supplied directly to the surface and a brushroll can remain stationary, thereby cleaning the surface with fluid and suction alone.

The wet and dry cleaning modes can rely on a vacuum assembly **400**. In an exemplary embodiment, the vacuum assembly **400** includes a motor assembly **410**, a recovery tank **420**, and hosing **230** coupled to an intake, such as a central intake **126**, which together can be operated to draw waste into the cleaning device **10**.

FIGS. 9A-9E depict a motor assembly **410** according to the exemplary embodiment. The motor assembly **410** is configured to be disposed within the body housing **210** beneath the fluid supply tank **610** and the handle stem **320** and above the recovery tank **420**. The illustrated motor assembly **410** includes a motor **412** and a fan **414** encased

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in a motor housing **416**. The motor housing **416** is divided into a lower motor housing portion **416a** and an upper motor housing portion **416b** coupled to or integrally formed with the lower motor housing **416a**, and the motor **412** and motor fan **414** are contained therebetween. At an upper extent of the upper motor housing portion **416b** are left and right air vents **418L**, **418R**, which allow for air drawn into the cleaning device **10** to exit out the rear side **210c** of the body housing **210**. When coupled together, the lower motor housing **416a** and the upper motor housing **416b** substantially surround and isolate the motor **412** and fan **414** from the rest of the cleaning device **10**. The motor assembly **410** sits atop the divider **216** disposed within the body assembly **200**. The divider **216** can be seen especially in FIG. 8D, and it includes a plurality of apertures, which allow for air to flow through the divider **216** to facilitate the various cleaning operations which rely on suction. Moreover, the divider **216** forms an upper extent of the first cavity **210e**, which receives the fluid recovery tank **420**.

With reference now to FIGS. 10A-10L, an exemplary embodiment of the recovery tank **420** is shown. The recovery tank **420** can be removably retained within the body housing **210** in the first cavity **210e**, as explained above. The illustrated recovery tank **420** generally includes a container **422**, a separator **440**, a lid **460**, and a latch assembly **470**.

FIGS. 10A-10D and 10L depict the container **422** in relation to the remainder of the recovery tank **420**. The container **422** has a bottom surface **422a** and a sidewall **422b** extending upward from the bottom surface **422a**. As mentioned previously, the container sidewall **422b** can have a rounded front face **422c** to conform with the overall substantially cylindrical shape of the body housing **210**. A rear face **422d** of the sidewall **422b** can be substantially flat. At an upper extent, the container can have a top end **422e** that is open and is able to receive the separator **440** therein. The top end **422e** can be sloped from the front and extending downward toward the rear face **422d** of the sidewall **422b**. The container **422** can also include an inlet on the bottom surface in the form of a hollow standpipe **424** that extends nearly the entirety of the container **422** height. The upper extent of the hollow standpipe is open to allow for fluid to pass into the container **422**. The hollow standpipe **424** can be disposed rearward of a center of the bottom surface **422a**, closer to the rear face **422d** of the sidewall **422b**.

FIGS. 10E-10H show the separator **440** separated from the rest of the recovery tank **420**. The separator **440** can be received within the container top end **422e** and can extend downward into the container **422** from the top end **422e**, such that a lower end **440a** of the separator **440** extends downward beyond an upper end of the standpipe **424** to a distance above the bottom surface **422a** of the container **422**. The lower end **440a** of the separator **440** can be shaped to allow the standpipe **424** to extend through an opening **440b**, while surrounding the standpipe **424**. The lower end **440a** of the separator **440** can also be sloped, similar to the top end **422e** of the container **422**, however the lower end **440a** of the separator **440** can slope downward from the rear face **422d** of the container sidewall **422b** to the front face **422c**, bottoming out to a drain **442** some distance from the front face **422c**. A secondary slope **444** can extend from the front face **422c** of the sidewall **422b** to the drain **442**. The drain **442** itself, shown in a bottom view in FIG. 10G, can be in the form of a slot in the lower end **440a** of the separator **440** that extends substantially the entire width thereof. On either side of the drain **442** are a plurality of ridges **446** defining channels **447** therebetween, which can act to catch and hold large debris, yet still allowing for fluids to pass through the

drain 442. FIG. 10H is a cross-sectional view of the separator 440, and the plane of the cross section is located within the drain 442 to provide a view of the plurality of ridges 446. In this view, it can be seen that the ridges 446 form a wave pattern, so larger debris would be unable to fully block a pathway to the drain as fluid and smaller particles remain able to pass into the channels 447. A person skilled in the art will appreciate that the drain and the ridges can have a variety of other configurations, and the separator can include any number of drain holes therein.

The illustrated separator 440 also includes first and second deflectors 448a, 448b which extend downward in the container 422. The first deflector 448a extends downward to partially cover the upper portion of the standpipe 424. The first deflector 448a is curved and shaped like a quarter-pipe to extend frontward, over the standpipe 424 and past an upper extent of the standpipe 424. The second deflector 448b extends at a downward angle from a front side of the standpipe 424 and out over the drain 442. The first and second deflectors 448a, 448b are configured to mitigate the effects of splash back and to prevent fluids and debris from getting near the top end 422e of the container 422. In other embodiments, the deflectors 448a, 448b may take on other forms, and they may extend in different angles, shapes, or regions of the container 422 as needed.

On an underside of the separator 440 are first and second fluid level detectors 449a, 449b which extend downward from the separator 440. The fluid level detectors 449a, 449b are configured to sense when fluid has reached a predetermined threshold, and upon the fluid level reaching the predetermined threshold, to send a signal to the cleaning device 10 to display an alert. In an exemplary embodiment, the first and second level detectors 449a, 449b have exposed electrical contacts which, when submerged in fluid, complete a circuit and send a signal to the cleaning device 10 to display an alert message signifying that the recovery tank 420 can be emptied. In other embodiments, other level detector arrangements can be used, such as, for example, a float, a displacer, or others.

The separator 440 near the top end 422e of the container 422 can also include two openings, namely a spout 450 and a lid opening 452. The illustrated spout 450 is disposed in a rear-most region of the separator 440 and has a curved lip 454 in a rear region thereof to allow for the controlled disposal of fluid captured by the recovery tank 420. The spout 450 is aligned with a channel 451, shown in FIG. 10D, formed in the back side of the separator such that a fluid flow pathway is defined between the back of the separator 440 and the container 422 for allowing fluid to flow there-through. When the recovery tank 420 is retained within the body housing 210, the spout 450 is pressed against the seal 214 disposed on the body housing 210 next to the divider 216 to prevent the premature expulsion of fluid from recovery tank 420. When the recovery tank 420 is removed from the body housing 210, the spout 450 is open and allows a user to invert the container 422 to pour fluid out of the spout. The lid opening 452 is positioned next to the spout 450, closer the front sidewall 422b of the container 422. The lid opening 452 is sized to securely receive a lid 460, which can be removable from the lid opening 452 when the recovery tank 420 is not retained within the body housing 210.

The lid 460, shown most clearly in FIGS. 10I and 10J, can be shaped to fit within the lid opening 452. In the illustrated embodiment, the lid 460 is shaped approximately like a half ellipse and is skewed to align with the slope of the lid opening 452. The lid 460 includes a frame 462 having a top support 462a, an open bottom 462b, an inner surface 462c,

and an outer surface 462d. The outer surface 462d of the body frame 462 includes multiple grooves 464 that align with ridges 452a of the lid opening 452 and prevent the lid 460 from being over-inserted into the lid opening 452. A mesh structure 466 can be hinged over the open bottom 462b of the frame 462. The mesh structure 466 can have a first porosity which can act to prevent large particles from passing through the lid 460, while still allow air to pass through the lid 460. The mesh 466 can be made of plastic, but in other embodiments, the mesh 466 can be made from various materials, including metals, rubbers, or other materials known in the art. Contained within the frame 462 is a filter material 467 sized to fill the entirety of the frame 462. The filter material 467, in the exemplary embodiment, is made from a foam-like material, which has a second porosity that is smaller than the first porosity of the mesh 466. In other embodiments, the filter material 467 can be made other materials which can act as filters, such as various pulps, plastics, sponges, or other materials known in the art. The filter material 467 can additionally have different porosities, which may be smaller, greater, or equal to the porosity of the mesh 466, and this porosity can be varied depending upon the type of matter to be filtered. In the illustrated embodiment, the smaller porosity of the filter material 467 prevents the escape of other particles from the recovery tank 420 which may not be stopped by the mesh 466. Covering the top of the frame 462 is a top support 462a, which, in the exemplary embodiment, is a scaffold that prevents the filter material 467 from being misaligned in the frame 462. A central portion of the top support 462a includes a handle 463 to assist in removal of the lid 460 from the lid opening 452. Surrounding a perimeter of the top support 462a and extending beyond the bounds of the frame is a gasket 468 which helps to seal the lid 460 into the lid opening 452. In this way, fluid cannot pass around the lid 460 when it is seated in the lid opening 462, and instead it must pass through the mesh 466 and the filter material 467. When the recovery tank 420 is retained within the body housing 210, such as depicted in FIGS. 6A-6D, the lid 460 is pressed against the divider 216, such that the container 422 and the motor assembly 410 are in fluid communication with each other.

A latch assembly 470 extends from the front sidewall 422b of the recovery tank 420 and forms a part of the container top 422e. The latch assembly 470 functions to secure the recovery tank 420 within body housing 210, and it can be actuated to allow for removal of the recovery tank 420 from the first area 410e of the housing body 410. The illustrated latch assembly 470 includes a latch seat 472, acting as a kind of housing, which extends from the container sidewall 422b in the form of an arc-shaped protrusion. In this way, the latch is disposed entirely outside of the container 422. The latch seat 472 is hollow and defines a depression 472a, seen in FIG. 10D, which receives a latch 474 and a spring 475. The spring 475 is positioned centrally in the latch seat 472, and the latch 474 is seated in the latch seat 472 over the spring 475 such that the latch 474 is biased to a raised position, being lifted by the spring force of the spring 475.

The latch 474 itself, shown in FIGS. 10A-10D and 10K-10L, and especially FIG. 10K, has an upper arc-shaped protrusion 476 which extends out of the latch seat 472 and above an upper extent of the container 422 and separator 440. Inward from the arc-shaped protrusion 476 and also extending from the latch 474 out of the latch seat 472 is an engagement feature 478, which can be received in the complimentary slot 218 on the body housing 210 in order to enable retention of the recovery tank 420. The shape of the

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both the latch 474 and the latch seat 474 are curved to align with a contour of the container 422, and, in turn, the cleaning device 10. Actuation of the latch 474 counters the spring force and drives the latch 474 into the latch seat 472 in a downward direction. When the latch 474 is no longer actuated, the spring force returns the latch 474 to an upward position to extend above the latch seat 472 and above the container 422, thereby also extending the engagement feature 478 into the complimentary slot 218. Therefore, the latch is spring-biased to the locked position. In this way, the latch 474 moves vertically upward and downward, without moving laterally within the latch seat 472, all the while remaining parallel, or substantially parallel, to the top of the container 422. Further, in either the upward or downward position, the latch 474 extends above the top of the container 422. Although the exemplary embodiment depicts a latch 474 as described above, other embodiments may use alternative structures to secure the recovery tank 420 in the body housing 210, for example, a sliding mechanism, a clip, a knob, or another device known in the art.

When the recovery tank 420 is mounted within the body housing 210, hosing 230 extending between the body assembly and the head assembly can allow fluid and debris to be delivered to the recovery tank 420. In particular, the hosing 230 can be located in a lower portion of the body housing, centrally disposed beneath the recovery tank 420, and it can be configured to fluidly coupled to the outside of the standpipe 424 and to provide a fluid communication path between the container 422 and the central intake 126 in the head assembly 100. The hosing 230 is flexible so as to not inhibit full articulation of the head assembly and body assembly at the articulator. When the recovery tank 420 is retained in the body housing 210, an upper end of the hosing 230 contacts the lower end of the container 422 and creates a substantial seal around the inlet. Accordingly, the central intake 126 is in fluid communication with the recovery tank 420 when the recovery tank 420 is retained in the body housing 210.

When the recovery tank 420 is retained within the body housing 210, the lid aligns with the divider 216 and is therefore in fluid communication with the motor and the suction path by virtue of the apertures contained within the divider.

When operated in the dry cleaning modes, the sub-assemblies of the vacuum assembly 400 work together to enable debris to be drawn into the cleaning device 10 for disposal. In a dry cleaning mode, the motor assembly 410, via the motor fan 414, spins to draw in air through the central intake 126 located in the head assembly 100. Air flows into the central intake 126 in the head assembly 100, up through the hosing 230, introduced above with respect to FIG. 2E, and into the container 422 of the recovery tank 420. When the cleaning device 10 is passed near waste and debris, the suction generated by the motor assembly 410 will draw the waste and debris through the vacuum assembly where it will enter the container 422 in the recovery tank 420. Afterward, the airflow leaves the recovery tank 420 through the lid 460 passing through the mesh 466 and the filter material 467, where it enters the motor assembly 410. However, the mesh 466 and filter material 467 of the lid 460 does not allow the waste and debris to pass through, so instead it will become trapped in the container 422 until disposal. Finally, the airflow is vented out of the rear exhausts 418L, 418R of the motor assembly 410 through the rear side 210c of the body housing 210.

A person skilled in the art will appreciate that the recovery tank can have variety of other configurations. FIGS. 11A-

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11H and 17A-20 depicted embodiments of recovery tanks having various configurations, features, and arrangements. Features similar to those described above for the embodiment of FIG. 10A-10L will not be described again.

FIGS. 11A-11C show a recovery tank 520 that generally includes a container 522 and a separator 540. The recovery tank 520 can be retained within a body assembly 210' of a cleaning device 10 so that it can be used during cleaning processes, as explained above. In this embodiment, the separator 540 is sized to be seated around a standpipe 524 while occupying an entire width of the container 522. The standpipe 524 is received within a central shaft 540b built in to the separator 540 itself, which leads to an upper deflector 548 in the form of a quarter-pipe structure. Located on opposing sides of the central shaft are fluid detecting electrodes 549a, 549b, similar to those described previously. The separator 540 includes a flat bottom 540a and a porous rear side 540c that allows for fluid and smaller particles to pass through, while retaining larger particles.

In other aspects, the recovery tank 520 may lack a built-in lid or filter system, and instead those components can be retained directly within the body assembly 210' of the cleaning device 10. When the recovery system 520 is retained in the body assembly 210', the components can interact in order to enable the capture of dirt, debris, and waste, while allowing for air to flow freely through the system and facilitate suction.

During a cleaning process, when a fluid level within the recovery tank 510 rises to a predetermined threshold to contact the electrodes 549a, 549b, the cleaning device 10 can measure a drop in resistance across the electrodes 549a, 549b and alert a user to the detected fluid level. The cleaning device 10 can also interrupt a cleaning process and prevent further cleaning until the electrodes 549a, 549b no longer detect a fluid level exceeding the predetermined threshold. Similarly to the embodiment described above with respect to FIGS. 10A-10L, when disposing of captured dirt, debris, and waste, the separated liquid slurry can be emptied through a built-in pour spout 550 located at the top of the recovery tank 520. Then, the separator 540 can be removed, and with it, captured particles too big to pass through the separator 540 can be easily disposed of.

FIG. 11D shows the recovery tank 520 retained within the body assembly 210'. A handle 570 extends outward from the container 522 for allowing the recovery tank 520 to be pulled and removed from its retained position. When retained, an upper end of the separator 540 of the recovery tank 520 interfaces with a retention feature located within the body assembly 210', as shown in FIG. 11E. At a lower end of the separator 540, the body assembly 210' interacts with the container 522 in order to create a fluid pathway into the container 522 through the standpipe 524, as shown in FIG. 11F.

FIGS. 11G-11I show a retention and removal process for a filter 560. The illustrated filter 560 is retained in a slot 552 so as to be disposed above a retained recovery tank 520. The filter 560 interfaces with a filter retention feature 562, depicted in FIG. 11H, in order to secure it in place. The filter 560 further includes a front rim 564 to aid in removal, which, when the recovery tank 520 is retained in the body assembly 210', is blocked by an extension 521 located on the recovery tank 520, as shown in FIG. 11D. When the recovery tank 520 is not retained, this slot 552 is accessible by a user and can be used to remove the filter 560, as shown in FIG. 11I.

FIGS. 17A-17D depict another embodiment of a recovery tank 810 having the same configuration as recovery tank

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520, but that includes a peripheral seal **814** extending around an outer perimeter region thereof of the separator **812**. The peripheral seal **814** can be made from a variety of materials, such as rubber, plastic, elastic, or other materials. When placed within a container **816** having a lid **815**, the peripheral seal **814** can close off any gap left between the separator **812** and the sidewalls of the container **816**, thereby preventing the passage of fluid and/or debris around the separator **812**. In other words, a water-tight seal is formed. In operation, fluid and debris drawn into the container **816** must pass through the separator **812** via the drain **818**, rather than avoiding the separation process. Similar to embodiments described above, the separator **812** also includes a deflector **819** in the form of a wedge-shaped protrusion. While the deflector **819** can have any form or orientation, the deflector **819** is shown angled downward toward the area of the drain **818**. As fluid and debris are drawn into the recovery tank **810**, the deflector **819** can assist in preventing this fluid and debris from impacting unwanted areas of the recovery tank **810**, e.g., the lid **815**.

FIGS. 18A-18D illustrate other embodiments of recovery tanks **820a-d**, each having a configuration similar to recovery tank **420**, but that have a hollow standpipe positioned at various locations. When coupled to a cleaning device (e.g., cleaning device **10**) for various cleaning operations, such as those described herein, the recovery tanks **820a-d** can be received by a complimentary structure located on the cleaning device (not shown). In this configuration, fluid and debris can still enter the recovery tank via the hollow standpipe, and the overall operation of the recovery assemblies **820a-d** can occur in a substantially similar manner as described previously with respect to various embodiments detailed above.

FIG. 18A depicts a recovery tank **820a** having a slanted lower sidewall **824a**. The hollow standpipe **822a** extends upward from the slanted sidewall **824a**. FIG. 18B depicts a recovery tank **820b** having a rear carve-out portion in a rear sidewall **824b** thereof. The hollow standpipe **822b** extends laterally from the rear sidewall **824b** and protrudes into the container **826b** in an upward manner, curving to compensate for the initial lateral extension. FIG. 18C depicts a recovery tank **820c** having an irregularly shaped hollow standpipe **822c**. The hollow standpipe **822c** is shown extending into the container **828c** from a bottom surface **824c** thereof, before curving rearward at an approximate right angle. The hollow standpipe **822c** extends along a rear sidewall **826c** of the container **828c** and then is depicted curving frontward before extending upward in a central portion of the container **828c**. FIG. 18D depicts the recovery tank **820d** having a similar configuration as recovery tank **820a**, depicted in FIG. 18A, except that instead of a slanted lower sidewall, recovery tank **820d** has a curved lower sidewall **824d** from which the hollow standpipe **822d** extends.

FIGS. 19A-19B depict another embodiment of a recovery tank **830** having an external hollow standpipe **832**. The hollow standpipe **832** is shown extending upward at a rear portion of the recovery tank **830** and then curving laterally such that it extends adjacent a rearward side of a lid **834**. The hollow standpipe **832**, like those described previously, can be in fluid communication with the interior of recovery tank **830**, such that fluid and debris can be drawn therethrough to be deposited within the container **836**. Although not shown in FIGS. 19A-19B, the recovery tank **830** can include a separator (e.g., separator **812**), and the hollow standpipe **832** can be configured to deposit drawn-in fluid and debris onto the separator, to allow the drawn-in debris to be separated from the fluid.

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FIG. 20 depicts another embodiment of a recovery tank **840**. In this embodiment, the separator **842** is fixedly coupled to (e.g., unitary or integral with) the container **844** such that it cannot be removed from the container. As with prior embodiments, a hollow standpipe **846** can extend through the separator **842**. However, the hollow standpipe **846** can be fixed to or integral with the separator **842**. In this manner, the container **844**, the separator **842**, and the hollow standpipe **846** are integral with one another. The recovery tank **840** can also include a spout **845** disposed in the rear of the container **844**, which can allow for the disposal of captured fluid and small debris from the container **844**. A lid **848** can be removably coupled to the container **844** and can include a lower extension **848a** extending downward over the hollow standpipe **846**. The extension **848a** can redirect fluid and debris drawn into the container **844** toward the hollow standpipe **846**, thus preventing fluid and debris from flowing through the lid **848**. The container **844** can also include a removable bottom **847** that can allow for access to the region of the container **844** beneath the separator **842**. The removable bottom **847** can be attached to the container **844** in any number of ways, including a friction fit, a hinge, etc. The removable bottom **847** can include an intake port **847a** with an upward extending collar **847b**. The collar **847b** can receive the hollow standpipe **846** therein and allow for the introduction of fluid and debris into the container therethrough.

In addition to dry cleaning modes, the cleaning device is also operable in wet cleaning modes which employ the use of a fluid supply tank, a fluid pump, spray nozzles, and deflectors, in addition to the vacuum assembly with sub-assemblies described previously. FIGS. 12A-15D illustrate components of a fluid assembly for use in wet cleaning modes and operations, such as a fluid supply tank **610**, tubing **620**, a fluid pump **622**, a fluid application face **624**, and spray nozzles **630**.

As explained above, the body housing **210** includes a second cavity on the upper front face **210b** that is shaped to receive and retain a fluid supply tank **610**. In the exemplary embodiment, as shown in FIGS. 12A-12F, the fluid supply tank **610** includes a valve cap **612** removably threaded to a fluid tank **614**. The fluid tank **614** is divided into an upper tier **614a** and a lower tier **614b**, and each of the tiers **614a**, **614b** has a substantially hemi-cylindrical shape. The upper tier **614a** is shaped to conform with the overall form of the body housing **210**, providing an outer limit for the upper front face **210b** of the body housing **210**. The lower tier **614b** is smaller than the upper tier **614a** and is received internally within the body housing **210**, occupying part of the second area. The fluid tank **614** further defines a hollow interior, which receives fluid to be supplied by the cleaning device **10** during a wet cleaning operation. The valve cap **612** of the fluid supply tank **610** is threadably coupled to the lower tier **614a**, and it permits one-way flow of fluid therethrough—from the hollow interior of the fluid tank **614** to externally thereof. The valve cap **612** is sized to be received in the second cavity **210f** of the body housing **210** in a complementary recess. When the valve cap **612** is properly seated in the second cavity, fluid is able to flow therethrough, and when the valve cap **612** is not properly seated, the valve cap **612** can act to seal the fluid within the fluid supply tank **610**.

As shown in FIGS. 12E and 12F, the lower tier **614b** can further include a bleeder valve **616** and a retention depression **618**. As the fluid tank **614** empties of fluid, the bleeder valve **616** can allow for an equalization of pressure in the hollow interior to facilitate a constant supply of fluid to the cleaning device **10**, without creating a vacuum within the

hollow interior. The retention depression **618**, in the exemplary embodiment, is a depression disposed in the lower tier **614b** which is shaped to receive the tank engagement feature **211** on the body housing **210**, introduced above and depicted in FIGS. **8A** and **8B**. As explained above, actuation of the fluid tank switch **212** will allow for the fluid supply tank **610** to be removed from the second area, and specifically, in the exemplary embodiment, actuation of the fluid tank switch **212** retracts the tank engagement feature **211** into the body housing **210** so that it no longer engages the retention depression **618**.

FIGS. **13A-13I** and **23** depict various components used in wet cleaning modes, including tubing **620**, a fluid pump **622**, and fluid application face **624**, and spray nozzles **630**. During a wet cleaning mode, fluid contained within the fluid supply tank **610** is expelled from the cleaning device **10** and onto a surface to be cleaned. When fluid leaves the fluid supply tank **610**, it is transported through the cleaning device **10** in tubing **620**. The tubing **620** connects to the fluid supply tank **610**, travels down the body assembly **200**, and then into the head assembly **100**.

FIG. **13A** depicts an internal view of the head assembly **100** without the top side **110b** of the head housing **110**. FIG. **13B** depicts only the tubing **620**, the fluid pump **622**, and the spray nozzles **630**, which are all contained within the head assembly **100**. The fluid pump **630** is configured to pump fluid from the fluid supply tank **610** through the cleaning device **10**. The tubing **630** connects the fluid supply tank **610** to the pump **622** and then leaves the pump **622** before splitting and finally connecting to the left and right spray nozzles **630L**, **630R** disposed on the fluid application face **624** of the head assembly **100**, as discussed above. Therefore, in this configuration, the fluid supply tank **610** is in fluid communication with the left and right spray nozzles **630L**, **630R** by way of the fluid pump **622** and tubing **620**. Operation of the cleaning device **10** during wet cleaning modes will be described in greater detail below.

As previously explained and as shown in FIG. **13C**, the head assembly **100** includes a fluid application face **624** disposed on a front side **110e** thereof. The fluid application face **624** is mounted to an upper portion of the front side **110e** of the head housing **110**, and it spans a substantial width of the head housing between the left and right spray nozzles **630L**, **630R**. The face **624** is substantially cylindrical and concave in shape, protruding slightly toward a middle of the face **624**. When viewed from a side perspective, such as seen in a partial cross-sectional view of FIG. **13I**, the face **624** is concave enough to allow for the left and right spray nozzles **630L**, **630R** to be fully contained within the resulting depression.

Also contained on the fluid application face **624** are left and right deflectors **640L**, **640R**. Each deflector can be in the form of a projection extending generally perpendicular to the application face **624**. Both of the left and right deflectors **640L**, **640R** are shown as having a curved configuration, curving away from the spray nozzles **630L**, **630R**, however the deflectors can be planar or they can have alternative shapes and orientations in other embodiments. In the illustrated embodiment, the left and right deflectors **640L**, **640R** are molded directly onto the fluid application face **624**. In other embodiments, these deflectors could be removably attached to the fluid application face **624**, or may be attached—either molded or removably so—to another feature. The left and right deflectors **640L**, **640R** are at least partially disposed in the spray paths of the left and right spray nozzles **630L**, **630R** respectively so that, during a cleaning operation, the deflectors **640L**, **640R** are impacted

by sprayed fluid. In an exemplary embodiment, the deflectors **640L**, **640R** are positioned a distance from each spray nozzle **630L**, **630R** that is sufficient to allow fluid spray from the spray nozzles to encounter the deflectors and to be deflected back toward the end of the brushroll, thus aiding in delivering fluid to the ends of the brushroll.

The left and right spray nozzles **630L**, **630R** protrude from the fluid application face **624** and are generally aimed inward at each other, while also being aimed toward the brushroll **150**. In an exemplary embodiment, each of the left and right spray nozzles **630L**, **630R** is substantially similar, and as such, description will be made to only one spray nozzle **630** in a general manner. FIGS. **13D-13F** show the right side of the head assembly **100**, including the fluid application face **624**, right spray nozzle **630R**, and tubing **620**, while FIGS. **13G-13I** show the left side of the head assembly.

FIG. **23** depicts a portion of a fluid application face **624'** according to another embodiment. The fluid application face **624'** can function similarly to the fluid application face **640**, through the cooperation of the right spray nozzle **630R'** and the right deflector **640R'**, however the fluid application face **624'** can include a secondary right deflector **641R'** mounted thereon. The secondary right deflector **641R'** is depicted in FIG. **23** as being larger than the right deflector **640R'** and is also depicted as being positioned closer to the right spray nozzle **630R'** than the right deflector **640R'**. However, the specific form and location of the secondary right spray nozzle **640R'** could vary. In operation, the right spray nozzle **630R'** and the secondary right spray nozzle **641R'** can operate to redirect fluid emitted by the right spray nozzle **630R'**. The secondary right spray nozzle **641R'** can be configured to direct additional fluid to a periphery of a brushroll.

FIGS. **14A-14D** depict a spray nozzle **630** according to an embodiment. The spray nozzle **630** has a generally cylindrical form and includes a discharge port **632** in fluid communication with the fluid supply tank **610** and through which fluid is emitted. The discharge port **632** itself is wedge-shaped with a central hole **634**, and this arrangement creates a planar, fan-shaped spray pattern when fluid is emitted from the discharge port **632**. This planar, fan-shaped pattern can, in some embodiments, be between 5°-60°. In other embodiment, the fan-shaped pattern can be between 10°-50°. In still other embodiments, the fan-shaped pattern can be between 15°-45°.

In some embodiments, the discharge port **632** can be rotated such that a spray angle θ is offset from a vertical axis by a few degrees. This offset can be anywhere from 1°-30° in either the clockwise or counter-clockwise orientation. In some embodiments, the offset is between 5°-25°. In still other embodiments, the offset is between 10°-20°. In the exemplary embodiment, the discharge port of the left spray nozzle is shown rotated approximately 15° counterclockwise from a vertical orientation. The spray nozzle **630** can also be aligned with a plane tangential to the surface of the brushroll **150** at a point on the brushroll **150** closest to the spray nozzle **630**, such that the angle of offset and the tangential plane would be substantially parallel, in order to maximize fluid coverage onto the brushroll **150**. The angle of offset ϕ could also vary by several degrees in either direction, and in some embodiments may vary by as many as 1°, 2°, 3°, 4°, or 5° from the tangential plane. An exemplary view of the spray nozzle **630** with a spray angle skewed from the tangential plane is shown in FIG. **14D**.

FIGS. **15A-15D** depict a fluid application face **650** with built-in spray nozzles **652** according to another embodiment.

The fluid application face **650** is shown in FIG. **15A**, and it includes a single outlet **654** extending nearly the entire width of the face. At a rear side of the fluid application face **650** is a port **656** that is able to connect with tubing (not pictured) in order to fluidly couple the fluid application face **650** to a fluid path. Nested within the outlet **654** on the front side are spray nozzles **652** spaced evenly at the outlet **654**, and they are configured to output fluid along a path as shown in FIG. **15C**. In an exemplary embodiment, four spray nozzles **652** are dispersed evenly across the width of the outlet, however in other embodiments, the number of spray nozzles **652** and their positions can vary so that they are evenly or unevenly spaced. The spray nozzle **652** introduces fluid into the outlet, and the fluid flows in a path defined by the rounded contours of the outlet **654** before being emitted in a direction aimed toward a brushroll **150**. The supplied fluid volume can be large enough that fluid flows laterally within the outlet **654** and is therefore applied to a brushroll **150** from an entirety of the width of the outlet **654** in order to evenly coat the brushroll **150** with fluid, thereby preventing streaking during a cleaning operation.

FIG. **15D** depicts a partial cross-section of a head assembly having the fluid application face **650** therein. In the illustrated embodiment, a comb **658** is disposed proximate to the brushroll **150**. During a cleaning operation, as the brushroll **150** rotates, the brushroll **150** may capture and pick up fibrous material. As the brushroll **150** rotates, if allowed, the fibrous material can become entwined around the brushroll **150** and interfere with the cleaning capabilities of the cleaning system **10**. The presence of the comb **658** assists in capturing fibrous material to prevent entwinement with the brushroll **150**, which then allows an operator to remove the captured material. Also depicted in FIG. **15D** is another embodiment of a rubber guide **660**, similar to the embodiment described above. While not shown, the comb **658** can be disposed on the brushroll cover **140** of FIGS. **5A** and **5B**, or on any embodiment presented herein.

In another embodiment, the head assembly **100** can include at least one passive roller. FIG. **21** illustrates a first passive roller **161a** and a second passive roller **161b**, although one or the other may be used in place of both passive rollers **161a**, **161b**. The passive rollers **161a**, **161b** are substantially cylindrical and can be freely rotatable about central axes thereof. The first and/or second passive roller **161a**, **161b** can be configured to remove fluid from a surface and redirect it to a position where it can be drawn into the head assembly **100**. The first and/or second passive rollers **161a**, **161b** can also be configured to remove excess fluid from the brushroll **160**. The first passive roller **161a** is shown disposed near a floor surface, while the second passive roller **161b** is shown positioned near the central intake **126**. However, these locations can vary depending upon various arrangements of components as described herein.

In another embodiment, the brushroll cover can aid in removing excess fluid from the brushroll. FIG. **22** depicts the head assembly **100** including a brushroll cover **140'** having an arced inner wall extension **141**. The extension **141** can extend toward and into contact with the brushroll **160** to assist in the removal of excess fluid and debris trapped by the brushroll **160** itself during a cleaning operation. While the extension **141** is shown in FIG. **22** having leading and trailing edges **143a**, **143b**, the extension **141** can also gradually taper away from the brushroll **160**.

In operations employing the fluid mode, such as in the wet vacuum modes referenced above, fluid is applied to elements of the device, a surface to be cleaned, or a combination thereof, in order to aid in dirt and waste removal.

Wet cleaning modes can be activated through actuation of a wet cleaning switch **662**, which in one embodiment can be located on the top side of the body housing **210**. In order to prepare for cleaning in such modes, the fluid supply tank **610** is filled with a fluid and retained within the body housing **210**. The cleaning device **10** can operate normally in a dry cleaning mode in which the vacuum assembly **400** is employed to suction in dirt and debris, but upon actuation of the wet cleaning switch **662**, the cleaning device **10** will also begin to emit fluid to aid in the cleaning process.

In an exemplary embodiment, fluid is pumped out of the fluid supply tank **610** by the fluid pump **622** and is forced through the tubing **620** and out of the left and right spray nozzles **630L**, **630R**. The spray nozzles **630L**, **630R** are directed toward the brushroll **150** in order to spray a central region of the brushroll **150** with fluid. Some of the sprayed fluid is also directed at the deflectors, which as indicated above will deflect the sprayed fluid onto a different region of the brushroll **150**, such as an outer region on the left and right sides adjacent the central region. In this way, the brushroll **150** can be substantially and evenly coated by fluid emitted from the spray nozzles **630L**, **630R**. When the coated brushroll **150** spins over a cleaning surface, the rotation of the brushroll **150** and the emitted fluid can work together to loosen dirt and debris from the surface. Once the dirt and debris is loosened, it mixes with the emitted fluid and creates a slurry which can then be suctioned into the cleaning device **10** through the central intake **126**. From the central intake **126**, the slurry travels up the hosing **230** and into the container **422** of the recovery tank **420** via the standpipe **424** where it can then be separated into basic components with the separator **440**. The slurry will travel down the sloped portions of the separator and through the drain **442**. Larger debris will be captured by the ridges of the drain **442**, while smaller debris and fluid will fall to the bottom of the container **422**. As the slurry enters the container **422** through the standpipe **424**, any particulates and fluid which enter with excessive velocity will impact the deflector **640** and be redirected toward the rear of the container **422** and eventually toward the drain **442**. The slurry will undergo separation, with the fluid and smaller particles passing through the drain **442** and the larger particles being retained by the ridges **446**.

When the fluid level detectors **449a**, **449b** sense that the fluid in the container **422** has reached the max level, an indicator will appear on the body assembly **200** indicating that the fluid must be emptied. In one embodiment, the cleaning device **10** can indicate that the fluid level has reached the max level by providing a red water droplet on the cleaning device **10**. The indicator can appear anywhere on the cleaning device **10**, and, for example, may appear on the top side **210d** of the housing body **210**. At this point, the latch assembly **470** of the recovery tank **420** can be actuated, and the recovery tank **420** can be removed from the first cavity within the body housing **210**. Without needing to remove the lid, any waste in the bottom of the container **422** can be emptied through the spout **450** in the top of the recovery tank **420**. The container **422** can be tipped to guide the retained fluid toward the rear of the container **422**, closer to the channel **451**. To continue disposal, the container **422** can be tipped further to allow fluid to flow through the channel **451** and out the spout **450**, all while larger particles and debris retained by the separator **440** remain within the container **422**, unable to be disposed with the fluid. Once waste in the bottom of the container **422** has been emptied, the separator **440** can be removed from the container **422**. Larger particles captured by the separator **440**, which were

too large to end up in the bottom of the container **422**, will be removed with the separator **440**. These larger particles can then be disposed of.

During a wet cleaning operation, the fluid supply tank **610** can run low or run out of fluid. When the supply of fluid reaches a low point, such as when the fluid supply tank **610** is empty, the cleaning device **10** will be prevented from operating in a wet cleaning operation. If this happens, an alert can be presented on the device **10** that the fluid supply tank **610** must be refilled before a wet cleaning operation can begin, or, if one was in progress, can continue. In some embodiments, the alert can use the same water droplet as is used to indicate a max fluid level has been reached. In other embodiments, when indicating that the fluid supply tank **610** has run low, the water droplet can appear to pulse in a blue color, indicating that the fluid supply tank requires additional fluid. Once the fluid supply tank **610** is refilled, the cleaning device **10** can continue in the wet cleaning operation as before, or it may begin a new wet cleaning operation.

In other embodiments, as shown in FIGS. **16A-16G**, a charging mat **700** can be provided for use with the cleaning device **10**. The illustrated charging mat **700** includes a substantially square base **710** with a depressed region **720** in the center. The depressed region **720** can be sized to correspond to the bottom of the head assembly **100**, and it can support the cleaning device **10** therein. The depressed region **720** can also include areas to receive the large and small wheels **112**, **114**, the brushroll in use, such as brushroll **150** or brushroll **160**, and the overall head housing **110**. The charging mat **700** can also include accessory holders **730** at a rear thereof, which can act as receptacles to receive attachments for the cleaning device **10**. These attachments vary, and can include replacement brushrolls, such as brushroll **150** and brushroll **160**, and other tools for cleaning. In some embodiments, the accessory holders **730** may not be included.

Also located at the rear end of the charging mat **700** is a charging port **740** extending upwardly from the base **710**. The charging port **740** is configured to electrically couple to electrical contacts **220** located on the cleaning device **10**. The contacts **220** on the cleaning device **10** can be found at the rear of the body assembly **210**, as depicted in FIG. **16G**. Extending from the back of the charging mat **700** is a plug **750**, which can be inserted into an outlet to provide the entire charging mat **700**—and, in turn, the cleaning device **10**—with power. When connected to the charging mat **700**, the cleaning device **10** may turn on a battery life indicator to represent a simplified level of charge. In an embodiment, the battery life indicator is a battery image. As the cleaning device **10** charges to various levels, such as 25%, 50%, 75% and 100% capacity, the battery life indicator can proportionally indicate a relative charge level of the cleaning device. In other embodiments, the battery life indicator can be a different image, such as a pie chart, or a simple percentage indicator.

As the cleaning device **10** is used, the same battery indicator can deplete as the device **10** loses power. The battery indicator can indicate the same levels when in use as when charging, displaying the stages in a reverse order as the cleaning device **10** uses power. In some embodiments, the cleaning device **10** can rely on a separate indicator for use during operation of the device **10**, in addition to the indicator for use during charging.

Certain exemplary implementations have been described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the systems, devices, and methods disclosed herein. One or more

examples of these implementations have been illustrated in the accompanying drawings. Those skilled in the art will understand that the systems, devices, and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary implementations and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary implementation may be combined with the features of other implementations. Such modifications and variations are intended to be included within the scope of the present invention. Further, in the present disclosure, like-named components of the implementations generally have similar features, and thus within a particular implementation each feature of each like-named component is not necessarily fully elaborated upon.

Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise.

One skilled in the art will appreciate further features and advantages of the invention based on the above-described implementations. Accordingly, the present application is not to be limited by what has been particularly shown and described, except as indicated by the appended claims. All publications and references cited herein are expressly incorporated by reference in their entirety.

What is claimed is:

1. A fluid recovery tank for use on a cleaning device, comprising:
 - a container defining an inner chamber therein, the container having a suction outlet configured to couple to a suction source for allowing a suction force to be applied to the inner chamber, and the container having a fluid inlet for allowing fluid to be drawn into the chamber by the suction force;
 - a separator removably disposed within the container and having a separator opening formed therein and configured to allow fluid to pass therethrough while substantially preventing solid debris from passing therethrough such that solid debris is retained within the separator; and
 - a button extending forward from a front sidewall of the container such that the button is housed external to and forward of the container and configured to move between a locked configuration in which the button is configured to engage the cleaning device to retain the container on the cleaning device, and an unlocked configuration in which the button is configured to disengage from the cleaning device to allow the container to be removed from the cleaning device.
2. The fluid recovery tank of claim 1, wherein a top of the button is substantially parallel to a top of the container when the button is in the locked configuration and the unlocked configuration.
3. The fluid recovery tank of claim 1, wherein the button is biased to the locked configuration.

4. The fluid recovery tank of claim 1, wherein a top of the button extends above a top of the container when the button is in the locked configuration.

5. The fluid recovery tank of claim 4, wherein the top of the button extends above the top of the container when the button is in the unlocked configuration. 5

6. The fluid recovery tank of claim 1, wherein the button is disposed within a housing formed on an external surface of the container, and wherein the housing includes a biasing element disposed therein and biases the button to the locked configuration. 10

7. The fluid recovery tank of claim 1, wherein the button includes a protrusion configured to be received by a complementary depression on the cleaning device.

8. The fluid recovery tank of claim 7, wherein the button is curved to align with a contour of the cleaning device. 15

9. The fluid recovery tank of claim 1, further comprising a lid removably disposed within a top opening in the separator, the lid defining the suction outlet.

10. The fluid recovery tank of claim 9, wherein the lid includes at least one filter disposed therein and extending across the suction outlet. 20

11. The fluid recovery tank of claim 10, wherein the at least one filter is a first filter material having a first porosity and a second filter material having a second porosity greater than the first porosity. 25

12. The fluid recovery tank of claim 1, wherein the fluid inlet comprises a hollow standpipe extending into the inner chamber.

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